

Huawei Access Certification Training

HCIP-Access

Lab Guide for Access Engineers

ISSUE:2.5



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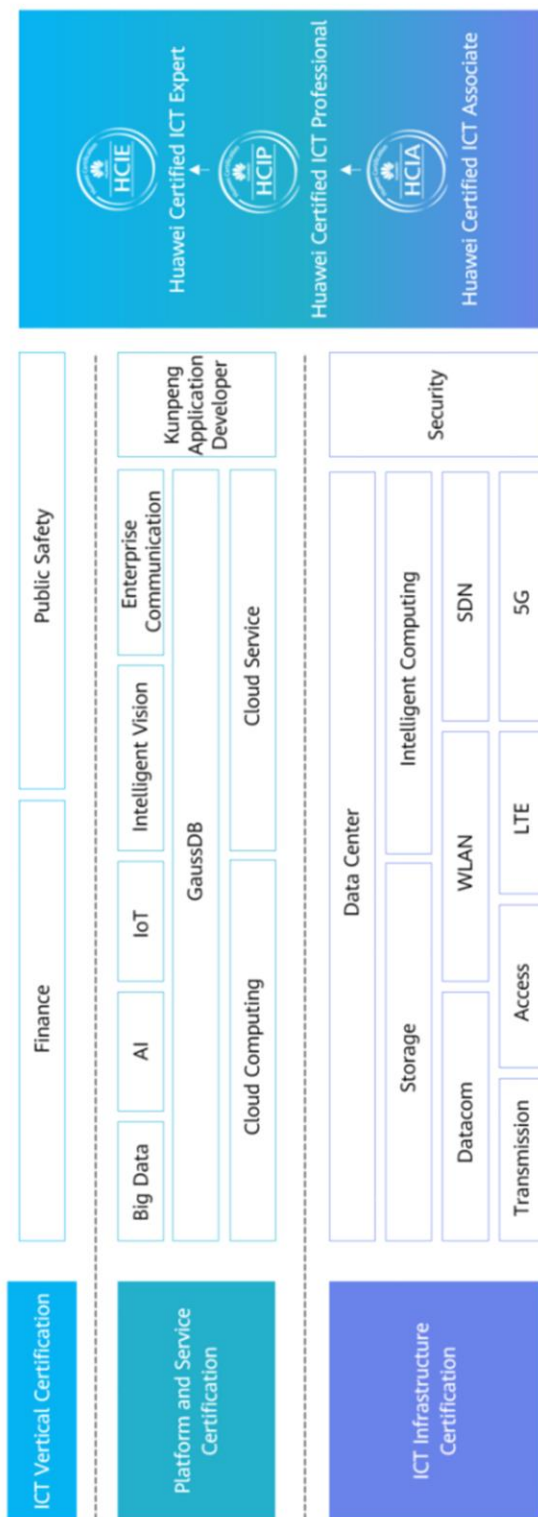
HCIP-Access (Huawei Certified ICT Professional-Access, Huawei Certified Network Professional-Access) is intended for Huawei representative offices, frontline engineers of representative offices, and other personnel who want to learn Huawei access products. HCIP-Access authentication covers VoIP principles and configuration, multicast principles and configuration, PON security basics, QoS basics, PON networking protection, NMS service configuration, access network planning and design, and typical PON troubleshooting.

The HCIP-Access certificate system introduces you to the industry and market, helps you in innovation, and enables you to stand atop the access frontiers.

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Huawei Certification



About This Document

Introduction

This document is a training course for HCIP-ACCESS certification. It is intended for trainees who are going to take the HCIP-ACCESS exam or readers who want to understand access network technologies such as features of Huawei access network products and related service configurations.

Description

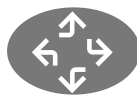
This document consists of five exercises, which describe the configuration of the voice service, multicast service, QoS, security protection, and networking protection in the access network one by one.

- Experiment 1: Configuring the SIP/H.248 VoIP service in the GPON scenario. This experiment helps you understand the configuration roadmap and common test methods of the GPON SIP/H.248 VoIP service on the ONU.
- Experiment 2: The basic configuration of the IPTV multicast service in the GPON scenario. By performing operations and configurations, you can get familiar with the configuration roadmap and common query commands of the IPTV service in the GPON scenario.
- Experiment 3: PON QoS configuration. This exercise aims to help you understand how to configure traffic classification, priority application, traffic shaping, and traffic adjustment.
- Experiment 4: PON security feature configuration. This exercise aims to help readers understand how to defend against DoS/ICMP/IP attacks, filter MAC addresses, and manage address segments.
- Experiment 5: GPON protection configuration. This exercise aims to help you understand how to configure and verify GPON downstream and upstream protection, including GPON type B protection, GPON type C protection, Ethernet link aggregation, and Ethernet port protection groups.

Background Knowledge Required

- This course is a basic course for Huawei certification. To better understand this document, the readers of this document must meet the following requirements:

Common Icons

**NMS****ONT****Splitter****PC****BRAS/Router****MDU****Switch****IPTV Server****OLT****Soft switch****Telephone**

Experiment Environment Overview

Networking Introduction

This lab environment is intended for access network engineers who are preparing for the HCIP-Access exam. 2 OLTs, 2 to 16 ONUs/MDUs, 1 core switch, 1 eSight/NCE-FAN server, 1 AR/BRAS device (terminating PPPoE dialup and functioning as a network gateway), 1 soft switch, and 1 IPTV server, each lab environment is suitable for 1 to 16 trainees.

Version Description

This document is also compatible with the MA5680T, MA5800, and EA5800. The CLI and NMS operations described in this document are based on the MA5680T. If the MA5800 or EA5800 is used, perform the operations according to the instructions of the trainer.

Device Introduction

To meet the HCIP-Access lab requirements, you are advised to configure each lab environment as follows:

Device Name	Device Model	Software Version
Core switch	S3700-28TP-PWR-EI or S5700-28C-PWR-EI	Version 5.70 (S3700 V100R005C01SPC100) or later Version 5.130 (S5700 V200R003C00SPC300) or later
OLT device	MA5680T or MA5800 or EA5800	MA5680T: V800R013C00 and later/EA5800: V100R019C20
ONT device	HG8247H9 (EG8145V5)	AP4030DN V200R007C10SPC100
MxU	EA5821 and EA5822 (P612E/ P805E)	V800R018C00
optical splitter	1:16, 2:16 or higher	
NMS server	eSight PON	V300R010C00
NMS server	NCE-FAN	R19C00
SIP server	Huawei SoftX3000	Other SIP server software can be used instead, such as TeKSIP.
Router/BRAS	AR1220/ME60	AR V200R005

Experiment Environment Preparation

Checking Whether All Devices Are Available

Before starting the experiment, check whether all required devices are ready. The following table lists the required devices.

Device Name	Quantity	Remarks
eSight&NCE-FAN	1 PCS	Shared by all experimental groups
SIP server	1 PCS	Shared by all experimental groups
Huawei S3700 or Huawei 5700 switch	All groups share one	Shared by all experimental groups
AR1220 router/ME60	1 for all groups	Shared by all experimental groups

OLT	2 in all groups	Shared by all experimental groups
ONU	Two for each group	One desktop ONT and one MDU
optical splitter	Two to four	Shared by all experimental groups Ensure that the split ratio is proper.
Notebook or desktop computer	One for each group	A desktop computer must have two network adapters.
RJ11 ordinary phone	Two for each group	All groups can share two devices for testing.
Twisted pair	One for each group	Used for data service tests
Console cable	One for each group	Alternatively, use Telnet.

Each group must check whether the following devices are ready:

- Prepare one desktop ONT and one MDU.
- Prepare one laptop or desktop computer.
- One twisted pair cable.
- One console cable (or a remote Telnet device)
- A telephone set (which can be shared by all groups).

Clearing OLT Configurations

To avoid the impact of residual configurations on the experiment, the environment builder must clear the configuration information saved on the device after the experiment is complete and before the device is shut down. At the beginning of the experiment, ensure that the device starts from the empty configuration. Otherwise, clear the configuration and restart the device.

Note: If the trainee logs in to the OLT through Telnet, do not perform this operation. Otherwise, the account data created on the OLT will be lost and the trainee will fail to log in to the OLT.

You need to enter the user name and password to log in to the OLT. In this experiment, the default user name and password are root and admin respectively.

```
>>User name:root
>>User password:
Huawei Integrated Access Software (MA5680T).
Copyright(C) Huawei Technologies Co., Ltd. 2002-2013. All rights reserved.
-----
User last login information:
-----
Access Type: Serial
IP-Address: --
Login Time: 2018-04-17 10:30:32+08:00
Logout Time: 2018-04-17 11:00:51+08:00
-----
-----
All user fail login information:
-----
Access Type IP-Address Time Login Times
-----
Telnet 10.175.195.116 2018-04-17 10:46:06+08:00 1
Serial -- 2018-04-17 10:30:28+08:00 1
-----
User root has used a default password. Change the password in time.
huawei>
```

To clear the configuration, run the following command:

```
huawei>enable
huawei#
huawei#erase
{flash<K>}:flash
{data<K>}:data
{Options<E><active,standby>}:active
Command:
erase flash data active
This command will clear the active board data that has been loaded or saved
Please remember to backup the system configuration data
Are you sure to continue? (y/n)[n]:y
Command executes successfully, and will take effect after active board
rebooted
```

To restart an OLT, run the following command:

```
huawei#reboot
{Options<E><system,active,standby>}:active
Command:
reboot active
Please check whether data has saved, the unsaved data may lose if reboot
active board, are you sure to reboot active board? (y/n)[n]:y
Standby board failure or not exist, reboot active will cause system reboot,
```

```
are you sure to reboot active board? (y/n)[n]:y
huawei#
Platform System start...
Memory Data Bus Test.....pass.
Memory Address Bus Test.....pass.
Press Ctrl+T to start the memory test.
Trace code: 0x00000000
Copy from ROM to RAM..... OK!
Uncompress from ROM to RAM..... OK!
Update fpga...OK.
Copyright (c) 1998 - 2013 by Huawei Technologies Co.,Ltd.
All Rights Reserved.
The last update date of base BIOS B is: Jan 15 2013 16:39:10
=====
BOARD INFORMATION:
CPU: MPC8548E - Security Engine. Processor #1.(Rev 2.02) running at 1333Mhz
512M bytes DDR(ECC), 128M bytes flash installed on board
Baudrate of serial channel is 9600bps
Board's LAN IP address is 10.11.104.6
Board's LAN Subnet mask is 255.255.255.0
Board' s LAN MAC address is 08:19:A6:C1:6A:52
=====
Base BIOS version is 307
Save extended BIOS disable start flag...OK!
Transferring control to the extended BIOS...OK!
System Vfs init...OK
The last update date of extended BIOS is: Feb 28 2013 16:23:52
System is booting from extended BIOS...
Extended BIOS version is 311
Press <D> key to stop auto-boot 0
Now system will boot from flash memory.
Testing Flash memory...OK!
Select area information: Program Area A ,Data Area A.
Begin to expand program... OK!
Save extended BIOS enable start flag...OK!
Transferring control to the loaded program...OK!
Starting system application init.....successfully!
Starting system configuration data init...successfully!
Press any key to get started
>>User name:
```

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1 GPON ONT SIP Voice Service Configuration

1.1 Experiment Introduction

1.1.1 About This Experiment

In this experiment, data is configured on the OLT and ONT so that two users connected to different ONTs connected to the same PON port can make voice calls.

1.1.2 Objectives

- Understand the basic principles of GPON.
- Understand the basic principles of VoIP.
- Understand the GPON data profile.
- Configure voice service profiles.
- Understand the overall configuration process of the voice service.

1.1.3 Experiment Networking

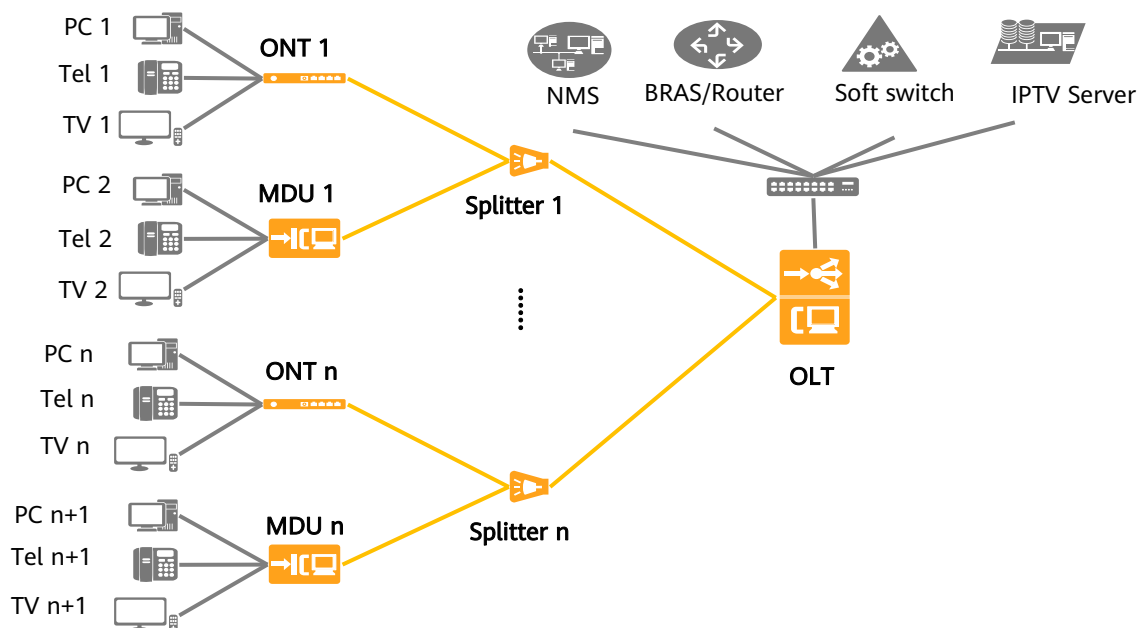


Figure 1-1 Topology of the GPON ONT SIP configuration experiment

1.1.4 Experiment Planning

According to the VoIP configuration scenario of the access network, phone 1 and phone 2 are common phones, and PC1 and PC2 run the Windows operating system.

Table 1-1 SIP data plan for a GPON ONT

Configuration Item	ONT1 data	Configuration Item	ONT2 data
OLT service VLAN	172. Smart	OLT service VLAN	172. Smart
OLT upstream port	0/20/0	OLT upstream port	0/20/0
OLT downstream port	0/3/0	OLT downstream port	0/3/0
ONT ID,SN	1, SN 48575443B205EC07	ONT ID,SN	2, SN 48575443B203E407
TCONT,DBA	1. DBA profile ID: 10	TCONT,DBA	2. DBA profile ID: 20
GEM Port	1	GEM Port	2
ONT user VLAN	172	ONT user VLAN	172

Table 1-2 Data plan for the GPON ONT SIP voice server

Configuration Item	Data	Configuration Item	Data
SIP server	200.200.200.200	Voice gateway	17.0.0.1/8
Port number of the SIP server	5060	Phone Number	7727033/7727034
IP address of the SIP local terminal	17.1.1.33/8(ONT1) 17.1.1.34/8(ONT2)	Username/Password	7727033/7727000, 7727034/7727000

1.2 Experiment Task Configuration

1.2.1 Configuration Roadmap

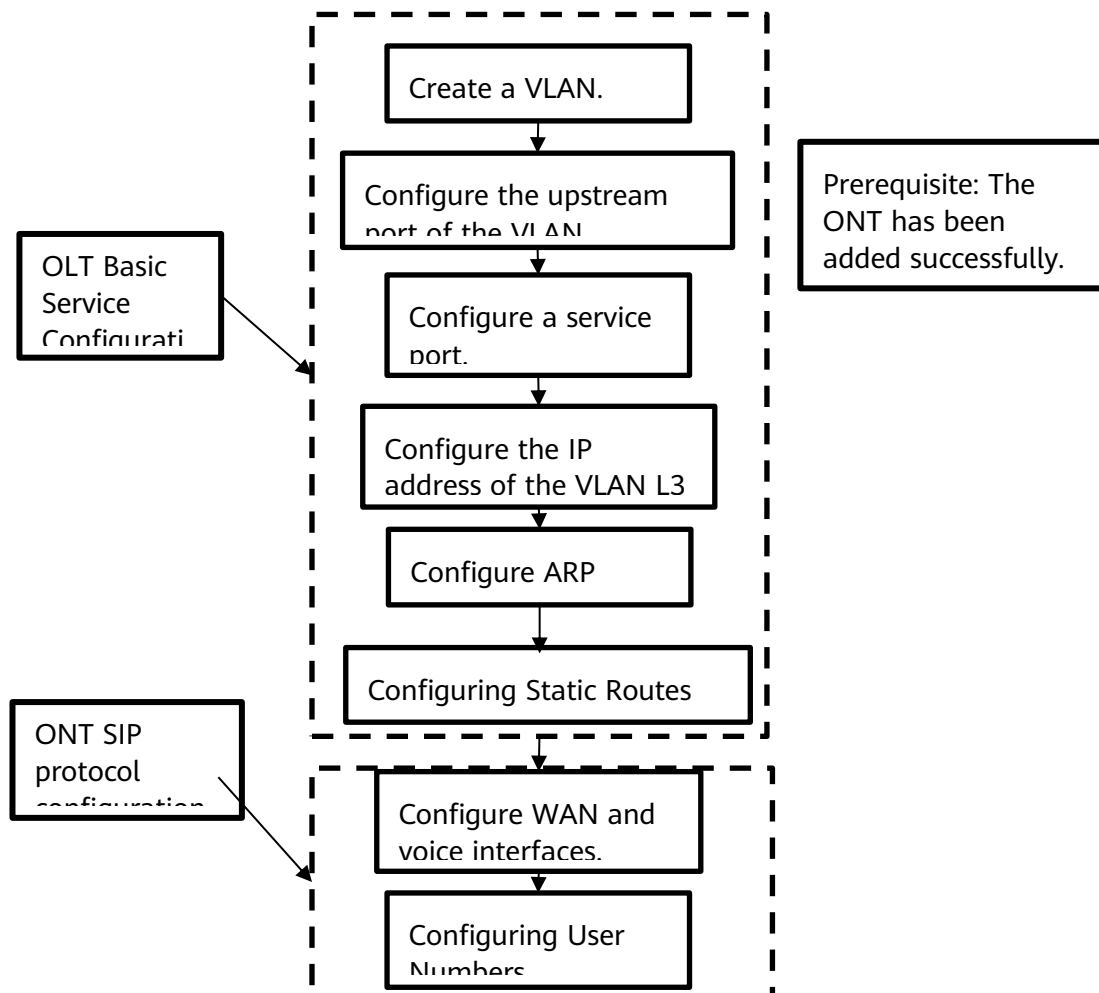


Figure 1-2 Flowchart for configuring SIP for a GPON ONT

1.2.2 Configuration Procedure (OLT Side)

Step 1 DBA Profile Configuration

Run the display dba-profile command to query the existing DBA profiles in the system. If the existing DBA profile in the system cannot meet the requirements, run the dba-profile add command to add a DBA profile. Set the profile ID to 10, bandwidth type to type1, and bandwidth value to 2048 kbit/s.

```

Huawei_OLT(config)#dba-profile
{add<K>|delete<K>|modify<K>}:add
{profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K>}:profile-id
{profile-id<U><10,512>}:10
  
```

```
{profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K>}:type1
{fix<K>}:fix
{fix-bandwidth<U><128,10000000>}:2048
{<cr>|bandwidth_compensate<K>}:
Command:
dba-profile add profile-id 10 type1 fix 2048
Adding a DBA profile succeeded
Profile ID: 10
Profile name: dba-profile_10
```

Display the DBA profile status.

```
Huawei_OLT(config)#display dba-profile profile-id 10
-----
Profile-name: dba-profile_10
Profile-ID: 10
type: 1
Bandwidth compensation: No
Fix-delay: No
Fix(kbps): 2048
Assure(kbps): 0
Max(kbps): 0
Additional-bandwidth: -
Best-effort-priority: -
Best-effort-weight: -
bind-times: 1
-----
```

Step 2 Configuring an ONT Line Profile

Set the line profile ID to 10 and profile name to 1hg8247.

```
Huawei_OLT(config)#ont-lineprofile
{epon<K>|gpon<K>}:gpon
{<cr>|profile-id<K>|profile-name<K>}:profile-id
{profile-id<U><0,8192>}:10
{<cr>|profile-name<K>}:profile-name
{profile-name<S><Length 1-32>}:1hg8247
Command:
ont-lineprofile gpon profile-id 10 profile-name 1hg8247
```

Bind T-CONT 1 to DBA profile 10.

```
HUAWEI(config-gpon-lineprofile-10)#tcont
{tcont-list<S><Length 1-13>}:1
{<cr>|dba-profile-id<K>|dba-profile-name<K>}:dba-profile-id
{profile-id<U><0,512>}:10
Command:
tcont 1 dba-profile-id 10
```

Add GEM port 1 to carry ETH service flows, and bind GEM port 1 to T-CONT 1.

```
HUAWEI(config-gpon-lineprofile-10)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:add
{gem-index<U><0,1023>}:1
```

```
{service-type<E><eth,tdm>}:eth
{tcont<K>}:tcont 1
{<cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K>}:
Command:
gem add 1 eth tcont 1
```

Map the service flow of CVLAN 172 to GEM port 1. The mapping index is 1.

```
HUAWEI(config-gpon-lineprofile-10)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:mapping
{gem-index<U><0,1023>}:1
{mapping-index<U><0,7>}:1
{e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K>}:vlan
{untag<K>|vlan-id<U><0,4095>}:172
{<cr>|flow-car<K>|priority<K>|transparent<K>}:
Command:
gem mapping 1 1 vlan 172
```

After the configuration is complete, run the commit command to make the configuration take effect.

```
HUAWEI(config-gpon-lineprofile-10)#commit
HUAWEI(config-gpon-lineprofile-10)#quit
```

Check information about the line profile.

```
Huawei_OLT(config)#display ont-lineprofile gpon
{all<K>|bound-info<K>|profile-id<K>|profile-name<K>}:profile-id
{profile-id<U><0,8192>}:10
Command:
display ont-lineprofile gpon profile-id 10
```

```
-----
Profile-ID :10
Profile-name :1hg8247
Access-type :GPON
-----
```

```
FEC upstream switch :Disable
OMCC encrypt switch :Off
Qos mode :PQ
Mapping mode :VLAN
TR069 management :Disable
TR069 IP index :0
-----
```

```
<T-CONT 0> DBA Profile-ID:1
<T-CONT 1> DBA Profile-ID:10
<Gem Index 1>
-----
```

```
|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
|Upstream-priority-queue:0 |Downstream-priority-queue:- |
-----
```

```
Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR
-----
```

```
1 172 - - - - -
```

Notes: Run the display traffic table ip command to query traffic table configuration

Binding times :0

Step 3 Configuring a Service Profile

Set the service profile ID to 10 and profile name to 1hg8247.

```
Huawei_OLT(config)#ont-srvprofile
{epon<K>|gpon<K>}:gpon
{<cr>|profile-id<K>|profile-name<K>}:profile-id
{profile-id<U><0,8192>}:10
{<cr>|profile-name<K>}:profile-name
{profile-name<S><Length 1-32>}:1hg8247
Command:
ont-srvprofile gpon profile-id 10 profile-name 1hg8247
```

Configure the port type and number of ports on the ONT. The number of LAN ports is 4, and the number of POTS ports is 2.

```
HUAWEI(config-gpon-srvprofile-10)#ont-port
{catv<K>|eth<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K>}:eth
{adaptive<K>|eth-port<U><0,8>}:4
{<cr>|catv<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K>}:pots
{adaptive<K>|pots-port<U><0,32>}:2
{<cr>|catv<K>|moca<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K>}:catv
{adaptive<K>|catv-port<U><0,8>}:1
{<cr>|moca<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K>|vdsl<K>}:
Command:
ont-port eth 4 pots 2 catv 1
```

After the configuration is complete, run the commit command to make the configuration take effect.

```
HUAWEI(config-gpon-srvprofile-10)#commit
HUAWEI(config-gpon-srvprofile-10)#quit
```

Check the service profile status.

```
Huawei_OLT(config)#display ont-srvprofile gpon profile-id 10
```

```
Profile-ID: 10
Profile-name: 1hg8247
Access-type: GPON
```

```
Port-type Port-number
```

```
POTS 2
ETH 4
VDSL 0
TDM 0
MOCA 0
```

CATV 1

TDM port type: E1
TDM service type: TDMoGem
MAC learning function switch: Enable
ONT transparent function switch: Disable
Ring check switch: Disable
Ring port auto-shutdown: Enable
Ring detect frequency: 8 (pps)
Ring resume interval: 300 (s)
Multicast forward mode: Unconcern
Multicast forward VLAN: -
Multicast mode: Unconcern
Upstream IGMP packet forward mode: Unconcern
Upstream IGMP packet forward VLAN: -
Upstream IGMP packet priority: -
Native VLAN option: Concern
Upstream PQ color policy: -
Downstream PQ color policy: -

Port-type Port-ID QinQmode PriorityPolicy Inbound Outbound

ETH 1 unconcern unconcern unconcern unconcern
ETH 2 unconcern unconcern unconcern unconcern
ETH 3 unconcern unconcern unconcern unconcern
ETH 4 unconcern unconcern unconcern unconcern

Notes: Run the display traffic table ip command to query
traffic table configuration

Port-type Port-ID DownstreamMode MismatchPolicy

ETH 1 operation discard
ETH 2 operation discard
ETH 3 operation discard
ETH 4 operation discard

Port-type Port-ID Dscp-mapping-table-index

ETH 1 0
ETH 2 0
ETH 3 0
ETH 4 0
IHOST 1 0

Port-type Port-ID IGMP-mode IGMP-VLAN IGMP-PRI Max-MAC-Count

ETH 1 - - - Unlimited
ETH 2 - - - Unlimited
ETH 3 - - - Unlimited
ETH 4 - - - Unlimited

Binding times: 0

Step 4 Enable the auto-discovery function of the PON port.

Enter the PON board view and enable the auto-discovery function of the PON port in the network topology.

```
Huawei_OLT(config)#interface
{adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|h
248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vds<K>|vlanif<K>|vrgw<K>}:gpon
{frameid/slotid<S><Length 3-15>}:0/3
Command:
interface gpon 0/3
HUAWEI(config-if-gpon-0/3)#port
{dba<K>|optical-alarm-profile<K>|portid<U><0,7>}:0
{fec<K>|laser-switch<K>|ont-auto-find<K>|ont-password-renew<K>|range<K>}:ont-auto-find
{switch<E><enable,disable>}:enable
Command:
port 0 ont-auto-find enable
```

Step 5 Querying the ONTs that are automatically discovered by a PON port

```
HUAWEI(config-if-gpon-0/3)#display ont autofind 0
```

```
-----
Number: 1
F/S/P: 0/3/0
Ont SN: 48575443B205EC07 (HWTC-B203E407)
Password: 0x00000000000000000000
Loid:
Checkcode:
VendorID: HWTC
Ont Version: 120D4600
Ont SoftwareVersion: V1R006C01S220
Ont EquipmentID: 247
Ont autofind time: 2018-04-13 15:56:02+08:00
-----
```

Step 6 ONT registration

Register an ONT, set the authentication mode to SN authentication and management type to OMCI, and bind line profile 10 and service profile 10 to the ONT.

```
HUAWEI(config-if-gpon-0/3)#ont
{activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K>}:confirm
{portid<U><0,7>}:0
{all<K>|loid-auth<K>|ontid<K>|password-auth<K>|sn-auth<K>}:ontid
{ontid<U><0,127>}:1
{loid-auth<K>|password-auth<K>|sn-auth<K>}:sn-auth
{sn-value<S><Length 12-16>}:48575443B205EC07
{omci<K>|password-auth<K>|snmp<K>}:omci
{<cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K>|ont-srvprofile-id<K>|ont-srvprofile-
name<K>}:ont-lineprofile-id
```

```
{profile-id<U><0,8192>}:10
{<cr>|desc<K>|ont-srvprofile-id<K>|ont-srvprofile-name<K>}:ont-srvprofile-id 10
{<cr>|desc<K>}:
Command:
ont confirm 0 ontid 1 sn-auth 48575443B205EC07 omci ont-lineprofile-id 10 ont-srvprofile-id 10
Number of ONTs that can be added: 1, success: 1
PortID :0, ONTID :1
```

Display the ONT status.

```
HUAWEI(config-if-gpon-0/3)#display ont info 0 all
-----
F/S/P ONT SN Control Run Config Match Protect
ID flag state state state side
-----
0/ 3/0 1 48575443B205EC07 active online normal match no
-----
F/S/P ONT-ID Description
-----
0/ 3/0 1 ONT_NO_DESCRIPTION
-----
In port 0/ 3/0, the total of ONTs are:1, online: 1
```

Run the display ont info 0 1 command to query the detailed information about the ONT.

Step 7 Configure a voice service VLAN.

Configure voice VLAN 172.

```
Huawei_OLT(config)#vlan
{attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-
policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093>}:172
{<cr>|to<K>|vlantype<E><mux,standard,smart,super>}:smart
Command:
vlan 172 smart
```

Add an upstream interface to the voice VLAN.

```
Huawei_OLT(config)#port
{desc<K>|mode<K>|vlan<K>}:vlan
{name<K>|vlan-list<S><Length 1-256>|vlanid<U><1,4093>}:172
{frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K>}:0/20
{portlist<S><Length 1-256>}:0
Command:
port vlan 172 0/20 0
```

Display information about VLAN 172.

```
Huawei_OLT(config)#display vlan 172
{<cr>|inner-vlan<K>|to<K>}:
Command:
display vlan 172
VLAN ID: 172
VLAN name: VLAN_0172
```

```

VLAN type: smart
VLAN attribute: common
VLAN description:
VLAN forwarding mode in control board: VLAN-MAC
VLAN forwarding mode: VLAN-MAC
VLAN broadcast packet forwarding policy: forward
VLAN unknown multicast packet forwarding policy: forward
VLAN unknown unicast packet forwarding policy: forward
VLAN bind service profile ID: -
VLAN bind RAIO profile index: -
VLAN priority: -
-----
F /S /P Native VLAN State
-----
0 /20/0 1 up
-----
Standard port number: 1
Service virtual port number: 0

```

Step 8 Configuring a VLAN Layer 3 Interface

Set the IP address of the Layer 3 interface to 17.1.1.100/8.

```

Huawei_OLT(config)#interface
{adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|h
248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vdsl<K>|vlanif<K>|vrgw<K>}:vlanif
{<1-4093>}:172
Command:
interface vlanif 172
HUAWEI(config-if-vlanif172)#ip
{address<K>|binding<K>|verify<K>}:address
{ip_addr<I><X.X.X.X>}:17.1.1.100
{integer<U><0,32>|ip_addr<I><X.X.X.X>}:8
{<cr>|description<K>|sub<K>}:
Command:
ip address 17.1.1.100 8

```

Step 9 Configure a service port in the voice VLAN.

Set the VLAN to which the service flow belongs to 172, the PON port to which the service flow belongs to 0/3/0, the ONT ID to 1, the GEM port ID to 1, and the C-VLAN to 172.

```

Huawei_OLT(config)#service-port
{desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K>}:vlan
{aoe<K>|name<K>|vlanid<U><1,4093>}:172
{adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K>}:gpon
{frameid/slotid/portid<S><Length 5-18>}:0/3/0
{ont<K>}:ont
{ontid<U><0,127>}:1
{eth<K>|gemport<K>|iphost<K>}: gemport
{gemindex<U><0,1023>}:1
{<cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K>}:multi-service
{user-8021p<K>|user-encap<K>|user-vlan<K>}:user-vlan
{other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095>}:172

```

```
{<cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K>}:
Command:
```

```
service-port vlan 172 gpon 0/3/0 ont 1 gempport 1 multi-service user-vlan 172
```

Check the status of the service virtual port.

```
Huawei_OLT(config)#display service-port all
```

```
{<cr>|sort-by<K>||<K>}:
Command:
```

```
display service-port all
```

```
Switch-Oriented Flow List
```

```
-----
INDEX VLAN VLAN PORT F/ S/ P VPI VCI FLOW FLOW RX TX STATE
ID ATTR TYPE TYPE PARA
-----
```

```
0 172 common gpon 0/3 /0 1 1 vlan 172 - - up
```

```
-----
Total: 1 (Up/Down: 1/0)
```

The status of the service virtual port is up.

Step 10 Check voice VLAN 172.

Run the display vlan 172 command to check the status of the upstream and downstream interfaces of VLAN 172.

```
Huawei_OLT(config)#display vlan 172
```

```
{<cr>|inner-vlan<K>|to<K>}:
Command:
```

```
display vlan 172
```

```
VLAN ID: 172
```

```
VLAN name: VLAN_0172
```

```
VLAN type: smart
```

```
VLAN attribute: common
```

```
VLAN description:
```

```
VLAN forwarding mode in control board: VLAN-MAC
```

```
VLAN forwarding mode: VLAN-MAC
```

```
VLAN broadcast packet forwarding policy: forward
```

```
VLAN unknown multicast packet forwarding policy: forward
```

```
VLAN unknown unicast packet forwarding policy: forward
```

```
VLAN bind service profile ID: -
```

```
VLAN bind RAIO profile index: -
```

```
VLAN priority: -
```

```
-----
F /S /P Native VLAN State
-----
```

```
0 /20/0 1 up
```

```
-----
Standard port number: 1
-----
```

```
INDEX TYPE STATE F /S /P VPI VCI FLOWTYPE FLOWPARA
-----
```

```
0 gpon up 0 /3 /0 1 1 vlan 172
```

```
-----
Service virtual port number: 1
```

Note: F--Frame, S--Slot, P--Port,
VPI indicates ONT ID for PON, VCI indicates GEM index for GPON,
v/e--vlan/encap, pritag--priority-tagged

The status of the upstream port 0/20/0 and the service virtual port is normal.

Step 11 Enable global ARP proxy.

```
Huawei_OLT(config)#arp
{aging-mode<K>|aging-time<K>|conflict<K>|gratuitous-
arp<K>|ip_addr<I><X.X.X.X>|learning<K>|proxy<K>}:proxy
{disable<K>|enable<K>}:enable
Command:
arp proxy enable
```

Step 12 Enable ARP proxy on the Layer 3 interface.

```
Huawei_OLT(config)#interface vlanif 172
HUAWEI(config-if-vlanif172)#arp
{proxy<K>}:proxy
{disable<K>|enable<K>}:enable
{<cr>|subvlan<K>}:
Command:
arp proxy enable
```

Step 13 Configuration of ONT 2

The configuration of ONT2 is similar to that of ONT1.

Step 14 Common Commands

Table 1-3 Common Commands

Common Commands	Function
display dba-profile	This command is used to query the information about the dynamic bandwidth assignment (DBA) profile in the system.
display ont-lineprofile	This command is used to query the information about the GPON ONT line profile that has taken effect, including the binding relationship between the T-CONT and the DBA profile, the QoS mode of the traffic stream, and the mapping between the GEM port and the ONT-side service.
display ont-srvprofile	This command is used to query the information about a GPON ONT service profile.
display ont autofind	This command is used to query the basic information about the currently auto-found ONT in the system.
display ont info	This command is used to query the information about an ONT, including the current status of the ONT, the related

	configuration of the ONT, and the information about the ONT T-CONT.
display vlan	This command is used to query the VLAN information.
display service-port	This command is used to query the information about a service virtual port.

1.2.3 Configuration Procedure (ONT Side)

Step 1 Log in to the HG8247.

Use a network cable to connect the PC to the network port of the HG8247 and set the IP address of the PC to 192.168.100.100/24.

Enter HYPERLINK "http://192.168.100.1/" http://192.168.100.1/ in the address box of the browser on the PC.

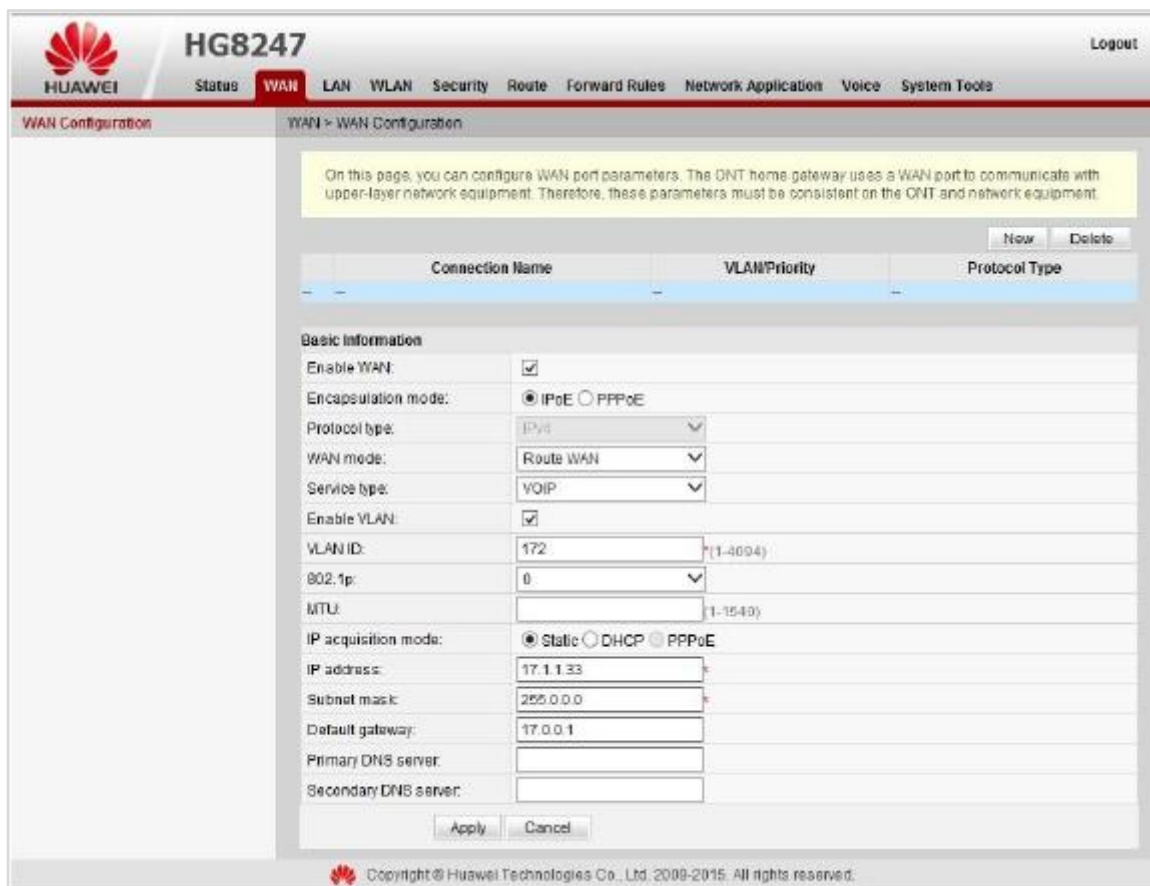
Login account: telecomadmin/admintelecom

The login page is displayed, as shown in the following figure.



Step 2 ONT WAN Port Configuration

Set the WAN port type to route WAN port, service type to VoIP, VLAN ID to 172, IP address of the ONT to 17.1.1.33, subnet mask to 255.0.0.0, and default gateway to 17.0.0.1. Click Apply.



HG8247 Logout

Status **WAN** LAN WLAN Security Route Forward Rules Network Application Voice System Tools

WAN Configuration WAN > WAN Configuration

On this page, you can configure WAN port parameters. The ONT home gateway uses a WAN port to communicate with upper-layer network equipment. Therefore, these parameters must be consistent on the ONT and network equipment.

New Delete

Connection Name	VLAN/Priority	Protocol Type

Basic Information

Enable WAN: ☒

Encapsulation mode: ☒ IPoE ☐ PPPoE

Protocol type: IPv4

WAN mode: Route WAN

Service type: VOIP

Enable VLAN: ☒

VLAN ID: 172 (1-4094)

802.1p: 0

MTU: (1-1540)

IP acquisition mode: ☒ Static ☐ DHCP ☐ PPPoE

IP address: 17.1.1.33

Subnet mask: 255.0.0.0

Default gateway: 17.0.0.1

Primary DNS server:

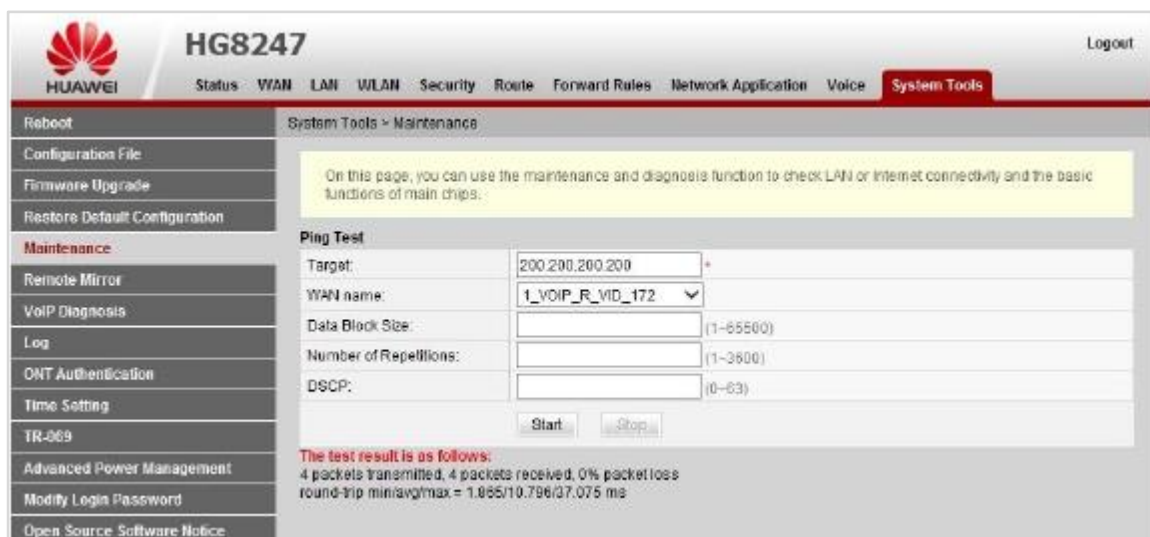
Secondary DNS server:

Apply Cancel

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Step 3 Connectivity test

Perform a ping test on the ONT to check whether the connectivity between the ONT and the softswitch is normal.



HG8247 Logout

Status WAN LAN WLAN Security Route Forward Rules Network Application Voice **System Tools**

System Tools > Maintenance

On this page, you can use the maintenance and diagnosis function to check LAN or Internet connectivity and the basic functions of main chips.

Ping Test

Target: 200.200.200.200

WAN name: 1_VOIP_R_VID_172

Data Block Size: (1-65500)

Number of Repetitions: (1-3600)

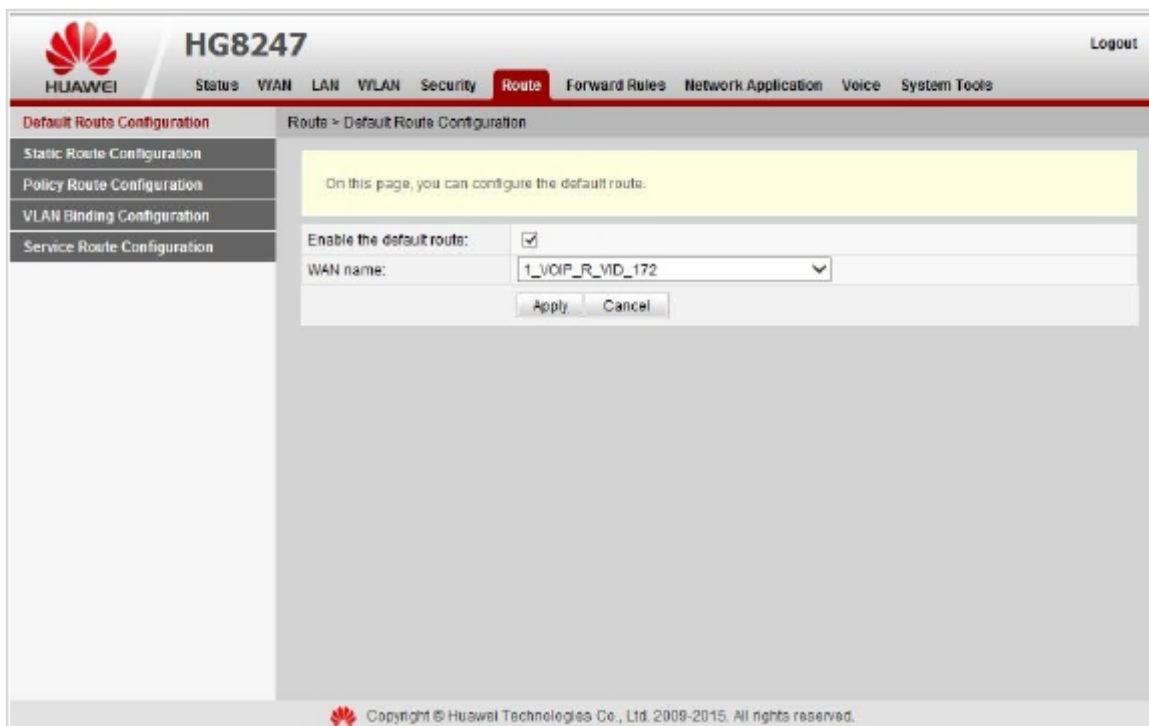
DSCP: (0-63)

Start Stop

The test result is as follows:
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.865/10.796/37.075 ms

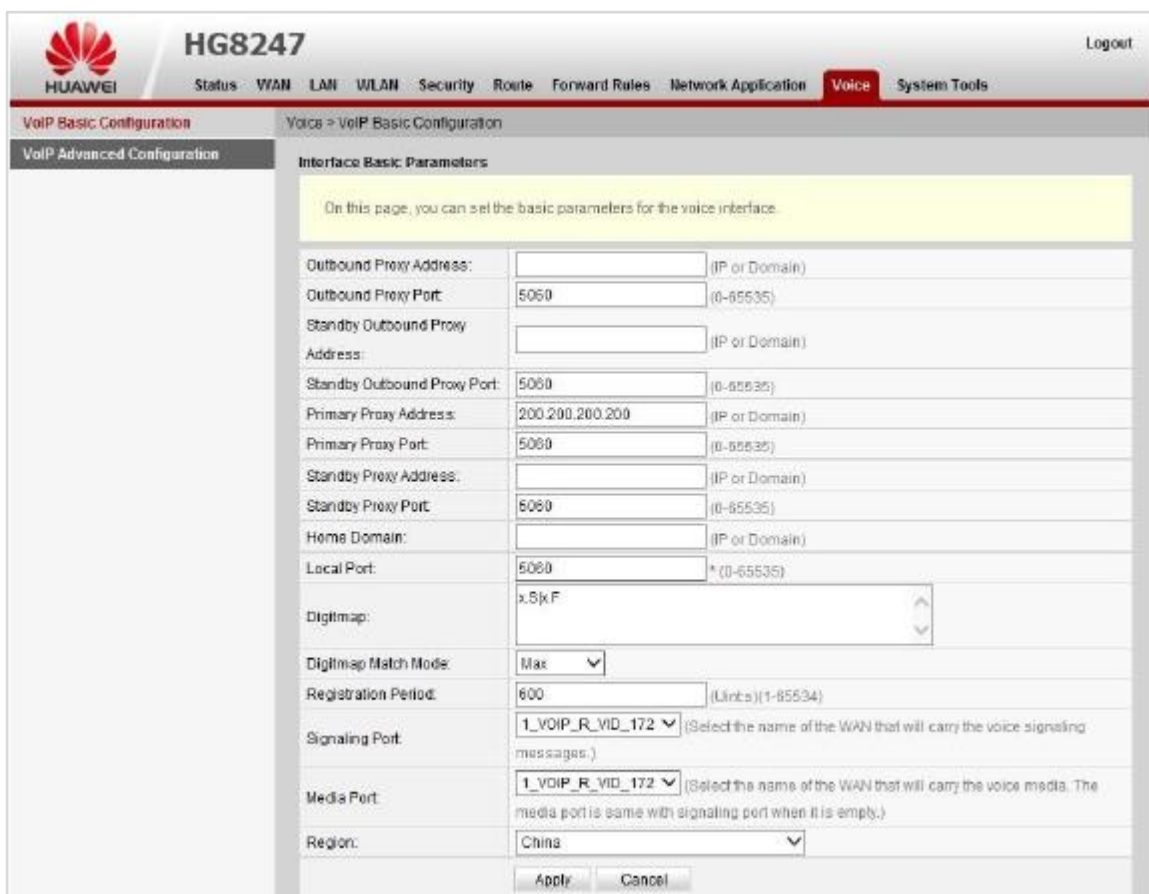
Reboot
Configuration File
Firmware Upgrade
Restore Default Configuration
Maintenance
Remote Mirror
VoIP Diagnosis
Log
ONT Authentication
Time Setting
TR-069
Advanced Power Management
Modify Login Password
Open Source Software Notice

Step 4 Route Configuration



The screenshot shows the Huawei HG8247 web interface. The top navigation bar includes links for Status, WAN, LAN, WLAN, Security, Route, Forward Rules, Network Application, Voice, and System Tools. The 'Route' tab is selected. On the left, there is a sidebar with 'Default Route Configuration' and 'Static Route Configuration'. The main content area is titled 'Route > Default Route Configuration'. It contains a yellow box with the text 'On this page, you can configure the default route.' Below this, there is a form with the following fields: 'Enable the default route:' with a checked checkbox, 'WAN name:' with a dropdown menu showing '1_VOIP_R_VID_172', and 'Apply' and 'Cancel' buttons. At the bottom, there is a copyright notice: 'Copyright © Huawei Technologies Co., Ltd. 2009-2015. All rights reserved.'

Step 5 ONT Voice Interface, Digitmap, and Digitmap Matching Mode Configuration

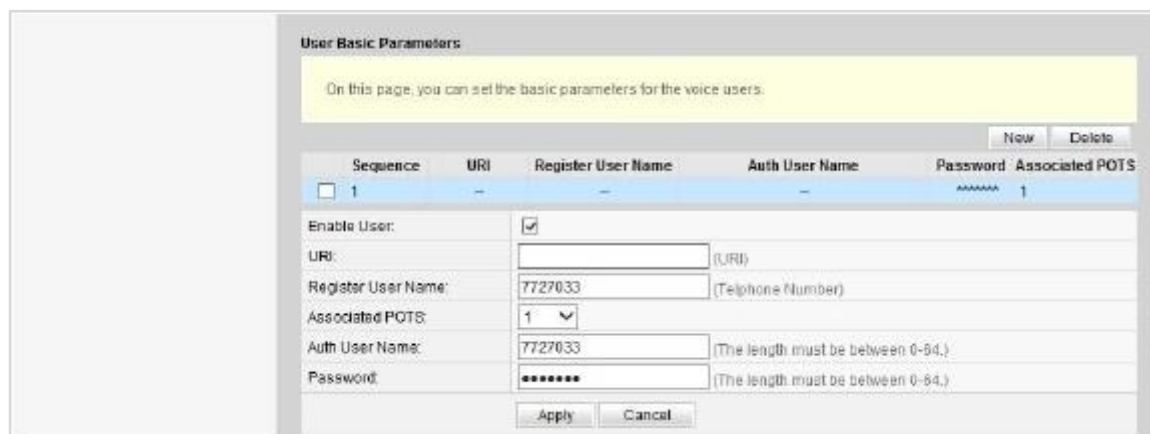


The screenshot shows the Huawei HG8247 web interface. The top navigation bar includes links for Status, WAN, LAN, WLAN, Security, Route, Forward Rules, Network Application, Voice, and System Tools. The 'Voice' tab is selected. On the left, there is a sidebar with 'VoIP Basic Configuration' and 'VoIP Advanced Configuration'. The main content area is titled 'Voice > VoIP Basic Configuration'. It contains a yellow box with the text 'On this page, you can set the basic parameters for the voice interface.' Below this, there is a form with the following fields: 'Outbound Proxy Address:' (IP or Domain), 'Outbound Proxy Port:' (5060, 0-65535), 'Standby Outbound Proxy Address:' (IP or Domain), 'Standby Outbound Proxy Port:' (5060, 0-65535), 'Primary Proxy Address:' (200.200.200.200, IP or Domain), 'Primary Proxy Port:' (5060, 0-65535), 'Standby Proxy Address:' (IP or Domain), 'Standby Proxy Port:' (5060, 0-65535), 'Home Domain:' (IP or Domain), 'Local Port:' (5060, * (0-65535)), 'Digitmap:' (x,5jKf), 'Digitmap Match Mode:' (Max), 'Registration Period:' (600, (Units) (1-65534)), 'Signaling Port:' (1_VOIP_R_VID_172, (Select the name of the WAN that will carry the voice signaling messages.)), 'Media Port:' (1_VOIP_R_VID_172, (Select the name of the WAN that will carry the voice media. The media port is same with signaling port when it is empty.)), 'Region:' (China), and 'Apply' and 'Cancel' buttons.

Table 1-4 Digitmap rules

Symbol	Meaning	Symbol	Meaning
T	Start timer	-	Range, for example, 2-4.
L	Long timer	.	Matches any structure that appears before it one or more times.
S	Short timer		Separator
E	Key (*)	Digit	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
F	Press #.	[]	Matches any number or character within the range of parentheses.
Wildcard x	Matches any number from 0 to 9.		

Step 6 Voice User Configuration



User Basic Parameters

On this page, you can set the basic parameters for the voice users.

Sequence URI Register User Name Auth User Name Password Associated POTS

1 - - - ***** 1

Enable User: ☒

URI: (URI)

Register User Name: 7727033 (Telephone Number)

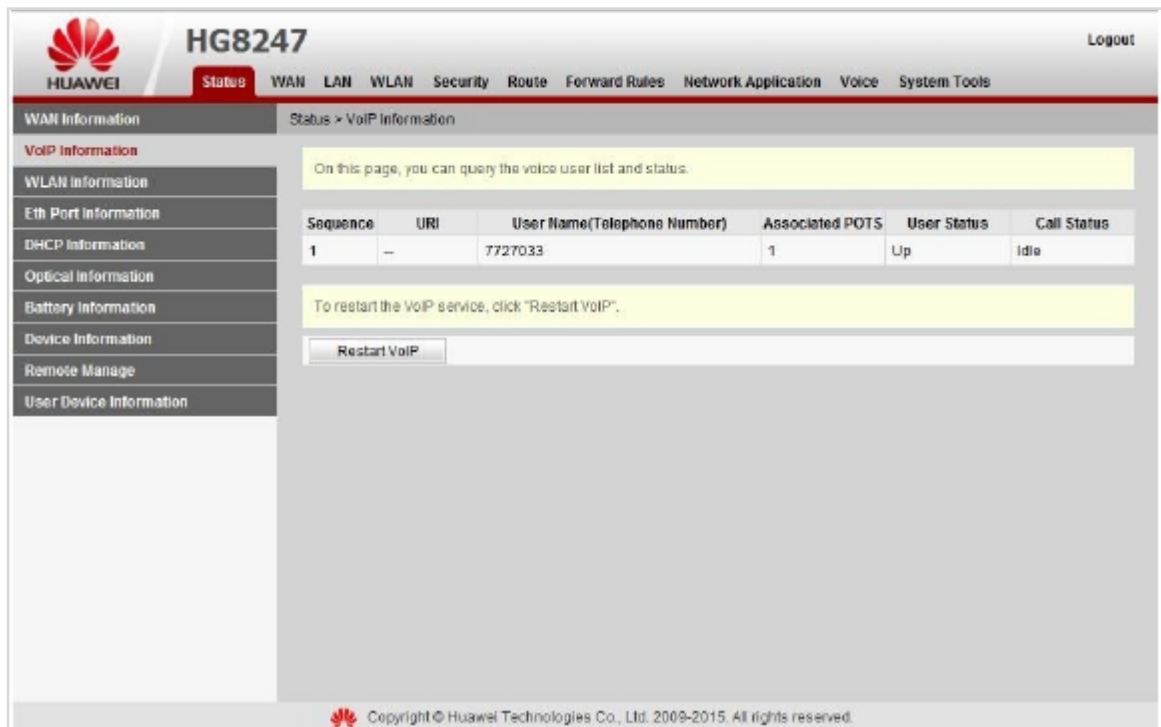
Associated POTS: 1

Auth User Name: 7727033 (The length must be between 0-64.)

Password: ***** (The length must be between 0-64.)

Apply Cancel

Step 7 View the configuration result.



HG8247 Logout

Status WAN LAN WLAN Security Route Forward Rules Network Application Voice System Tools

WAN Information Status > VoIP Information

VoIP Information

On this page, you can query the voice user list and status.

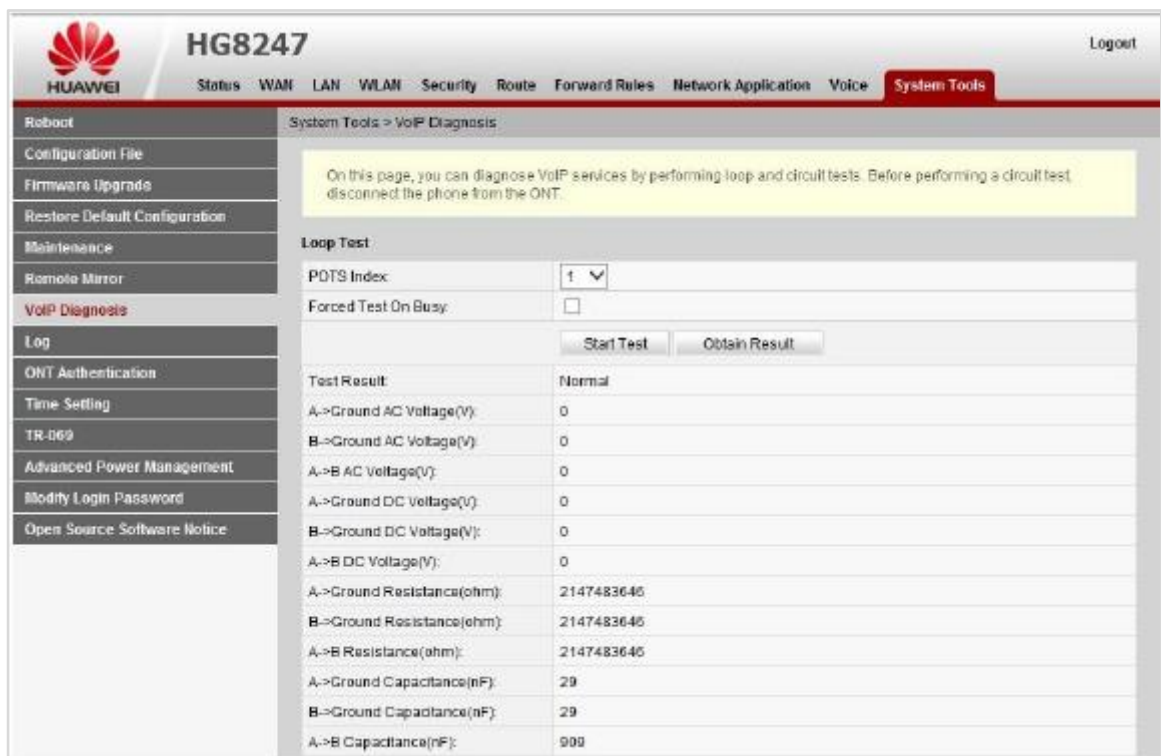
Sequence	URI	User Name(Telephone Number)	Associated POTS	User Status	Call Status
1	--	7727033	1	Up	Idle

To restart the VoIP service, click "Restart VoIP".

Restart VoIP

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Step 8 ONT loop line test



HG8247 Logout

Status WAN LAN WLAN Security Route Forward Rules Network Application Voice System Tools

Reboot Configuration File Firmware Upgrade Restore Default Configuration Maintenance Remote Mirror VoIP Diagnosis Log ONT Authentication Time Setting TR-069 Advanced Power Management Modify Login Password Open Source Software Notice

System Tools > VoIP Diagnosis

On this page, you can diagnose VoIP services by performing loop and circuit tests. Before performing a circuit test, disconnect the phone from the ONT.

Loop Test

POTS Index: 1

Forced Test On Busy: ☐

Start Test Obtain Result

Test Result: Normal

A->Ground AC Voltage(V):	0
B->Ground AC Voltage(V):	0
A->B AC Voltage(V):	0
A->Ground DC voltage(V):	0
B->Ground DC Voltage(V):	0
A->B DC Voltage(V):	0
A->Ground Resistance(ohm):	2147483645
B->Ground Resistance(ohm):	2147483645
A->B Resistance(ohm):	2147483645
A->Ground Capacitance(nF):	29
B->Ground Capacitance(nF):	29
A->B Capacitance(nF):	909

Step 9 ONT circuit line test

Circuit Test	
POTS Index:	1
Start Test Obtain Result	
Loop Current Test Result:	Normal
Feed Voltage Test Result:	Normal
Ringling Test Result:	Normal
On-hook/Off-hook Test Result:	Normal
Loop Current(mA):	26
Feed Voltage(mV):	-49833
Ringling Voltage(mV):	61590

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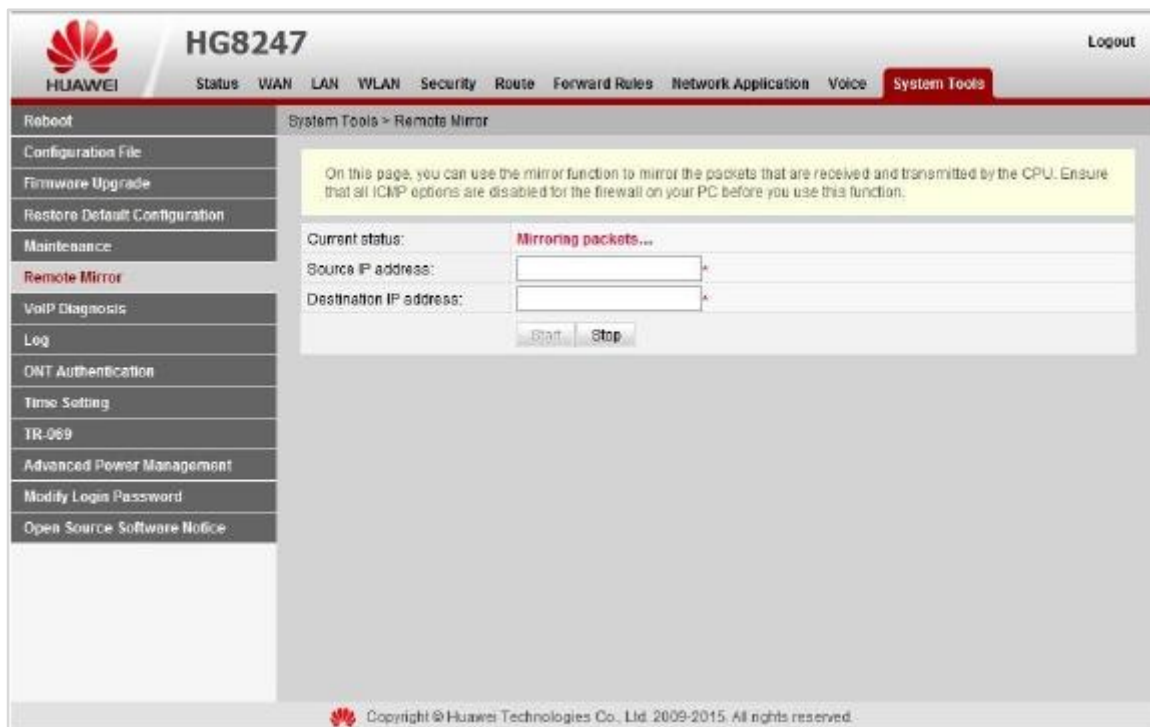
Step 10 (Optional) ONT remote mirroring

The source IP address is the management IP address of the ONT, and the destination IP address is the IP address of the PC connected to the LAN port.

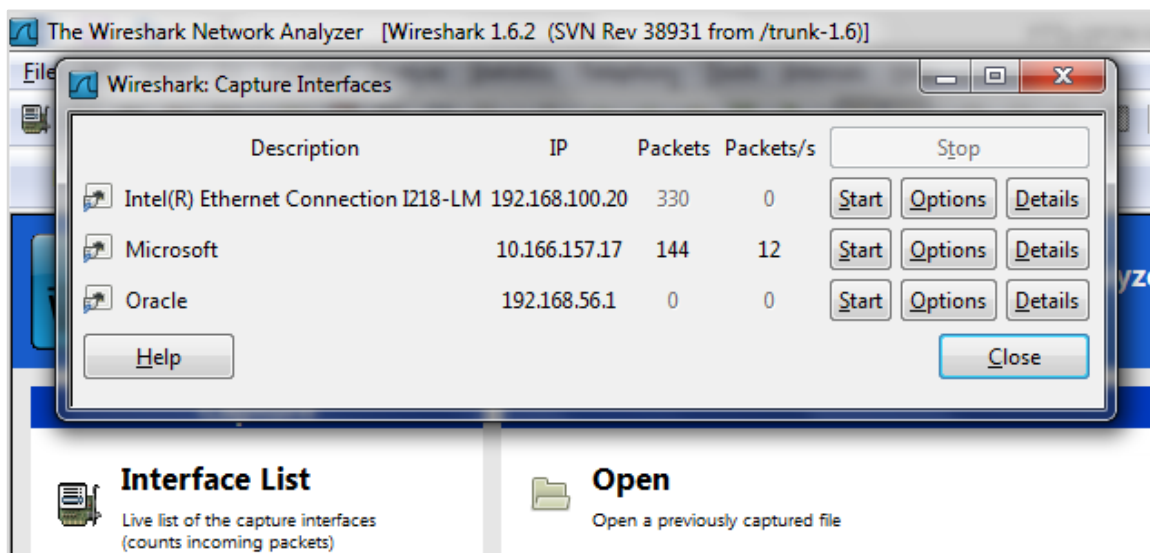
HG8247		Logout								
Status: WAN LAN WLAN Security Route Forward Rules Network Application Voice System Tools										
Reboot Configuration File Firmware Upgrade Restore Default Configuration Maintenance Remote Mirror VoIP Diagnosis Log ONT Authentication Time Setting TR.069 Advanced Power Management Modify Login Password Open Source Software Notice	System Tools > Remote Mirror On this page, you can use the mirror function to mirror the packets that are received and transmitted by the CPU. Ensure that all ICMP options are disabled for the firewall on your PC before you use this function. <table> <tr> <td>Current status:</td> <td>Stop</td> </tr> <tr> <td>Source IP address:</td> <td>192.168.100.1</td> </tr> <tr> <td>Destination IP address:</td> <td>192.168.100.20</td> </tr> <tr> <td colspan="2"> Start Stop </td> </tr> </table>		Current status:	Stop	Source IP address:	192.168.100.1	Destination IP address:	192.168.100.20	Start Stop	
Current status:	Stop									
Source IP address:	192.168.100.1									
Destination IP address:	192.168.100.20									
Start Stop										

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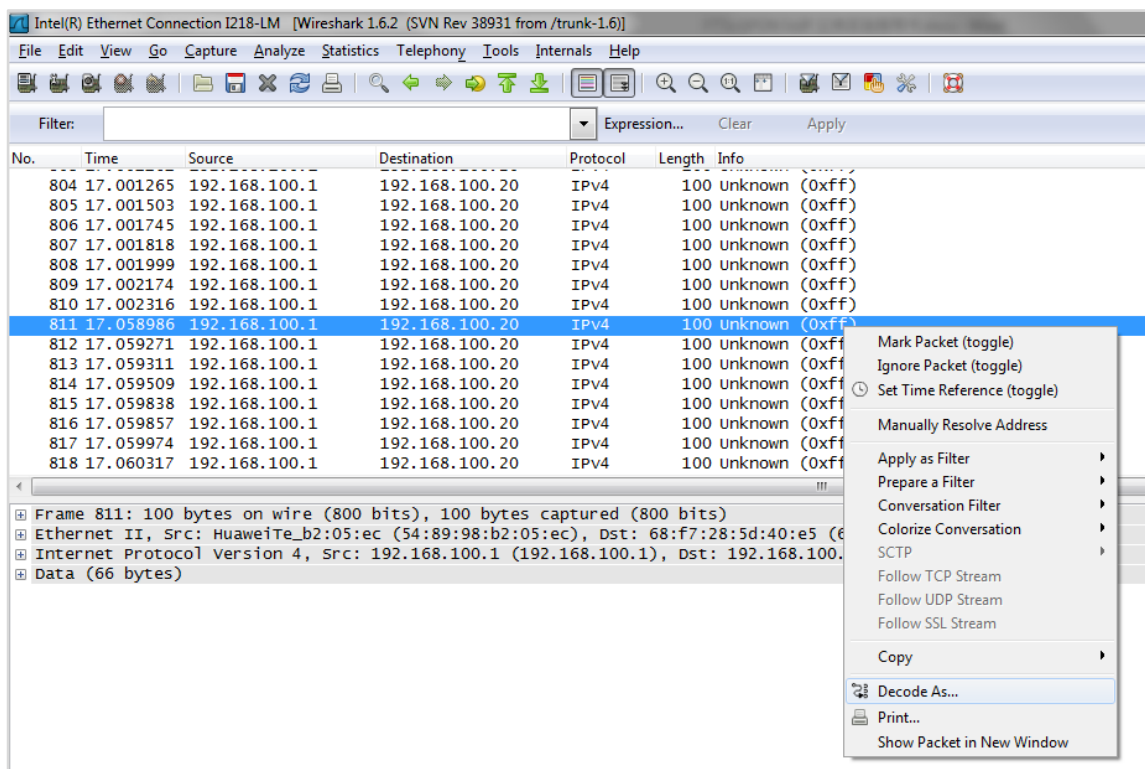
After setting the IP address, click Start.



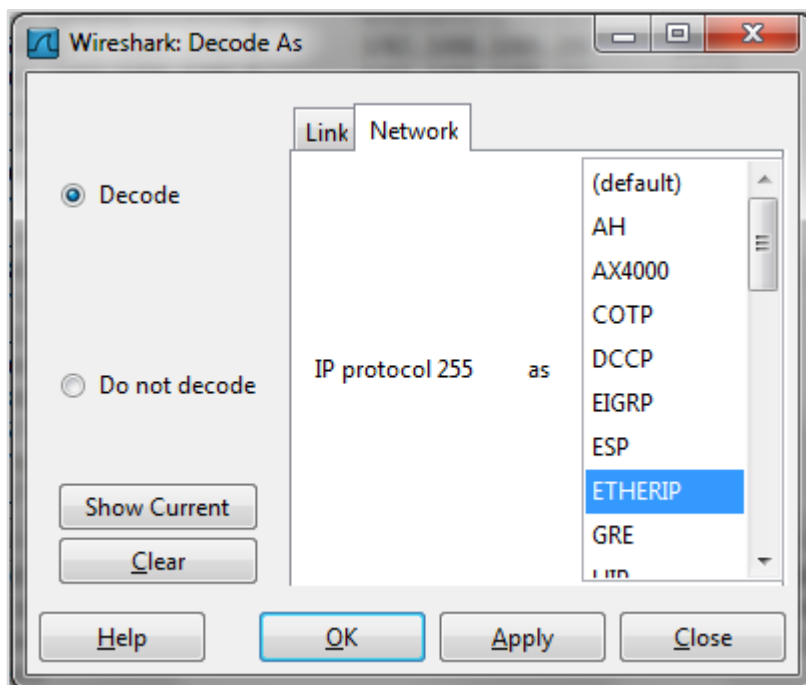
Run the Wireshark software on the PC to capture packets on the network adapter connected to the ONT LAN port.



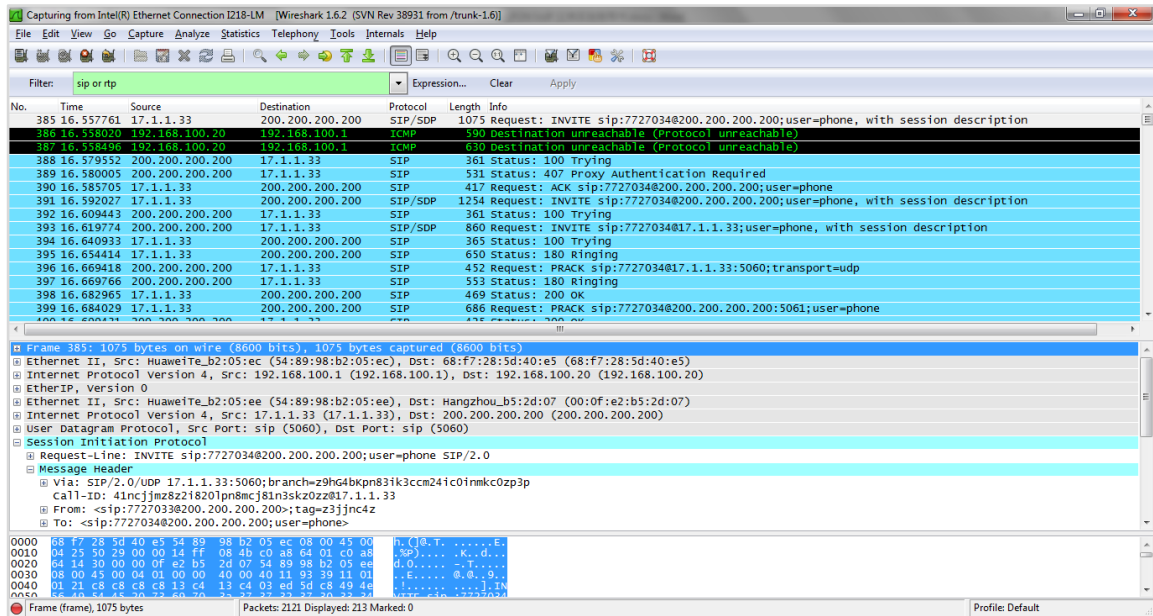
Right-click a packet whose Info is unknown and choose Decode As from the shortcut menu.



Select ETHERIP and click OK.



SIP and RTP packets can be obtained after the user picks up the phone and dials a number.



1.2.4 Experiment Verification

Verification method: Connect two phones to different POTS ports on the ONT and make calls to each other.

1.3 Configuration Reference

1.3.1 OLT Data Configuration

```
dba-profile add profile-id 10 type1 fix 2048
```

```
ont-lineprofile gpon profile-id 10 profile-name 1hg8247
tcont 1 dba-profile-id 10
gem add 1 eth tcont 1
gem mapping 1 1 vlan 172
commit
quit
```

```
ont-srvprofile gpon profile-id 10 profile-name 1hg8247
ont-port eth 4 pots 2catv 1
commit
quit
```

```
interface gpon 0/53
port 0 ont-auto-find enable
```

```
ont confirm 0 ontid 1 sn-auth 48575443B205EC07 omci ont-lineprofile-id 10 ont-srvprofile-id 10
quit
```

```
vlan 172 smart
port vlan 172 0/20 0
interface vlanif 172
```

```
ip address 17.1.1.100 8
quit
service-port 1 vlan 172 gpon 0/3/0 ont 1 gemport 1 multi-service user-vlan 172

arp proxy enable
interface vlanif 172
arp proxy enable
quit
```

1.4 Questions

1. If the ARP proxy is not enabled on the phones connected to two ONTs connected to the same PON port, can the phones communicate with each other? If not, please explain why?
2. Does ARP proxy need to be enabled for two ONTs connected to different PON ports?

2 GPON ONU SIP Voice Service Configuration

2.1 Experiment Introduction

2.1.1 About This Experiment

In this experiment, the data configuration of the OLT and the MDU is used to implement the voice call between two users of different MDUs connected to the same PON.

2.1.2 Objectives

- Understand the basic principles of GPON.
- Understand the basic principles of VoIP.
- Understand the GPON data profile.
- Master the configuration of the MDU data.
- Configure voice service profiles.
- Understand the overall configuration process of the voice service.

2.1.3 Experiment Networking

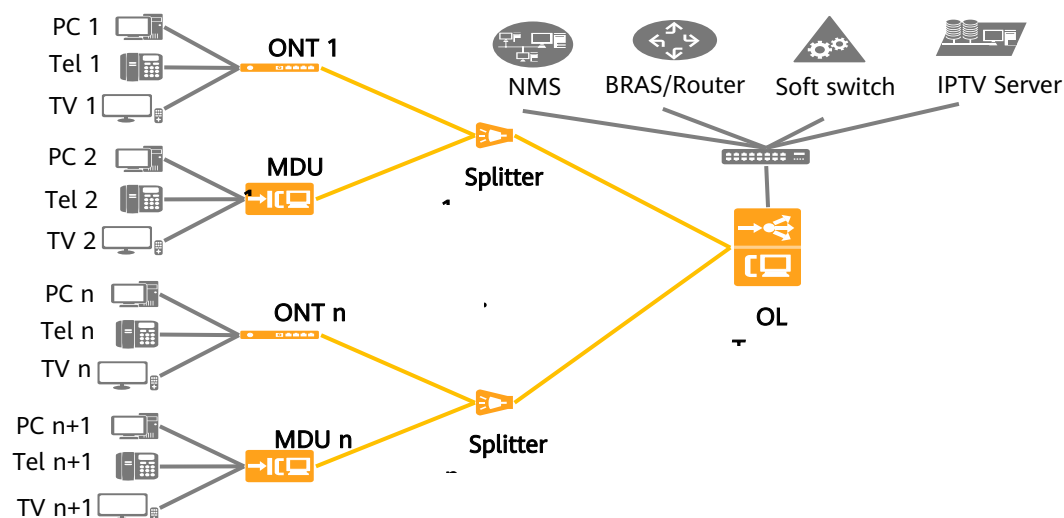


Figure 2-1 Topology for configuring the GPON ONU VoIP service

2.1.4 Experiment Planning

According to the VoIP configuration scenario of the access network, phone 1 and phone 2 are common phones, and PC1 and PC2 run the Windows operating system.

Table 2-1 GPON ONU VoIP data plan

Configuration Item	Data	Configuration Item	Data
OLT service VLAN	VLAN:172 Type: Smart IP:17.1.1.100/8	OLT service VLAN	VLAN:172 Type: Smart IP:17.1.1.100/8
OLT management VLAN	VLAN:500 Type: Smart IP:10.1.1.1/24	OLT management VLAN	VLAN:500 Type: Smart IP:10.1.1.1/24
OLT upstream port	0/20/0	OLT upstream port	0/20/0
OLT downstream port	0/3/0	OLT downstream port	0/3/0
MDU	ID:3 SN:4857544329FC9507 IP:10.1.1.3/24	MDU	ID:4 SN:48575443F2154B04 IP:10.1.1.4/24
TCONT,DBA	0 (management) 3 (service) DBA profile ID: 30	TCONT,DBA	0 (management) 4 (service) DBA profile ID: 40
GEM Port	0 (management) 3 (service)	GEM Port	0 (management) 4 (service)
ONT user VLAN	500 (management) 172 (service)	ONT user VLAN	500 (management) 172 (service)

Table 2-2 Data plan for the GPON ONU SIP voice server

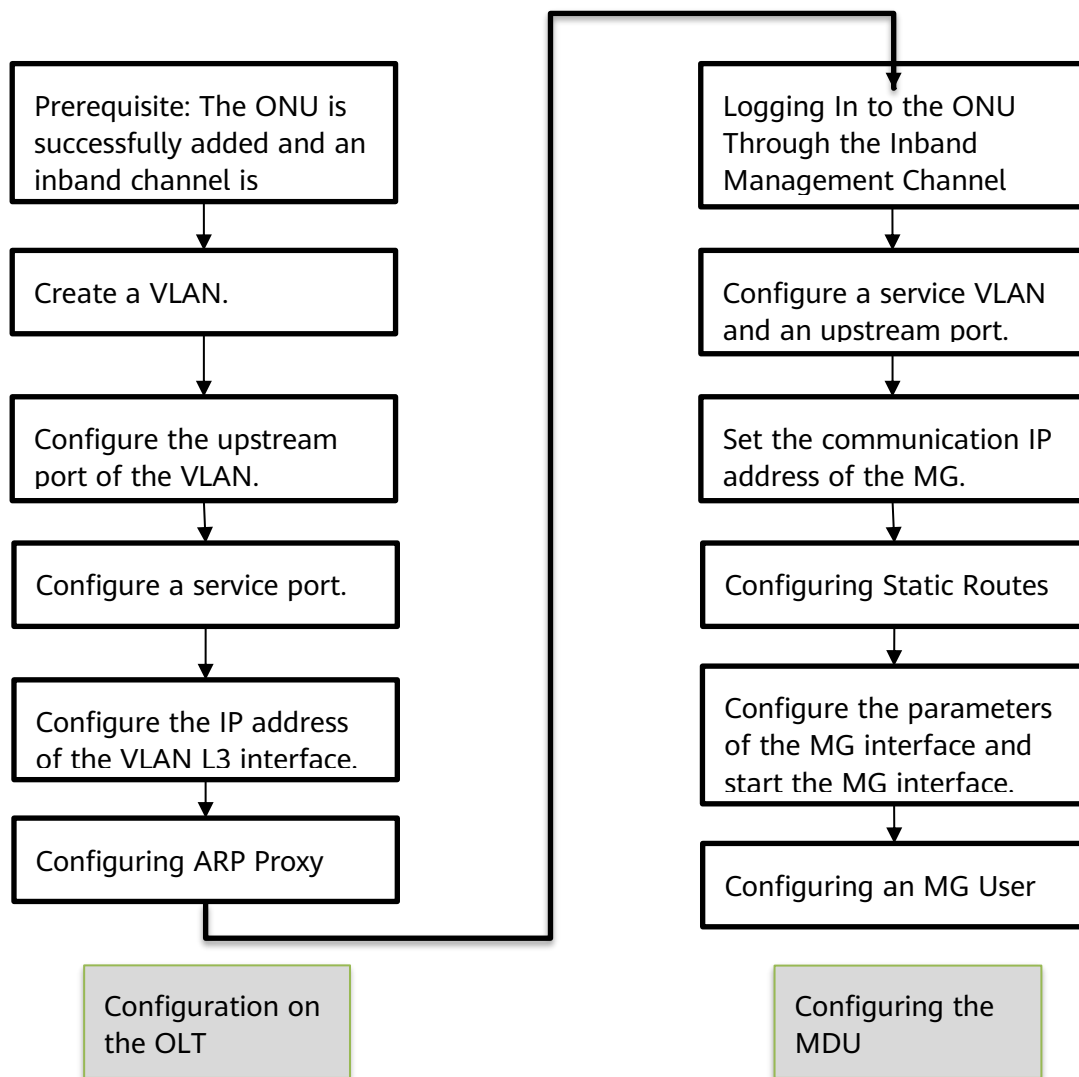
Configuration Item	Data	Configuration Item	Data
SIP server	200.200.200.200	Signaling IP address	17.1.1.43(MDU1) 17.1.1.44(MDU2)
Port number of the SIP server	5060	Media IP address	17.1.1.43(MDU1) 17.1.1.44(MDU2)

Transmission Mode	UDP	Domain Name	huawei
Voice gateway	17.0.0.1/8	Phone Number	7727043/7727044
VLANIF IP	17.1.1.43/8(MDU1) 17.1.1.44/8(MDU2)	Username/Password	7727043/7727000 7727044/7727000

2.2 Experiment Task Configuration

2.2.1 Configuration Roadmap

Figure 2-2 Flowchart for configuring the GPON ONU VoIP service



2.2.2 Configuration Procedure (OLT Side)

Step 1 DBA Profile Configuration

Run the **display dba-profile** command to query the existing DBA profiles in the system. If the existing DBA profile in the system cannot meet the requirements, run the **dba-profile add** command to add a DBA profile. Set the profile ID to 30, bandwidth type to type1, and bandwidth value to 1024 kbit/s.

```

Huawei_OLT(config)#dba-profile
{add<K>|delete<K>|modify<K>}:add
{profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K>}:profile-id
{profile-id<U><10,512>}:30
  
```

```
{profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K>}:type1
{fix<K>}:fix
{fix-bandwidth<U><128,10000000>}:1024
{<cr>|bandwidth_compensate<K>}:
Command:
dba-profile add profile-id 30 type1 fix 1024
Adding a DBA profile succeeded
Profile ID: 30
Profile name: dba-profile_30
```

Display the status of a DBA profile.

```
Huawei_OLT(config)#display dba-profile profile-id 30
-----
Profile-name: dba-profile_30
Profile-ID: 30
type: 1
Bandwidth compensation: No
Fix-delay: No
Fix(kbps): 1024
Assure(kbps): 0
Max(kbps): 0
Additional-bandwidth: -
Best-effort-priority: -
Best-effort-weight: -
bind-times: 0
-----
```

Step 2 Configuring a Line Profile

Set the line profile ID to 30 and profile name to EA5822.

```
Huawei_OLT(config)#ont-lineprofile
{epon<K>|gpon<K>}:gpon
{<cr>|profile-id<K>|profile-name<K>}:profile-id
{profile-id<U><0,8192>}:30
{<cr>|profile-name<K>}:profile-name
{profile-name<S><Length 1-32>}:EA5822
Command:
ont-lineprofile gpon profile-id 30 profile-name EA5822
```

Bind T-CONT 1 to DBA profile 30.

```
HUAWEI(config-gpon-lineprofile-30)#tcont
{tcont-list<S><Length 1-13>}:1
{<cr>|dba-profile-id<K>|dba-profile-name<K>}:dba-profile-id
{profile-id<U><0,512>}:30
Command:
tcont 1 dba-profile-id 30
```

Add GEM port 0 to carry ETH service flows, and bind GEM port 0 to T-CONT 0.

```
HUAWEI(config-gpon-lineprofile-30)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:add
{gem-index<U><0,1023>}:0
```

```
{service-type<E><eth,tdm>}:eth
{tcont<K>}:tcont
{tcont-id<U><0,127>}:0
{<cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K>}:
Command:
gem add 0 eth tcont 0
```

Add GEM port 3 to carry ETH service flows, and bind GEM port 3 to T-CONT 1.

```
HUAWEI(config-gpon-lineprofile-30)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:add
{gem-index<U><0,1023>}:3
{service-type<E><eth,tdm>}:eth
{tcont<K>}:tcont
{tcont-id<U><0,127>}:1
{<cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K>}:
Command:
gem add 3 eth tcont 1
```

Map the management flow of user-side VLAN 500 to GEM port 0. The mapping index is 0.

```
HUAWEI(config-gpon-lineprofile-30)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:mapping
{gem-index<U><0,1023>}:0
{mapping-index<U><0,7>}:0
{e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K>}:vlan
{untag<K>|vlan-id<U><0,4095>}:500
{<cr>|flow-car<K>|priority<K>|transparent<K>}:
Command:
gem mapping 0 0 vlan 500
```

Map the service flow of CVLAN 172 to GEM port 3 with the mapping index of 1.

```
HUAWEI(config-gpon-lineprofile-30)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:mapping
{gem-index<U><0,1023>}:3
{mapping-index<U><0,7>}:1
{e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K>}:vlan
{untag<K>|vlan-id<U><0,4095>}:172
{<cr>|flow-car<K>|priority<K>|transparent<K>}:
Command:
gem mapping 3 1 vlan 172
```

After the configuration is complete, run the commit command to make the configuration take effect.

```
HUAWEI(config-gpon-lineprofile-30)#commit
HUAWEI(config-gpon-lineprofile-30)#quit
```

Display information about the line profile.

```
Huawei_OLT(config)#display ont-lineprofile gpon profile-id 30
```

```
-----
Profile-ID :30
Profile-name :EA5822
Access-type :GPON
-----
FEC upstream switch :Disable
OMCC encrypt switch :Off
Qos mode :PQ
Mapping mode :VLAN
TR069 management :Disable
TR069 IP index :0
-----
<T-CONT 0> DBA Profile-ID:1
<Gem Index 0>
-----
|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
|Upstream-priority-queue:0 |Downstream-priority-queue:- |
-----
Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR
-----
0 500 - - - - -
-----
<T-CONT 1> DBA Profile-ID:30
<Gem Index 3>
-----
|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
|Upstream-priority-queue:0 |Downstream-priority-queue:- |
-----
Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR
-----
1 172 - - - - -
-----
Notes: Run the display traffic table ip command to query
traffic table configuration
-----
Binding times :0
-----
```

Step 3 Enable the auto-discovery function of the PON port.

Enter the PON board view and enable the auto-discovery function of the PON port in the network topology.

```
Huawei_OLT(config)#interface gpon
{frameid/slotid<S><Length 3-15>}:0/3
Command:
interface gpon 0/3
HUAWEI(config-if-gpon-0/3)#port
{dba<K>|optical-alarm-profile<K>|portid<U><0,7>}:0
{fec<K>|laser-switch<K>|ont-auto-find<K>|ont-password-renew<K>|range<K>}:ont-auto-find
{switch<E><enable,disable>}:enable
Command:
```

port 0 ont-auto-find enable

Step 4 Querying the MDUs that are automatically discovered by a PON port

```
HUAWEI(config-if-gpon-0/3)#display ont autofind 0
```

```
-----
Number: 1
F/S/P: 0/3/0
Ont SN: 4857544329FC9507 (HWTC-F2154B04)
Password: 0x30303030303030303030(0000000000)
Loid: 0000000000
Checkcode:
VendorID: HWTC
Ont Version: EA5822 VER.B
Ont SoftwareVersion: V8R018 C00
Ont EquipmentID: SmartAX EA5822
Ont autofind time: 2019-04-13 15:56:02+08:00
-----
```

Step 5 MDU registration

Register the MDU, set the authentication mode to SN authentication, set the management type to SNMP, and bind line profile 30 to the MDU.

```
HUAWEI(config-if-gpon-0/3)#ont
{activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K>}:confirm
{portid<U><0,7>}:0
{all<K>|loid-auth<K>|ontid<K>|password-auth<K>|sn-auth<K>}:ontid
{ontid<U><0,127>}:3
{loid-auth<K>|password-auth<K>|sn-auth<K>}:sn-auth
{sn-value<S><Length 12-16>}:4857544329FC9507
{omci<K>|password-auth<K>|snmp<K>}:snmp
{<cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K>}:ont-lineprofile-id
{profile-id<U><0,8192>}:30
{<cr>|desc<K>}:
Command:
Ont confirm 0 ontid 3 sn-auth 48575443F43FF304 snmp ont-lineprofile-id 30
Number of ONTs that can be added: 1, success: 1
PortID :0, ONTID :3
```

Display the ONT status.

```
HUAWEI(config-if-gpon-0/3)#display ont info 0 all
```

```
-----
F/S/P ONT SN Control Run Config Match Protect
ID flag state state state side -----
0/ 3/0 3 4857544329FC9507 active online normal match no
-----
F/S/P ONT-ID Description
-----
0/ 3/0 3 ONT_NO_DESCRIPTION
```

In port 0/ 3/0, the total of ONTs are: 1, online: 1

Step 6 Configure management VLAN 500.

Configure management VLAN 500, add the upstream interface to VLAN 500, and set the IP address of the interface in VLAN 500 to 10.1.1.1/24.

```
Huawei_OLT(config)#vlan
{attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-
policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093>}:500
{<cr>|to<K>|vlantype<E><mux,standard,smart,super>}:smart
Command:
vlan 500 smart
Huawei_OLT(config)#port
{desc<K>|mode<K>|vlan<K>}:vlan
{name<K>|vlan-list<S><Length 1-256>|vlanid<U><1,4093>}:500
{frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K>}:0/20
{portlist<S><Length 1-256>}:0
Command:
port vlan 500 0/20 0
Huawei_OLT(config)#interface
{adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|h
248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vdsl<K>|vlanif<K>|vrgw<K>}:vlanif
{<1-4093>}:500
Command:
interface vlanif 500
HUAWEI(config-if-vlanif500)#ip address
{ip_addr<I><X.X.X.X>}:10.1.1.1
{integer<U><0,32>|ip_addr<I><X.X.X.X>}:24
{<cr>|description<K>|sub<K>}:
Command:
ip address 10.1.1.1 24
```

Step 7 Configure a service channel for management VLAN 500.

Set the VLAN ID of the service port to 500, PON port to 0/3/0, ONT ID to 3, GEM port ID to 0, and C-VLAN ID to 500.

```
Huawei_OLT(config)#service-port
{desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K>}:vlan
{aoe<K>|name<K>|vlanid<U><1,4093>}:500
{adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K>}:gpon
{frameid/slotid/portid<S><Length 5-18>}:0/3/0
{ont<K>}:ont
{ontid<U><0,127>}:3
{eth<K>|gemport<K>|iphost<K>}:gemport
{gemindex<U><0,1023>}:0
{<cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K>}:multi-service
{user-8021p<K>|user-encap<K>|user-vlan<K>}:user-vlan
{other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095>}:500
{<cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K>}:
Command:
```

```
service-port vlan 500 gpon 0/3/0 ont 3 gempport 0 multi-service user-vlan 500
```

Display the status of a service port.

```
Huawei_OLT(config)#display service-port all
{<cr>|sort-by<K>||<K>}:
Command:
display service-port all
Switch-Oriented Flow List
-----
INDEX VLAN VLAN PORT F/ S/ P VPI VCI FLOW FLOW RX TX STATE
ID ATTR TYPE TYPE PARA
-----
1 500 common gpon 0/3 /0 3 0 vlan 500 - - up
-----
Total: 1 (Up/Down: 1/0)
```

Step 8 Configure the inband management VLAN and IP address of the MDU.

On the OLT, issue the IP address 10.1.1.3 24, gateway 10.1.1.1, and VLAN 500 to the MDU.

```
Huawei_OLT(config)#interface gpon 0/3
HUAWEI(config-if-gpon-0/3)#ont
{activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gempport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K>}:ipconfig
{portid<U><0,7>}:0
{ontid<U><0,127>}:3
{dhcp<K>|dscp-mapping-table<K>|ip-index<K>|pppoe<K>|static<K>}:static
{<cr>|gateway<K>|ip-address<K>|pri-dns<K>|slave-dns<K>|vlan<K>}:ip-address
{ip-address<I><X.X.X.X>}:10.1.1.3
{mask<K>}:mask
{mask<M><X.X.X.X>}:255.255.255.0
{<cr>|gateway<K>|pri-dns<K>|slave-dns<K>|vlan<K>}:gateway
{gateway<I><X.X.X.X>}:10.1.1.1
{<cr>|pri-dns<K>|slave-dns<K>|vlan<K>}:vlan
{vlanid<U><0,4095>}:500
{<cr>|pri-dns<K>|priority<K>|slave-dns<K>}:
Command:
ont ipconfig 0 3 static ip-address 10.1.1.3 mask 255.255.255.0 gateway 10.1.1.1 vlan 500
```

Step 9 Test the connectivity between the OLT and the MDU.

```
Huawei_OLT(config)#ping 10.1.1.3
PING 10.1.1.3:56 data bytes, press CTRL_C to break
Reply from 10.1.1.3: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.1.1.3: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.1.1.3: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.1.1.3: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.1.1.3: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.1.1.3 ping statistics ---
```

5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/1 ms

Step 10 Configure a voice service VLAN.

Configure voice service VLAN 172 and add an upstream port to the voice service VLAN.

```
Huawei_OLT(config)#vlan
{attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-
policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093>}:172
{<cr>|to<K>|vlantype<E><mux,standard,smart,super>}:smart
Command:
vlan 172 smart
Huawei_OLT(config)#port
{desc<K>|mode<K>|vlan<K>}:vlan
{name<K>|vlan-list<S><Length 1-256>|vlanid<U><1,4093>}:172
{frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K>}:0/20
{portlist<S><Length 1-256>}:0
Command:
port vlan 172 0/20 0
```

Step 11 Configure the Layer 3 interface of the service VLAN.

Set the IP address of the Layer 3 interface to 17.1.1.100 /8.

```
Huawei_OLT(config)#interface
{adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|h
248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vdsl<K>|vlanif<K>|vrgw<K>}:vlanif
{<1-4093>}:172
Command:
interface vlanif 172
HUAWEI(config-if-vlanif172)#ip address 17.1.1.100 8
{<cr>|description<K>|sub<K>}:
Command:
ip address 17.1.1.100 8
```

Step 12 Configure a service port in the voice VLAN.

```
Huawei_OLT(config)#service-port
{desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K>}:vlan
{aoe<K>|name<K>|vlanid<U><1,4093>}:172
{adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K>}:gpon
{frameid/slotid/portid<S><Length 5-18>}:0/3/0
{ont<K>}:ont
{ontid<U><0,127>}:3
{eth<K>|gemport<K>|iphos<K>}: gemport
{gemindex<U><0,1023>}:3
{<cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K>}:multi-service
{user-8021p<K>|user-encap<K>|user-vlan<K>}:user-vlan
{other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095>}:172
{<cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K>}:

```

Command:
service-port vlan 172 gpon 0/3/0 ont 3 gemport 3 multi-servi user-vlan 172
Huawei_OLT(config)#**display service-port all**
{<cr>|sort-by<K>||<K>}:
Command:
display service-port all
Switch-Oriented Flow List

INDEX	VLAN ID	VLAN TYPE	PORT	F	S	P	VPI	VCI	FLOW	FLOW RX	TX	STATE
1	500	common	gpon 0/3	/0	3	0	vlan 500	-	-	up		
2	172	common	gpon 0/3	/0	3	3	vlan 172	-	-	up		

Total: 2 (Up/Down: 2/0)

Step 13 Check the information about service VLAN 172.

Huawei_OLT(config)#**display vlan 172**
{<cr>|inner-vlan<K>|to<K>}:
Command:
display vlan 172
VLAN ID: 172
VLAN name: VLAN_0172
VLAN type: smart
VLAN attribute: common
VLAN description:
VLAN forwarding mode in control board: VLAN-MAC
VLAN forwarding mode: VLAN-MAC
VLAN broadcast packet forwarding policy: forward
VLAN unknown multicast packet forwarding policy: forward
VLAN unknown unicast packet forwarding policy: forward
VLAN bind service profile ID: -
VLAN bind RAIO profile index: -
VLAN priority: -

F	S	P	Native VLAN	State
0	/20	/0	1	up

Standard port number: 1

INDEX	TYPE	STATE	F	S	P	VPI	VCI	FLOWTYPE	FLOWPARA
3	gpon	up	0	/3	/0	3	3	vlan 172	

Service virtual port number: 2
Note: F--Frame, S--Slot, P--Port,
VPI indicates ONT ID for PON, VCI indicates GEM index for GPON,
v/e--vlan/encap, pritag--priority-tagged

Step 14 Enable global ARP proxy.

Huawei_OLT(config)#**arp**

```
{aging-mode<K>|aging-time<K>|conflict<K>|gratuitous-
arp<K>|ip_addr<I><X.X.X.X>|learning<K>|proxy<K>}:proxy
{disable<K>|enable<K>}:enable
Command:
arp proxy enable
```

Step 15 Enable ARP proxy on the Layer 3 interface.

```
Huawei_OLT(config)#interface vlanif 172
HUAWEI(config-if-vlanif172)#arp
{proxy<K>}:proxy
{disable<K>|enable<K>}:enable
{<cr>|subvlan<K>}:
Command:
arp proxy enable
```

Step 16 Configuration of MDU 2

The configuration of MDU2 on the OLT is similar to that of MDU1.

Step 17 Common Commands

Table 2-3 Common Commands

Common Commands	Function
display dba-profile	This command is used to query the information about the dynamic bandwidth assignment (DBA) profile in the system.
display ont-lineprofile	This command is used to query the information about the GPON ONT line profile that has taken effect, including the binding relationship between the T-CONT and the DBA profile, the QoS mode of the traffic stream, and the mapping between the GEM port and the ONT-side service.
display ont-srvprofile	This command is used to query the information about a GPON ONT service profile.
display ont autofind	This command is used to query the basic information about the currently auto-found ONT in the system.
display ont info	This command is used to query the information about an ONT, including the current status of the ONT, the related configuration of the ONT, and the information about the ONT T-CONT.
display vlan	This command is used to query the VLAN information.
display service-port	This command is used to query the information about a service virtual port.

2.2.3 Configuration Procedure on the MDU Side

Step 1 Log in to the MDU.

```
Huawei_OLT(config)#telnet 10.1.1.3
{<cr>|service-port<U><1,65535>}:
Command:
telnet 10.1.1.3
Press CTRL_] to quit telnet mode
Trying 10.1.1.3...
Connected to 10.1.1.3...
>>User name:root
>>User password:mduadmin
Huawei Integrated Access Software (EA5822).
Copyright(C) Huawei Technologies Co., Ltd. 2002-2012. All rights reserved.
EA5822>
User root has used a default password. Change the password in time.
```

Step 2 Check the current voice protocol. If the current voice protocol is switched to SIP, run the protocol support command to configure the voice protocol.

```
EA5822(config)#display protocol support
System support H.248 protocol
EA5822(config)#protocol support sip
Changing of the protocol configuration takes effect after you save the data
and then reboot the system. To ensure that the digitmap file is consistent with the protocol
configuration, reboot the system and then re-load the digitmap file or config the digitmap. Are you
sure to change the protocol configuration? (y/ n)[n]:y
EA5822(config)#save
EA5822(config)#reboot system
```

Step 3 Create SVLAN 172 and add the upstream port to VLAN 172.

```
EA5822(config)#vlan
{attrib<K>|bind<K>|desc<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093>}:172
{<cr>|to<K>|vlantype<E><mux,standard,smart>}:smart
Command:
vlan 172 smart
EA5822(config)#port
{desc<K>|mode<K>|vlan<K>}:vlan
{vlan-list<S><Length 1-256>|vlanid<U><1,4093>}:172
{frame/slot<S><Length 1-15>|to<K>}:0/0
{portlist<S><Length 1-256>}:1
Command:
port vlan 172 0/0 1
```

Step 4 Configure the Layer 3 interface of the service VLAN.

```
EA5822(config)#interface
{emu<K>|eponnni<K>|eth<K>|gponnni<K>|loopback<K>|meth<K>|null<K>|sip<K>|vlanif<K>}:vlanif
{<1-4093>}:172
EA5822(config-if-vlanif172)#ip
{address<K>|verify<K>}:address
```

```
{ip_addr<I><X.X.X.X>}:17.1.1.43 8
{<cr>|description<K>|sub<K>}:
Command:
ip address 17.1.1.43 8
EA5822(config-if-vlanif172)#
```

Step 5 Test the connectivity of the voice service between the MDU and the OLT.

```
EA5822(config)#ping 17.1.1.100
PING 17.1.1.100:56 data bytes, press CTRL_C to break
Reply from 17.1.1.100: bytes=56 Sequence=1 ttl=255 time=30 ms
Reply from 17.1.1.100: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 17.1.1.100: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 17.1.1.100: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 17.1.1.100: bytes=56 Sequence=5 ttl=255 time=10 ms

--- 17.1.1.100 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/8/30 ms
```

Step 6 Configure the default route from the MDU to the softswitch.

```
EA5822(config)#ip route-static 200.200.200.0 24 17.0.0.1
{<cr>|description<K>|permanent<K>|preference<K>|tag<K>}:
Command:
ip route-static 200.200.200.0 24 17.0.0.1
```

Step 7 Add an address pool in VoIP mode.

```
EA5822(config)#voip
EA5822(config-voip)#ip
{address<K>}:address
{media<K>|signaling<K>}:media
{media-ip-address<I><X.X.X.X>}:17.1.1.43
{gateway<I><X.X.X.X>}:17.0.0.1
Command:
ip address media 17.1.1.43 17.0.0.1
EA5822(config-voip)#ip
{address<K>}:address
{media<K>|signaling<K>}:signaling
{signaling-ip-address<I><X.X.X.X>}:17.1.1.43
Command:
ip address signaling 17.1.1.43
```

Step 8 Add a SIP interface.

```
EA5822(config)#interface
{emu<K>|eponnni<K>|eth<K>|gponnni<K>|loopback<K>|meth<K>|null<K>|sip<K>|vlanif<K>}:sip
{mgid<U><0,16777215>}:30

Command:
interface sip 30
```

Are you sure to add the SIP interface?(y/n)[n]:y

Step 9 Configuring the Attributes of a SIP Interface

```
EA5822(config-if-sip-30)#if-sip
{attribute<K>}:attribute
{basic<K>|optional<K>}:basic
{home-domain<K>|media-ip<K>|primary-proxy-domain<K>|primary-proxy-ip1<K>|primary-proxy-
ip2<K>|primary-proxy-port<K>|proxy-addr-mode<K>|secondary-proxy-domain<K>|secondary-proxy-
ip1<K>|secondary-proxy-ip2<K>|secondary-proxy-port<K>|server-dhcp-option<K>|signal-ip<K>|signal-
port<K>|sipprofile-index<K>|srvlogic-index<K>|transfer<K>}:media-ip
{media-addr-name<K>|media-dhcp-vlan<K>|media-ip-value<I>}:17.1.1.43
{<cr>|home-domain<K>|primary-proxy-domain<K>|primary-proxy-ip1<K>|primary-proxy-
ip2<K>|primary-proxy-port<K>|proxy-addr-mode<K>|secondary-proxy-domain<K>|secondary-proxy-
ip1<K>|secondary-proxy-ip2<K>|secondary-proxy-port<K>|server-dhcp-option<K>|signal-ip<K>|signal-
port<K>|sipprofile-index<K>|srvlogic-index<K>|transfer<K>}:signal-ip
{signal-addr-name<K>|signal-dhcp-vlan<K>|signal-ip-value<I>}:17.1.1.43
{<cr>|home-domain<K>|primary-proxy-domain<K>|primary-proxy-ip1<K>|primary-proxy-
ip2<K>|primary-proxy-port<K>|proxy-addr-mode<K>|secondary-proxy-domain<K>|secondary-proxy-
ip1<K>|secondary-proxy-ip2<K>|secondary-proxy-port<K>|server-dhcp-option<K>|signal-
port<K>|sipprofile-index<K>|srvlogic-index<K>|transfer<K>}:signal-port
{signal-port-value<U><5000,5999>}:5060
{<cr>|home-domain<K>|primary-proxy-domain<K>|primary-proxy-ip1<K>|primary-proxy-
ip2<K>|primary-proxy-port<K>|proxy-addr-mode<K>|secondary-proxy-domain<K>|secondary-proxy-
ip1<K>|secondary-proxy-ip2<K>|secondary-proxy-port<K>|server-dhcp-option<K>|sipprofile-
index<K>|srvlogic-index<K>|transfer<K>}:transfer
{transfermode<E><tcp,udp,sctp,tls>}:udp
{<cr>|home-domain<K>|primary-proxy-domain<K>|primary-proxy-ip1<K>|primary-proxy-
ip2<K>|primary-proxy-port<K>|proxy-addr-mode<K>|secondary-proxy-domain<K>|secondary-proxy-
ip1<K>|secondary-proxy-ip2<K>|secondary-proxy-port<K>|server-dhcp-option<K>|sipprofile-
index<K>|srvlogic-index<K>}:primary-proxy-ip1
{primary-proxy-ip1-value<I><X.X.X.X>}:200.200.200.200
{<cr>|home-domain<K>|primary-proxy-domain<K>|primary-proxy-ip2<K>|primary-proxy-
port<K>|proxy-addr-mode<K>|secondary-proxy-domain<K>|secondary-proxy-ip1<K>|secondary-proxy-
ip2<K>|secondary-proxy-port<K>|server-dhcp-option<K>|sipprofile-index<K>|srvlogic-
index<K>}:primary-proxy-port
{primary-proxy-port-value<U><1,65535>}:5060
{<cr>|home-domain<K>|primary-proxy-domain<K>|primary-proxy-ip2<K>|proxy-addr-
mode<K>|secondary-proxy-domain<K>|secondary-proxy-ip1<K>|secondary-proxy-ip2<K>|secondary-
proxy-port<K>|server-dhcp-option<K>|sipprofile-index<K>|srvlogic-index<K>}:home-domain
{home-domain-name<S><Length 1-64>}:huawei
{<cr>|primary-proxy-domain<K>|primary-proxy-ip2<K>|proxy-addr-mode<K>|secondary-proxy-
domain<K>|secondary-proxy-ip1<K>|secondary-proxy-ip2<K>|secondary-proxy-port<K>|server-dhcp-
option<K>|sipprofile-index<K>|srvlogic-index<K>}:
Command:
if-sip attribute basic media-ip 17.1.1.43 signal-ip 17.1.1.43 signal-port 5060 transfer udp primary-
proxy-ip1 200.200.200.200 primary-proxy-port 5060 home-domain huawei
```

Step 10 Enable the SIP interface.

```
EA5822(config-if-sip-30)#reset
Are you sure to reset the SIP interface?(y/n)[n]:y
EA5822(config-if-sip-30)#
Resetting SIP interface 30 succeeded
```

Step 11 Configure the data of the PSTN subscriber in slot 0/2.

```
EA5822(config)#esl
{user<K>}:user
Command:
esl user
EA5822(config-esl-user)#sippstnuser
{add<K>|attribute<K>|auth<K>|batadd<K>|batdel<K>|del<K>|modify<K>|rightflag<K>|servicedata<K>}:
add
{frameid/slotid/portid<S><Length 1-15>}:0/2/1
{mgid<U><0,16777215>}:30
{<cr>|telno<K>}:telno
{telno-value<S><Length 1-64>}:7727043
Command:
sippstnuser add 0/2/1 30 telno 7727043
Command processing is completed. User data has been added successfully

EA5822(config-esl-user)#display sippstnuser
{attribute<K>|authinfo<K>|call-state<K>|frameid/slotid/portid<S><Length 1-15>|frameid/slotid<S><Length 1-15>|reg-state<K>|rightflag<K>|servicedata<K>}:0/2/1
{<cr>|endframeid/slotid/portid<S><Length 1-15>}:
Command:
display sippstnuser 0/2/1
-----
F /S /P MGID TelNo
-----
0 /2 /1 30 7727043
-----
```

Step 12 Configuring Authentication Data for PSTN Subscribers

```
EA5822(config-esl-user)#sippstnuser
{add<K>|attribute<K>|auth<K>|batadd<K>|batdel<K>|del<K>|modify<K>|rightflag<K>|servicedata<K>}:
auth
{set<K>}:set
{frameid/slotid/portid<S><Length 1-15>}:0/2/1
{telno<K>}:telno 7727043
{<cr>|password-mode<K>}:password-mode
{password-mode-value<E><password,ha1>}:password

Command:
sippstnuser auth set 0/2/1 telno 7727043 password-mode password
User Name(<=64 characters, "-" indicates deletion):7727043
User Password(<=64 characters, "-" indicates deletion): 7727000
```

Step 13 Adding a local SIP digitmap

```
EA5822(config)#local-digitmap
{add<K>|append<K>|delete<K>|modify<K>}:add
{name-value<S><Length 1-32>}:SIPmap
{type-value<E><normal,emergency,scc,direct-centrex,second-centrex>}:normal
{body-value<S><Length 1-256>}:7727xxx|x.S|x.F

Command:
local-digitmap add SIPmap normal 7727xxx|x.S|x.F
```

Table 2-4 Digitmap rules

Symbol	Meaning	Symbol	Meaning
T	Start timer	-	Range, for example, 2-4.
L	Long timer	.	Matches any structure that appears before it one or more times.
S	Short timer		Separator
E	Key (*)	Digit	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
F	Press #.	[]	Matches any number or character within the range of parentheses.
Wildcard x	Matches any number from 0 to 9.		

Run the display local-digitmap command to check whether the created digitmap rule is correct.

```
EA5822(config)#display local-digitmap all
```

```
-----
Name: SIPmap
Type: normal
Body: 7727xxx|X.S|X.F
-----
```

Step 14 Modify the digitmap matching mode.

The default matching mode is 0, that is, the minimum matching mode is used. You are advised to change the value to 1, that is, the maximum matching mode is used. When there are multiple digitmap rules, the minimum matching mode may cause the system to report an incorrect number due to incorrect matching rules. As a result, the call fails.

```
EA5822(config)#interface sip 30
EA5822(config-if-sip-30)#mg-software
{parameter<K>|strpara<K>}:parameter
{name<U><0,27>}:13
{{value<L>}}:1
```

```
Command:
mg-software parameter 13 1
```

```
EA5822(config)#display mg-software parameter
{mgid<U><0,16777215>}:30
{<cr>|name<U><0,27>}:13
```

```
Command:
```

display mg-software parameter 30 13

```
-----
MGID:30 para index:13 value:1
-----
```

APPENDIX:

```
-----
Parameter Index: Interface software parameter name:
13: Digitmap matching mode
0: Minimum
1: Maximum
```

Step 15 (Optional) Configuring the Standalone Mode

The default value is 0, indicating that the emergency standalone function is not supported. You can change the value to 1, indicating that the emergency standalone function is supported. When the connectivity between the MDU and the softswitch is faulty, the MDU can automatically process the calls between the users connected to the same MDU.

```
EA5822(config)#interface sip 30
```

```
EA5822(config-if-sip-30)#mg-software
{parameter<K>|strpara<K>}:parameter
{name<U><0,27>}:2
{{value<L>}}:1
```

Command:

```
mg-software parameter 2 1
```

```
EA5822(config)#display mg-software parameter
{mgid<U><0,16777215>}:30
{<cr>|name<U><0,27>}:2
```

Command:

```
display mg-software parameter 30 2
```

```
-----
MGID:30 para index:2 value:1
-----
```

APPENDIX:

```
-----
Parameter Index: Interface software parameter name:
2: SAL Support
0: No
1: Yes
```

Step 16 (Optional) MDU Packet Mirroring

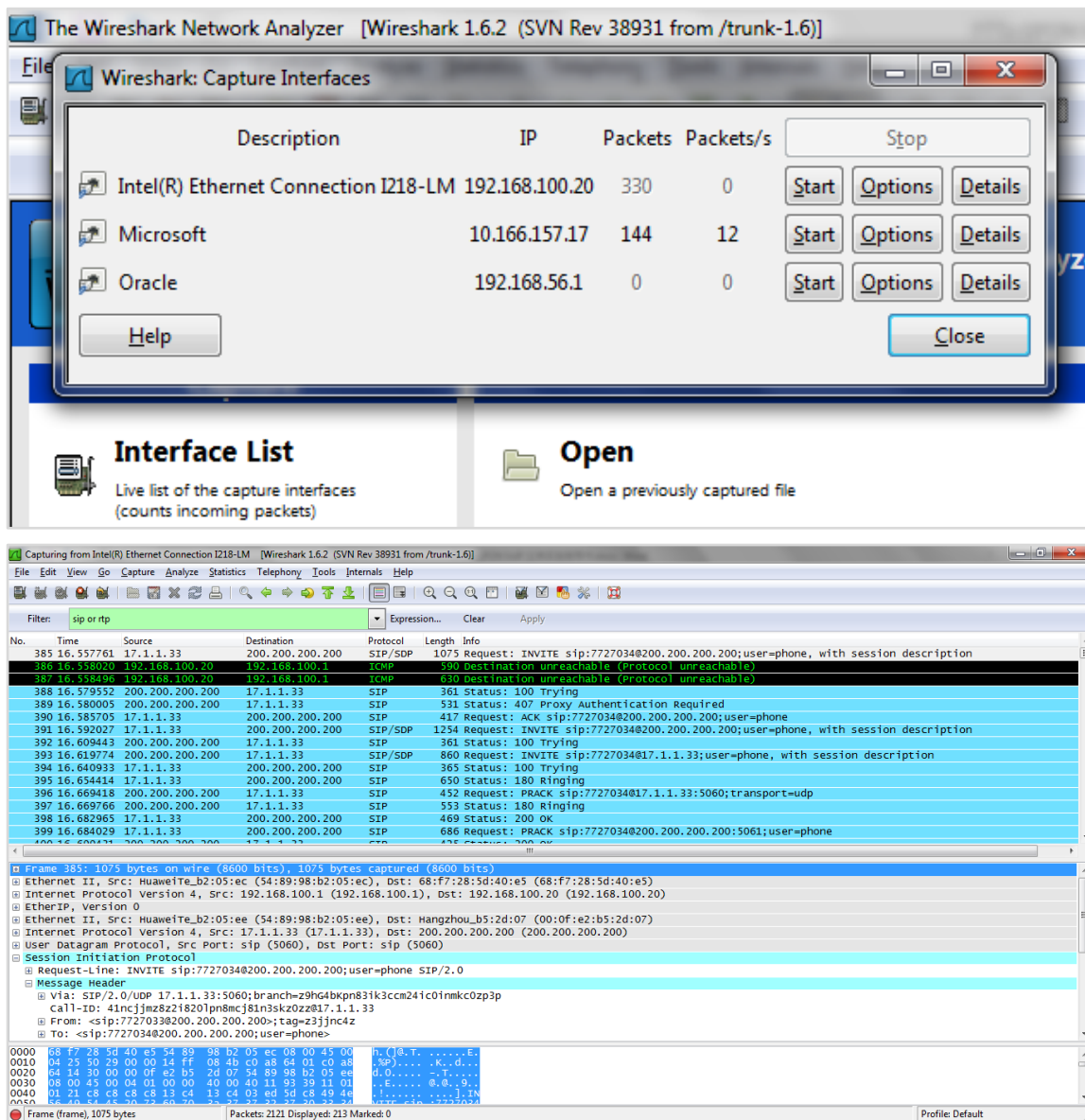
Mirror upstream and downstream packets passing through PON port 0/0/1 of the MDU to PON port 0/0/2. A GE copper module is installed on port 0/0/2, and the PC is connected to port 0/0/2 using a network cable.

```
EA5822(config)#diagnose
EA5822(diagnose)%%mirror port
{frameid/slotid/portid<S><Length 1-15>}:0/0/1
{frameid/slotid/portid<S><Length 1-15>}:0/0/2
{mirror-direction<E><ingress,egress,all>}:all
```

Command:

mirror port 0/0/1 0/0/2 all

Run the Wireshark software on the PC to capture packets on the network adapter connected to MDU port 0/0/2.



Step 17 Configuration of MDU 2

The configuration of MDU2 is similar to that of MDU1.

2.2.4 Experiment Verification

Test method: Connect two phones to different POTS ports on the MDU and make a call to each other.

Step 1 When the two parties are in conversation, run the `display esl online-info` command to query the information about the MG online users.

```
EA5822(config)#display esl online-info all
```

```
-----
UserPort DSPChannel Codec PT EC VAD Gain(I/O) Duration MIX
-----
```

```
0/ 2/ 1 0/ 2/0/ 9 G711A 20 ms On Off 0/0 dB 00:00:08 No
```

```
0/ 2/ 2 0/ 2/0/ 10 G711A 20 ms On Off 0/0 dB 00:00:08 No
-----
```

```
EA5822(config)#display esl online-info startuser 0/2/1
```

```
{<cr>|enduser<K>}:

```

Command:

```
display esl online-info startuser 0/2/1
```

```
-----
User Port: 0/2/1
```

```
DSP Channel: 0/2/0/9
```

```
MIX: No
```

```
Codec: G711A
```

```
PT(ms): 20
```

```
EC: On
```

```
VAD: Off
```

```
Gain(I/O)(dB): 0/0
```

```
Duration(hh:mm:ss): 00:00:44
```

```
Local IP:UDP Port: 17.1.1.43:58080
```

```
Remote IP:UDP Port: 17.1.1.43:58076
```

```
RTCP State: On
```

```
Remote PT(ms): 20
```

```
Vlan ID: 172
```

```
Vlan Priority: 6
```

```
QoS Strategy: DSCP
```

```
QoS Value: 56
```

```
Receive Package Count: 2185
```

```
Send Package Count: 2042
```

```
Receive Octet Count: 349600
```

```
Send Octet Count: 326720
```

```
Loop Delay(ms): 2
```

Local Jitter: 0
Remote Jitter: 0
Local Loss Rate(%): 0
Remote Loss Rate(%): 0
Network Packet Loss Rate(%): 0
Jitter Buffer Discard Rate(%): 0
RTCP Round-Trip Delay: 2
End System Delay: 33
Signal Level: -
Noise Level: 220
Residual Echo Return Loss: 30
R Factor: 77
Estimated MOSLQ: 4.0
Estimated MOSCQ: 3.9
Burst Loss Density: 0
Burst Duration: 0
Gap Loss Density: 0
Gap Duration: 39360

Step 2 Subscriber loop line test

```
EA5822(config)#test
EA5822(config-test)#pots
{circuit-test<K>|emulational-call<K>|emulational-callee<K>|emulational-
caller<K>|emulational<K>|loop-line-test<K>|loop-line-threshold<K>|path-test<K>|search-
tone-test<K>|test-para<K>}:loop-line-test
{frameid/slotid/portid<S><Length 1-15>|mgid<K>|telno<K>}:0/2/1
{<cr>|busy<K>}:

```

Command:

pots loop-line-test 0/2/1

Frame 0 slot 2 port 1 (telno 7727043 mgid 30 terminalid -)
under testing, Please wait.....

EA5822(config-test)#

Testing port: **0/2/1**

Telno: **7727043**

MGid: **30**

Terminalid: -

Test conclusion:

Line state Normal

Line length

Line A 0 m

Line B 0 m

PPA No PPA between A,B line

Termination type R-C network

Test parameter item Value Validity

Primary test result parameter:

A->ground AC voltage 0.000 V Valid

B->ground AC voltage 0.000 V Valid

A->B AC voltage 0.000 V Valid

A->ground AC Frequency 0 Hz Invalid

B->ground AC Frequency 0 Hz Invalid

A->B AC Frequency 0 Hz Invalid

A->ground DC voltage 0.053 V Valid

B->ground DC voltage 0.003 V Valid

A->B DC voltage 0.053 V Valid

A->ground insulation resistance >10 MΩ Valid

B->ground insulation resistance >10 MΩ Valid

A->B insulation resistance(Low voltage) 5.555 MΩ Valid

B->A insulation resistance(Low voltage) 5.094 MΩ Valid

A->B insulation resistance(High voltage) 2.546 KΩ Valid

B->A insulation resistance(High voltage) - Not support

A->ground capacitance 0 nF Valid

B->ground capacitance 0 nF Valid

A->B capacitance(Low voltage) 966 nF Valid

A->B capacitance(High voltage) 33 nF Invalid

Secondary test result parameter:

A->ground conductance 0.457 uS Valid

B->ground conductance 0.599 uS Valid

A->B conductance(Low voltage) 5.305 uS Valid
A->B conductance(High voltage) - Not support
A->ground admittance -1.638 uS Valid
B->ground admittance -8.512 uS Valid
A->B admittance(Low voltage) 24.350 uS Valid
A->B admittance(High voltage) - Not support
A->ground DC current 0 uA Valid
B->ground DC current 0 uA Valid
A->B DC current 0 uA Valid
B->A DC current 0 uA Valid
A->ground AC current 0 uA Valid
B->ground AC current 0 uA Valid
A->B AC current 0 uA Valid
B->A AC current 0 uA Valid

Test auxiliary parameter:

Test frequency for admittance - Not support
Resistance to test the abnormal voltage - Not support
Max. voltage to conduct the signature - Not support
A->ground max. voltage -2.544 V Valid
A->ground min. voltage -48.860 V Valid
B->ground max. voltage -2.569 V Valid
B->ground min. voltage -48.920 V Valid
A->B max. voltage 46.610 V Valid
A->B min. voltage -46.570 V Valid
A->ground max. current -36 uA Valid
A->ground min. current -87 uA Valid
B->ground max. current 17 uA Valid
B->ground min. current -34 uA Valid
A->B max. positive current 187 uA Valid
A->B max. negative current 121 uA Valid

Step 3 Subscriber circuit test.

EA5822(config-test)#**pots**

```
{circuit-test<K>|emulational-call<K>|emulational-callee<K>|emulational-  
caller<K>|emulational<K>|loop-line-test<K>|loop-line-threshold<K>|path-test<K>|search-  
tone-test<K>|test-para<K>}.circuit-test
```

```
{frameid/slotid/portid<S><Length 1-15>|mgid<K>|telno<K>}:0/2/1  
{<cr>|busy<K>}:  
-----
```

Command:

pots circuit-test 0/2/1

Frame 0 slot 2 port 1 (telno 7727043 mgid 30 terminalid -)
under testing, Please wait.....

EA5822(config-test)#

Testing port: **0/2/1**

Telno: **7727043**

MGid: **30**

Terminalid: -

Test item Result

Digital Voltage: Normal

Low Battery: Normal

High Battery: Normal

Positive Battery: Normal

Loop current: Normal

Feeder voltage: Normal

Ringing current voltage: Normal

Ringing current frequency: Normal

VAG: Normal

VBG: Normal

Feeder voltage(V): -50.760

Ringing current voltage(V): 50.690

Loop current(mA): 20.000

Step 4 In the calling party emulation test, the called party needs to connect to the phone to answer the call, press keys as instructed, and hang up to obtain the test result.

EA5822(config-test)#**pots**

```
{circuit-test<K>|emulational-call<K>|emulational-callee<K>|emulational-  
caller<K>|emulational<K>|loop-line-test<K>|loop-line-threshold<K>|path-test<K>|search-  
tone-test<K>|test-para<K>}:emulational-caller
```

```
{caller-port<K>}:caller-port
```

```
{frameid/slotid/portid<S><Length 1-15>}:0/2/1  
{callee-telno<K>}:callee-telno  
{telno-value<S><Length 1-32>}:7727044
```

Command:

pots emulational-caller caller-port 0/2/1 callee-telno 7727044

EA5822(config-test)#

Caller port: 0/2/1

Callee telno: 7727044

Test result: Test Succeed

Step 5 In a called party emulation test, the calling party needs to connect to a phone to dial a number, press a key as instructed, and then hang up to obtain the test result.

EA5822(config-test)#**pots**

```
{circuit-test<K>|emulational-call<K>|emulational-callee<K>|emulational-  
caller<K>|emulational<K>|loop-line-test<K>|loop-line-threshold<K>|path-test<K>|search-  
tone-test<K>|test-para<K>}:emulational-callee
```

```
{callee-port<K>}:callee-port
```

```
{frameid/slotid/portid<S><Length 1-15>}:0/2/1
```

Command:

pots emulational-callee callee-port 0/2/1

EA5822(config-test)#

Callee port: 0/2/1

Test result: Test Succeed

Step 6 In a call emulation test, no phone needs to be connected. After the test is complete, the device returns the test result.

EA5822(config-test)#**pots**

```
{circuit-test<K>|emulational-call<K>|emulational-callee<K>|emulational-  
caller<K>|emulational<K>|loop-line-test<K>|loop-line-threshold<K>|path-test<K>|search-  
tone-test<K>|test-para<K>}:emulational-call
```

```
{caller-port<K>}:caller-port
```

```
{frameid/slotid/portid<S><Length 1-15>}:0/2/2
```

```
{callee-port<K>}:callee-port
```

```
{frameid/slotid/portid<S><Length 1-15>}:0/2/1
```

```
{callee-telno<K>}:callee-telno
```

{telno-value<S><Length 1-32>}:7727043

Command:

pots emulational-call caller-port 0/2/2 callee-port 0/2/1 callee-telno 7727043

EA5822(config-test)#

Caller port: 0/2/2

Callee port: 0/2/1

Test result: Test Succeed

Step 7 Common Commands

Table 2-5 Common Commands

Common Commands	Function
display local-digitmap	This command is used to query the digitmap rules created locally.
display mg-software parameter	This command is used to query the software parameters of the MG interface that supports the SIP protocol.
display esl online-info	This command is used to query the information about the MG online users. When you need to query the information about the MG online users, such as the DSP port number, codec, and packet length, run this command.
pots loop-line-test	This command is used to perform an analog subscriber loop line test. The subscriber to be tested must be in the idle, locked, or blocked state.
pots circuit-test	This command is used to simulate the circuit test of a subscriber to check whether the functions of the subscriber circuit are normal.
pots emulational-caller	This command is used to start the caller emulation test of a PSTN port.
pots emulational-callee	This command is used to start the callee emulation test on a PSTN port.
pots emulational-call	Starts the call emulation test between two PSTN ports automatically. When you need to perform the call emulation test between PSTN ports without manual intervention, run this command.

2.2.5 Configuration Reference

#OLT side

```
dba-profile add profile-id 30 type1 fix 1024
ont-lineprofile gpon profile-id 30 profile-name EA5822
tcont 1 dba-profile-id 30
gem add 0 eth tcont 0
gem add 3 eth tcont 1
gem mapping 0 0 vlan 500
gem mapping 3 1 vlan 172
Commit
quit

interface gpon 0/3
port 0 ont-auto-find enable
ont add 0 3 sn-auth 48575443F43FF304 snmp ont-lineprofile-id 30
quit
vlan 500 smart
port vlan 500 0/20 0
interface vlanif 500
ip address 10.1.1.1 24
quit

service-port vlan 500 gpon 0/3/0 ont 3 gemport 0 multi-service user-vlan 500

interface gpon 0/3
ont ipconfig 0 3 static ip-address 10.1.1.3 mask 255.255.255.0 gateway 10.1.1.1 vlan 500

quit
vlan 172 smart
port vlan 172 0/20 0

interface vlanif 172
ip address 17.1.1.100 8
quit

service-port vlan 172 gpon 0/3/0 ont 3 gemport 3 multi-service user-vlan 172

telnet 10.1.1.3
>>User name:root
>>User password: mduadmin
```

#On the MDU side

```
vlan 172
port vlan 172 0/0 1
interface vlanif 172
ip address 17.1.1.43 8
quit

Voip
ip address media 17.1.1.43 17.0.0.1
```

```
ip address signaling 17.1.1.43
quit
```

```
ip route-static 200.200.200.0 24 17.0.0.1
interface sip 30
```

```
if-sip attribute basic media-ip 17.1.1.43 signal-ip 17.1.1.43 signal-port 5060 transfer udp primary-
proxy-ip1 200.200.200.200 primary-proxy-port 5060 home-domain huawei
```

```
sip-auth-parameter password-mode password auth-username "7727043" auth-password
*T1I=2A,*][OQ=^Q`MAF4<1!!*
```

```
reset
quit
esl user
sippstnuser add 0/2/1 30 telno 7727043
```

2.3 Questions

1. Can the call emulation test be used to test numbers of different devices?
2. How do I query the media port used by the calling and called parties in a call?

3

GPON ONT H.248 Voice Service Configuration

3.1 Experiment Introduction

3.1.1 About This Experiment

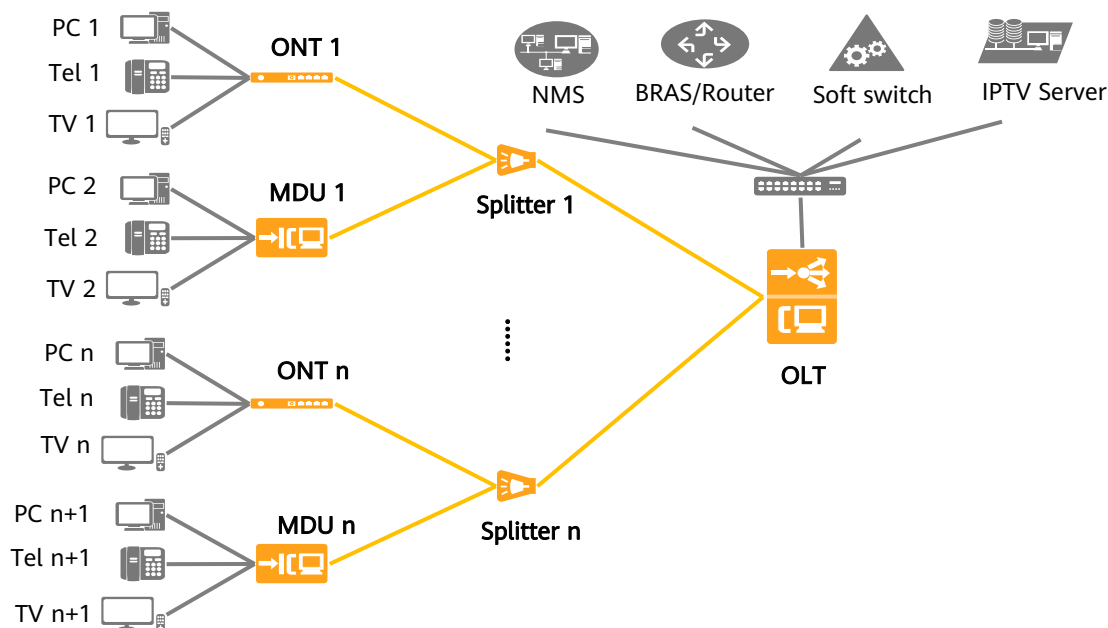
In this experiment, data is configured on the OLT and ONT so that two users connected to different ONTs connected to the same PON port can make voice calls.

3.1.2 Objectives

- Understand the basic principles of GPON.
- Understand the basic principles of VoIP.
- Understand the GPON data profile.
- Configure voice service profiles.
- Understand the overall configuration process of the voice service.

3.1.3 Experiment Networking

Figure 3-1 Topology of the GPON ONT H.248 configuration experiment



3.1.4 Experiment Planning

According to the VoIP configuration scenario of the access network, phone 1 and phone 2 are common phones, and PC1 and PC2 run the Windows operating system.

Table 3-1 H.248 data plan for a GPON ONT

Configuration Item	ONT1 data	Configuration Item	ONT2 data
OLT service VLAN	172. Smart	OLT service VLAN	172. Smart
OLT upstream port	0/20/0	OLT upstream port	0/20/0
OLT downstream port	0/3/0	OLT downstream port	0/3/0
ONT ID,SN	5, SN 48575443B205EC07	ONT ID,SN	6, SN 48575443B203E407
TCONT,DBA	1. DBA profile ID: 50	TCONT,DBA	2. DBA profile ID: 60
GEM Port	1	GEM Port	2
ONT user VLAN	172	ONT user VLAN	172

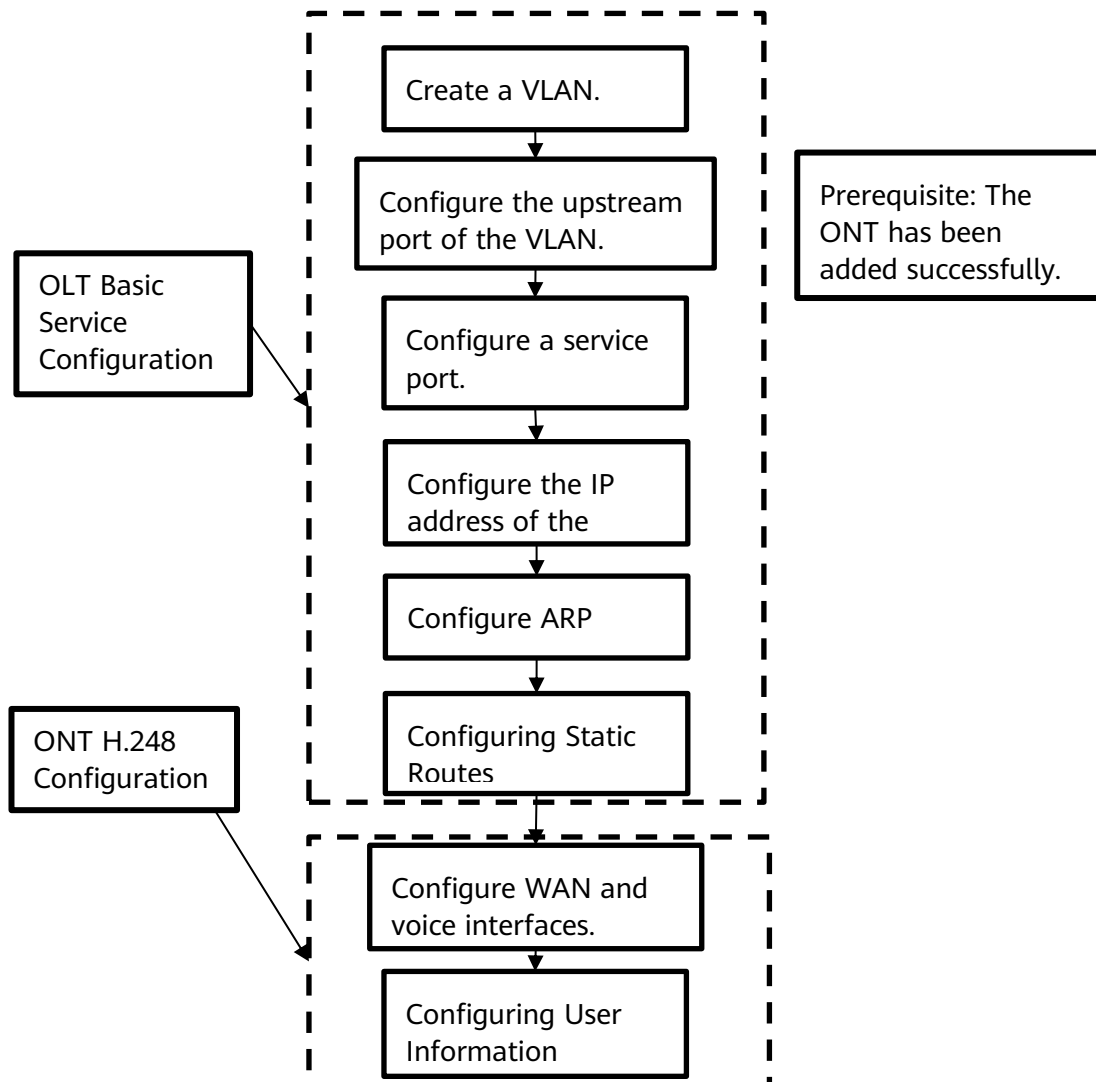
Table 3-2 Data plan for the GPON ONT H.248 voice server

Configuration Item	Data	Configuration Item	Data
H.248 server	200.200.200.200	Voice gateway	17.0.0.1/8
H.248 server port number	2944	Phone Number	18980100 (ONT1) 18980200 (ONT2)
MG IP	17.248.48.1/8 (ONT1) 17.248.48.2/8 (ONT2)	Terminal ID	0 (ONT1) 1 (ONT2)

3.2 Experiment Task Configuration

3.2.1 Configuration Roadmap

Figure 3-2 Flowchart for configuring GPON ONT H.248



3.2.2 Configuration Procedure (OLT Side)

Step 1 DBA Profile Configuration

Run the **display dba-profile** command to query the existing DBA profiles in the system. If the existing DBA profile in the system cannot meet the requirements, run the **dba-profile add** command to add a DBA profile. Set the profile ID to 50, bandwidth type to type1, and bandwidth value to 2048 kbit/s.

```

Huawei_OLT(config)#dba-profile
{add<K>|delete<K>|modify<K>}:add
{profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K>}:profile-id
{profile-id<U><10,512>}:50
{profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K>}:type1
{fix<K>}:fix
{fix-bandwidth<U><128,1000000>}:2048
{<cr>|bandwidth_compensate<K>}:
Command:
    
```

dba-profile add profile-id 50 type1 fix 2048

Adding a DBA profile succeeded

Profile ID: 50

Profile name: dba-profile_50

Display the DBA profile status.

Huawei_OLT(config)#**display dba-profile profile-id 50**

Profile-name: dba-profile_50

Profile-ID: 50

type: 1

Bandwidth compensation: No

Fix-delay: No

Fix(kbps): 2048

Assure(kbps): 0

Max(kbps): 0

Additional-bandwidth: -

Best-effort-priority: -

Best-effort-weight: -

bind-times: 0

Step 2 Configuring an ONT Line Profile

Set the line profile ID to 50 and profile name to 5hg8247.

Huawei_OLT(config)#**ont-lineprofile**

{epon<K>|gpon<K>}:**gpon**

{<cr>|profile-id<K>|profile-name<K>}:**profile-id**

{profile-id<U><0,8192>}:**50**

{<cr>|profile-name<K>}:**profile-name**

{profile-name<S><Length 1-32>}:**5hg8247**

Command:

ont-lineprofile gpon profile-id 50 profile-name 5hg8247

Bind T-CONT 1 to DBA profile 50.

HUAWEI(config-gpon-lineprofile-50)#**tcont**

{tcont-list<S><Length 1-13>}:**1**

{<cr>|dba-profile-id<K>|dba-profile-name<K>}:**dba-profile-id**

{profile-id<U><0,512>}:**50**

Command:

tcont 1 dba-profile-id 50

Add GEM port 1 to carry ETH service flows, and bind GEM port 1 to T-CONT 1.

HUAWEI(config-gpon-lineprofile-50)#**gem**

{add<K>|delete<K>|mapping<K>|modify<K>}:**add**

{gem-index<U><0,1023>}:**1**

{service-type<E><eth,tdm>}:**eth**

{tcont<K>}:**tcont 1**

{<cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K>}:
Command:

gem add 1 eth tcont 1

Map the service flow of CVLAN 172 to GEM port 1. The mapping index is 1.

```
HUAWEI(config-gpon-lineprofile-50)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:mapping
{gem-index<U><0,1023>}:1
{mapping-index<U><0,7>}:1
{e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K>}:vlan
{untag<K>|vlan-id<U><0,4095>}:172
{<cr>|flow-car<K>|priority<K>|transparent<K>}:
Command:
gem mapping 1 1 vlan 172
```

After the configuration is complete, run the commit command to make the configuration take effect.

```
HUAWEI(config-gpon-lineprofile-50)#commit
HUAWEI(config-gpon-lineprofile-50)#quit
```

Display information about the line profile.

```
Huawei_OLT(config)#display ont-lineprofile gpon
{all<K>|bound-info<K>|profile-id<K>|profile-name<K>}:profile-id
{profile-id<U><0,8192>}:50
Command:
display ont-lineprofile gpon profile-id 50
```

```
-----
Profile-ID :50
Profile-name :5hg8247
Access-type :GPON
-----
```

```
FEC upstream switch :Disable
OMCC encrypt switch :Off
Qos mode :PQ
Mapping mode :VLAN
TR069 management :Disable
TR069 IP index :0
-----
```

```
<T-CONT 0> DBA Profile-ID:1
<T-CONT 1> DBA Profile-ID:50
<Gem Index 1>
-----
```

```
|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
|Upstream-priority-queue:0 |Downstream-priority-queue:- |
-----
```

```
Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR
-----
```

```
1 172 - - - - -
-----
```

```
-----
Notes: Run the display traffic table ip command to query
traffic table configuration
-----
```

```
Binding times :0
```

Step 3 Configuring a Service Profile

Set the service profile ID to 50 and profile name to 5hg8247.

```
Huawei_OLT(config)#ont-srvprofile
{epon<K>|gpon<K>}:gpon
{<cr>|profile-id<K>|profile-name<K>}:profile-id
{profile-id<U><0,8192>}:50
{<cr>|profile-name<K>}:profile-name
{profile-name<S><Length 1-32>}:5hg8247
Command:
ont-srvprofile gpon profile-id 50 profile-name 5hg8247
```

Configure the port type and number of ports on the ONT. The number of LAN ports is 4, and the number of POTS ports is 2.

```
HUAWEI(config-gpon-srvprofile-50)#ont-port
{catv<K>|eth<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K>}:eth
{adaptive<K>|eth-port<U><0,8>}:4
{<cr>|catv<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K>}:pots
{adaptive<K>|pots-port<U><0,32>}:2
{<cr>|catv<K>|moca<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K>}: catv
{adaptive<K>|catv-port<U><0,8>}:1
{<cr>|moca<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K>|vdsl<K>}:
Command:
ont-port eth 4 pots 2 catv 1
```

After the configuration is complete, run the commit command to make the configuration take effect.

```
HUAWEI(config-gpon-srvprofile-50)#commit
HUAWEI(config-gpon-srvprofile-50)#quit
```

Display the service profile status.

```
Huawei_OLT(config)#display ont-srvprofile gpon profile-id 50
```

```
-----
Profile-ID: 50
Profile-name: 5hg8247
Access-type: GPON
-----
```

```
Port-type Port-number
-----
```

```
POTS 2
ETH 4
VDSL 0
TDM 0
MOCA 0
CATV 1
-----
```

```
TDM port type: E1
TDM service type: TDMoGem
MAC learning function switch: Enable
```

```

ONT transparent function switch: Disable
Ring check switch: Disable
Ring port auto-shutdown: Enable
Ring detect frequency: 8 (pps)
Ring resume interval: 300 (s)
Multicast forward mode: Unconcern
Multicast forward VLAN: -
Multicast mode: Unconcern
Upstream IGMP packet forward mode: Unconcern
Upstream IGMP packet forward VLAN: -
Upstream IGMP packet priority: -
Native VLAN option: Concern
Upstream PQ color policy: -
Downstream PQ color policy: -
-----
Port-type Port-ID QinQmode PriorityPolicy Inbound Outbound
-----
ETH 1 unconcern unconcern unconcern unconcern
ETH 2 unconcern unconcern unconcern unconcern
ETH 3 unconcern unconcern unconcern unconcern
ETH 4 unconcern unconcern unconcern unconcern
-----
Notes: Run the display traffic table ip command to query
traffic table configuration
-----
Port-type Port-ID DownstreamMode MismatchPolicy
-----
ETH 1 operation discard
ETH 2 operation discard
ETH 3 operation discard
ETH 4 operation discard
-----
Port-type Port-ID Dscp-mapping-table-index
-----
ETH 1 0
ETH 2 0
ETH 3 0
ETH 4 0
IPHOST 1 0
-----
Port-type Port-ID IGMP-mode IGMP-VLAN IGMP-PRI Max-MAC-Count
-----
ETH 1 - - - Unlimited
ETH 2 - - - Unlimited
ETH 3 - - - Unlimited
ETH 4 - - - Unlimited
-----
Binding times: 0
-----

```

Step 4 Enable the auto-discovery function of the PON port.

Enter the PON board view and enable the auto-discovery function of the PON port in the network topology.

Huawei_OLT(config)#**interface**

```
{adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|h
248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vds<K>|vlanif<K>|vrgw<K>}:gpon
```

```
{frameid/slotid<S><Length 3-15>}:0/3
```

Command:

interface gpon 0/3

HUAWEI(config-if-gpon-0/3)#**port**

```
{dba<K>|optical-alarm-profile<K>|portid<U><0,7>}:0
```

```
{fec<K>|laser-switch<K>|ont-auto-find<K>|ont-password-renew<K>|range<K>}:ont-auto-find
```

```
{switch<E><enable,disable>}:enable
```

Command:

port 0 ont-auto-find enable

Step 5 Querying the ONTs that are automatically discovered by a PON port

HUAWEI(config-if-gpon-0/3)#**display ont autofind 0**

Number: 1

F/S/P: 0/3/0

Ont SN: 48575443B205EC07 (HWTC-B203E407)

Password: 0x000000000000000000000000

Loid:

Checkcode:

VendorID: HWTC

Ont Version: 120D4600

Ont SoftwareVersion: V1R006C01S220

Ont EquipmentID: 247

Ont autofind time: 2018-08-13 16:45:50+08:00

Step 6 ONT registration

Register the ONT, set the authentication mode to SN authentication and management type to OMCI, and bind line profile 50 and service profile 50 to the ONT.

HUAWEI(config-if-gpon-0/3)#**ont**

```
{activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K>}:confirm
```

```
{portid<U><0,7>}:0
```

```
{all<K>|loid-auth<K>|ontid<K>|password-auth<K>|sn-auth<K>}:ontid
```

```
{ontid<U><0,127>}:5
```

```
{loid-auth<K>|password-auth<K>|sn-auth<K>}:sn-auth
```

```
{sn-value<S><Length 13-16>}:48575443B205EC07
```

```
{omci<K>|password-auth<K>|snmp<K>}:omci
```

```
{<cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K>|ont-srvprofile-id<K>|ont-srvprofile-
name<K>}:ont-lineprofile-id
```

```
{profile-id<U><0,8192>}:50
```

```
{<cr>|desc<K>|ont-srvprofile-id<K>|ont-srvprofile-name<K>}:ont-srvprofile-id 50
```

```
{<cr>|desc<K>}:
```

Command:

ont confirm 0 ontid 5 sn-auth 48575443B205EC07 omci ont-lineprofile-id 50 ont-srvprofile-id 50

Number of ONTs that can be added: 1, success: 1
PortID :0, ONTID :5

Display the ONT status.

```
HUAWEI(config-if-gpon-0/3)#display ont info 0 all
```

```
-----
F/S/P ONT SN Control Run Config Match Protect
ID flag state state state side
-----
```

```
0/ 3/0 5 48575443B205EC07 active online normal match no
-----
```

```
F/S/P ONT-ID Description
-----
```

```
0/ 3/0 5 ONT_NO_DESCRIPTION
-----
```

```
In port 0/ 3/0, the total of ONTs are: 1, online: 1
```

Run the display ont info 0 5 command to query the detailed information about the ONT.

Step 7 Configure a voice service VLAN.

Configure voice VLAN 172.

```
Huawei_OLT(config)#vlan
```

```
{attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-
policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093>}:172
```

```
{<cr>|to<K>|vlantype<E><mux,standard,smart,super>}:smart
```

```
Command:
```

```
vlan 172 smart
```

Add an upstream interface to the voice VLAN.

```
Huawei_OLT(config)#port
```

```
{desc<K>|mode<K>|vlan<K>}:vlan
```

```
{name<K>|vlan-list<S><Length 1-256>|vlanid<U><1,4093>}:172
```

```
{frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K>}:0/20
```

```
{portlist<S><Length 1-256>}:0
```

```
Command:
```

```
port vlan 172 0/20 0
```

Display information about VLAN 172.

```
Huawei_OLT(config)#display vlan 172
```

```
{<cr>|inner-vlan<K>|to<K>}:<cr>
```

```
Command:
```

```
display vlan 172
```

```
VLAN ID: 172
```

```
VLAN name: VLAN_0172
```

```
VLAN type: smart
```

```
VLAN attribute: common
```

```
VLAN description:
```

```
VLAN forwarding mode in control board: VLAN-MAC
```

```
VLAN forwarding mode: VLAN-MAC
```

```
VLAN broadcast packet forwarding policy: forward
VLAN unknown multicast packet forwarding policy: forward
VLAN unknown unicast packet forwarding policy: forward
VLAN bind service profile ID: -
VLAN bind RAIO profile index: -
VLAN priority: -
```

```
-----
F /S /P Native VLAN State
-----
```

```
0 /20/0 1 up
-----
```

```
Standard port number: 1
Service virtual port number: 0
```

Step 8 Configuring a VLAN Layer 3 Interface

Set the IP address of the Layer 3 interface to 17.1.1.100 /8.

```
Huawei_OLT(config)#interface
{adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|h
248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vdsl<K>|vlanif<K>|vrgw<K>}:vlanif
{<1-4093>}:172
Command:
interface vlanif 172
HUAWEI(config-if-vlanif172)#ip
{address<K>|binding<K>|verify<K>}:address
{ip_addr<I><X.X.X.X>}:17.1.1.100
{integer<U><0,32>|ip_addr<I><X.X.X.X>}:8
{<cr>|description<K>|sub<K>}:
Command:
ip address 17.1.1.100 8
```

Step 9 Configure a service port in the voice VLAN.

Set the VLAN ID of the service flow to 172, PON port to 0/3/0, ONT ID to 5, GEM port ID to 1, and C-VLAN ID to 172.

```
Huawei_OLT(config)#service-port
{desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K>}:vlan
{aoe<K>|name<K>|vlanid<U><1,4093>}:172
{adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K>}:gpon
{frameid/slotid/portid<S><Length 5-18>}:0/3/0
{ont<K>}:ont
{ontid<U><0,127>}:5
{eth<K>|gemport<K>|iphost<K>}: gemport
{gemindex<U><0,1023>}:1
{<cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K>}:multi-service
{user-8021p<K>|user-encap<K>|user-vlan<K>}:user-vlan
{other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095>}:172
{<cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K>}:
Command:
service-port vlan 172 gpon 0/3/0 ont 5 gemport 1 multi-service user-vlan 172
```

Display the status of the service virtual port.

```
Huawei_OLT(config)#display service-port all
{<cr>|sort-by<K>||<K>}:
Command:
display service-port all
Switch-Oriented Flow List
-----
INDEX VLAN VLAN PORT F/ S/ P VPI VCI FLOW FLOW RX TX STATE
ID ATTR TYPE TYPE PARA
-----
0 172 common gpon 0/3 /0 5 1 vlan 172 - - up
-----
Total: 1 (Up/Down: 1/0)
```

The status of the service virtual port is up.

Step 10 Check voice VLAN 172.

Run the display vlan 172 command to check the status of the upstream and downstream interfaces of VLAN 172.

```
Huawei_OLT(config)#display vlan 172
{<cr>|inner-vlan<K>|to<K>}:
Command:
display vlan 172
VLAN ID: 172
VLAN name: VLAN_0172
VLAN type: smart
VLAN attribute: common
VLAN description:
VLAN forwarding mode in control board: VLAN-MAC
VLAN forwarding mode: VLAN-MAC
VLAN broadcast packet forwarding policy: forward
VLAN unknown multicast packet forwarding policy: forward
VLAN unknown unicast packet forwarding policy: forward
VLAN bind service profile ID: -
VLAN bind RAIO profile index: -
VLAN priority: -
-----
F /S /P Native VLAN State
-----
0 /20/0 1 up
-----
Standard port number: 1
-----
INDEX TYPE STATE F /S /P VPI VCI FLOWTYPE FLOWPARA
-----
0 gpon up 0 /3 /0 5 1 vlan 172
-----
Service virtual port number: 1
Note: F--Frame, S--Slot, P--Port,
VPI indicates ONT ID for PON, VCI indicates GEM index for GPON,
v/e--vlan/encap, pritag--priority-tagged
```

The status of the upstream port 0/20/0 and the service virtual port is normal.

Step 11 Enable global ARP proxy.

```
Huawei_OLT(config)#arp
{aging-mode<K>|aging-time<K>|conflict<K>|gratuitous-
arp<K>|ip_addr<I><X.X.X.X>|learning<K>|proxy<K>}:proxy
{disable<K>|enable<K>}:enable
Command:
arp proxy enable
```

Step 12 Enable ARP proxy on the Layer 3 interface.

```
Huawei_OLT(config)#interface vlanif 172
HUAWEI(config-if-vlanif172)#arp
{proxy<K>}:proxy
{disable<K>|enable<K>}:enable
{<cr>|subvlan<K>}:
Command:
arp proxy enable
```

Step 13 Configuration of ONT 2

The configuration of ONT2 is similar to that of ONT1.

Step 14 Common Commands

Table 3-3 Common Commands

Common Commands	Function
display dba-profile	This command is used to query the information about the dynamic bandwidth assignment (DBA) profile in the system.
display ont-lineprofile	This command is used to query the information about the GPON ONT line profile that has taken effect, including the binding relationship between the T-CONT and the DBA profile, the QoS mode of the traffic stream, and the mapping between the GEM port and the ONT-side service.
display ont-srvprofile	This command is used to query the information about a GPON ONT service profile.
display ont autofind	This command is used to query the basic information about the currently auto-found ONT in the system.
display ont info	This command is used to query the information about an ONT, including the current status of the ONT, the related configuration of the ONT, and the information about the ONT T-CONT.
display vlan	This command is used to query the VLAN information.
display service-port	This command is used to query the information about a service virtual port.

3.2.3 Configuration Procedure (ONT Side)

Step 1 Log in to the HG8247.

Use a network cable to connect the PC to the network port of the HG8247 and set the IP address of the PC to 192.168.100.100/24.

Enter HYPERLINK "http://192.168.100.1/" http://192.168.100.1/ in the address box of the browser on the PC.

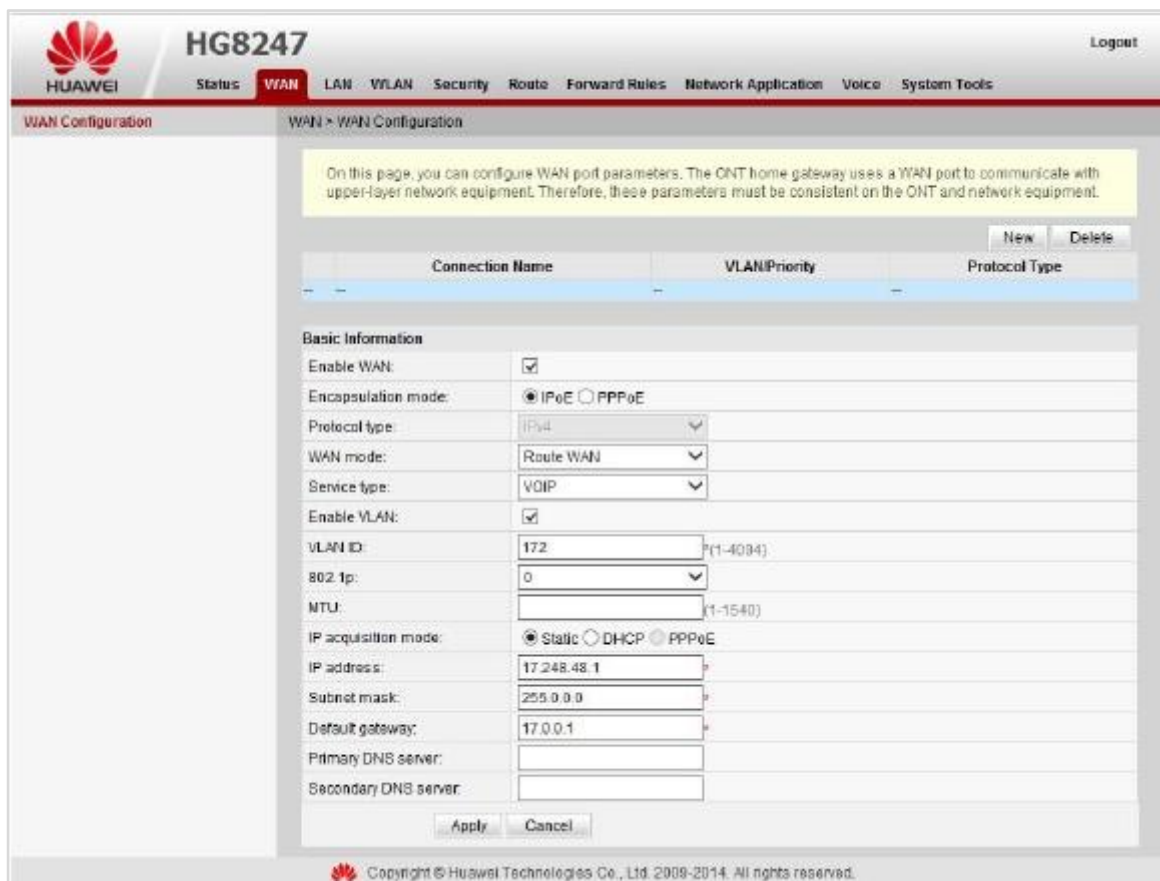
Login account: telecomadmin/admintelecom

The login page is displayed, as shown in the following figure.



Step 2 ONT WAN Port Configuration

Set the WAN port type to route WAN port, service type to VoIP, VLAN ID to 172, IP address of the ONT to 17.248.48.1/8, subnet mask to 255.0.0.0, and gateway to 17.0.0.1. Then, click Apply.



HG8247 Logout

Status: **WAN** LAN WLAN Security Route Forward Rules Network Application Voice System Tools

WAN Configuration WAN > WAN Configuration

On this page, you can configure WAN port parameters. The ONT home gateway uses a WAN port to communicate with upper-layer network equipment. Therefore, these parameters must be consistent on the ONT and network equipment.

New Delete

Connection Name	VLAN	Priority	Protocol Type

Basic Information

Enable WAN: ☒

Encapsulation mode: ☒ IPoE ☐ PPPoE

Protocol type: IPv4

WAN mode: Route WAN

Service type: VOIP

Enable VLAN: ☒

VLAN ID: 172 (1-4094)

802.1p: 0

MTU: (1-1540)

IP acquisition mode: ☒ Static ☐ DHCP ☐ PPPoE

IP address: 172.48.48.1

Subnet mask: 255.0.0.0

Default gateway: 172.0.0.1

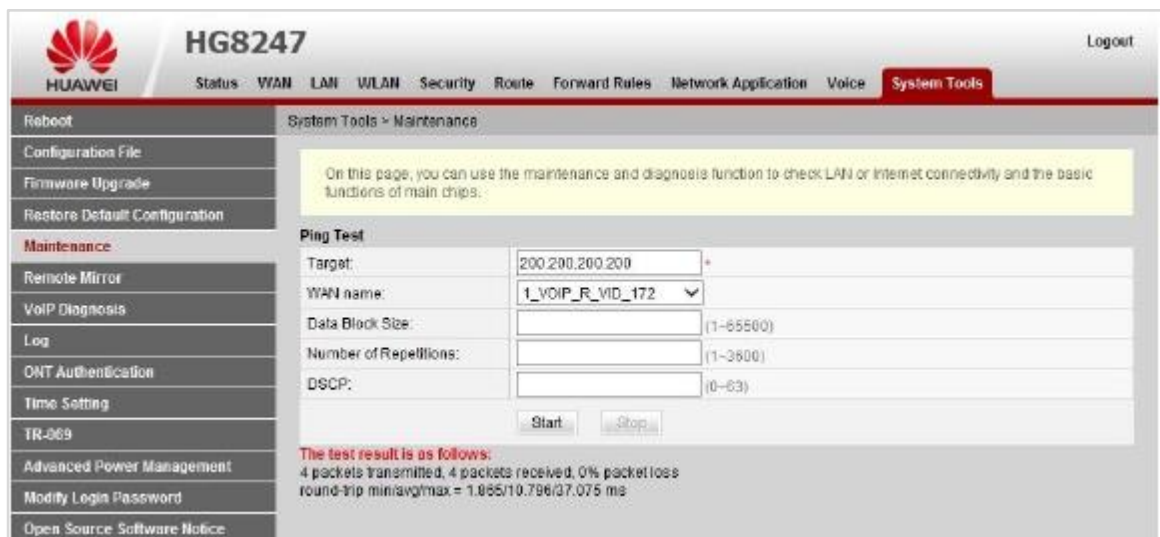
Primary DNS server:

Secondary DNS server:

Apply Cancel

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Step 3 Route Configuration



HG8247 Logout

Status: WAN LAN WLAN Security Route Forward Rules Network Application Voice **System Tools**

Reboot Configuration File Firmware Upgrade Restore Default Configuration

Maintenance

Remote Mirror VoIP Diagnosis Log ONT Authentication Time Setting TR-069 Advanced Power Management Modify Login Password Open Source Software Notice

System Tools > Maintenance

On this page, you can use the maintenance and diagnosis function to check LAN or Internet connectivity and the basic functions of main chips.

Ping Test

Target: 200.200.200.200 *

WAN name: 1_VOIP_R_VID_172

Data Block Size: (1-65500)

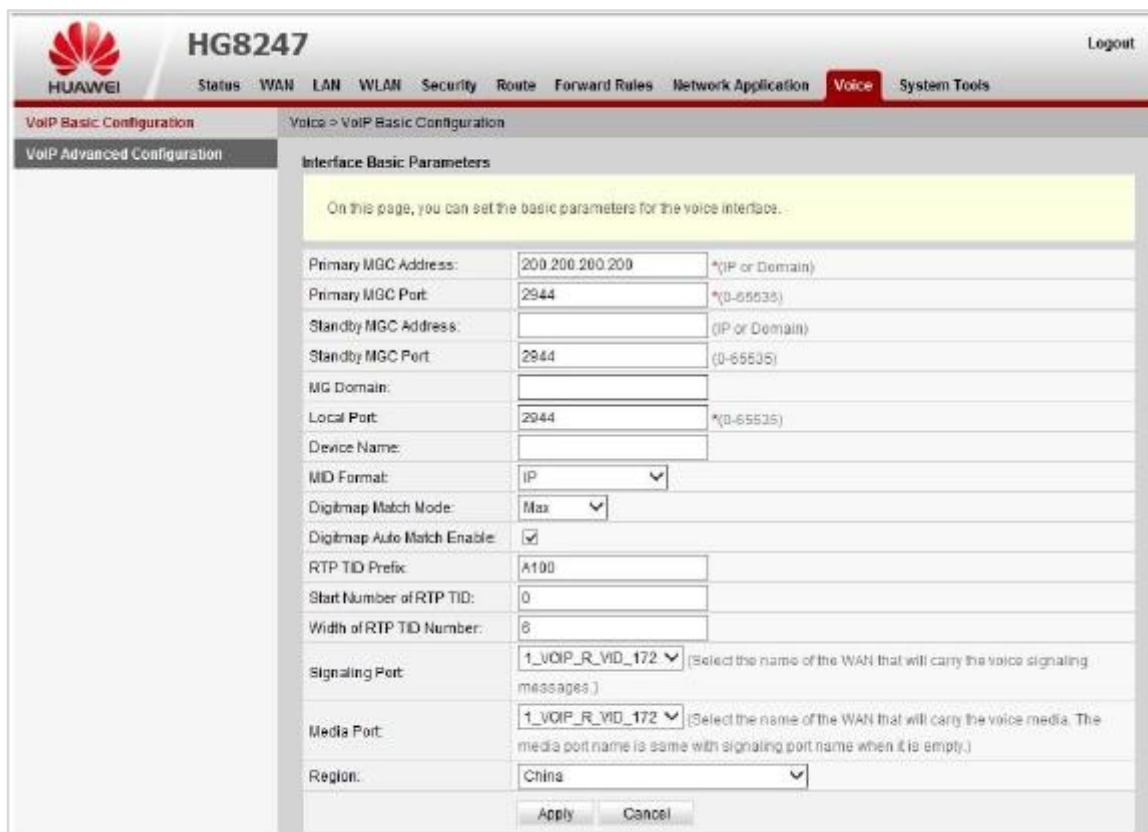
Number of Repetitions: (1-3600)

DSCP: (0-63)

Start Stop

The test result is as follows:
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.885/10.796/37.075 ms

Step 4 ONT Voice Interface Configuration



HG8247 Logout

Status WAN LAN WLAN Security Route Forward Rules Network Application **Voice** System Tools

Voice Basic Configuration Voice > Voice Basic Configuration

Voice Advanced Configuration

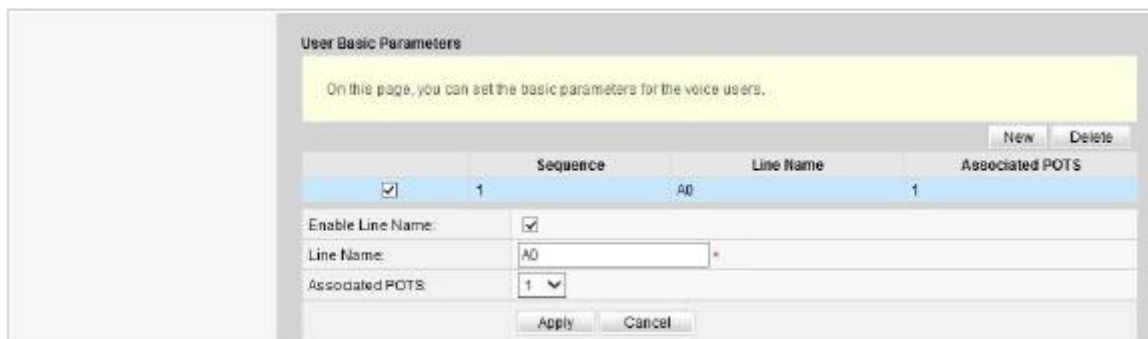
Interface Basic Parameters

On this page, you can set the basic parameters for the voice interface.

Primary MG Address:	200.200.200.200	*(IP or Domain)
Primary MG Port:	2944	*(0-65535)
Standby MG Address:		(IP or Domain)
Standby MG Port:	2944	(0-65535)
MG Domain:		
Local Port:	2944	*(0-65535)
Device Name:		
MID Format:	IP	
Digitmap Match Mode:	Max	
Digitmap Auto Match Enable:	☒	
RTP TID Prefix:	A100	
Start Number of RTP TID:	0	
Width of RTP TID Number:	6	
Signaling Port:	1_VOIP_R_VID_172	(Select the name of the WAN that will carry the voice signaling messages.)
Media Port:	1_VOIP_R_VID_172	(Select the name of the WAN that will carry the voice media. The media port name is same with signaling port name when it is empty.)
Region:	China	

Apply Cancel

Step 5 Voice User Configuration



User Basic Parameters

On this page, you can set the basic parameters for the voice users.

New Delete

	Sequence	Line Name	Associated POTS
☒	1	A0	1

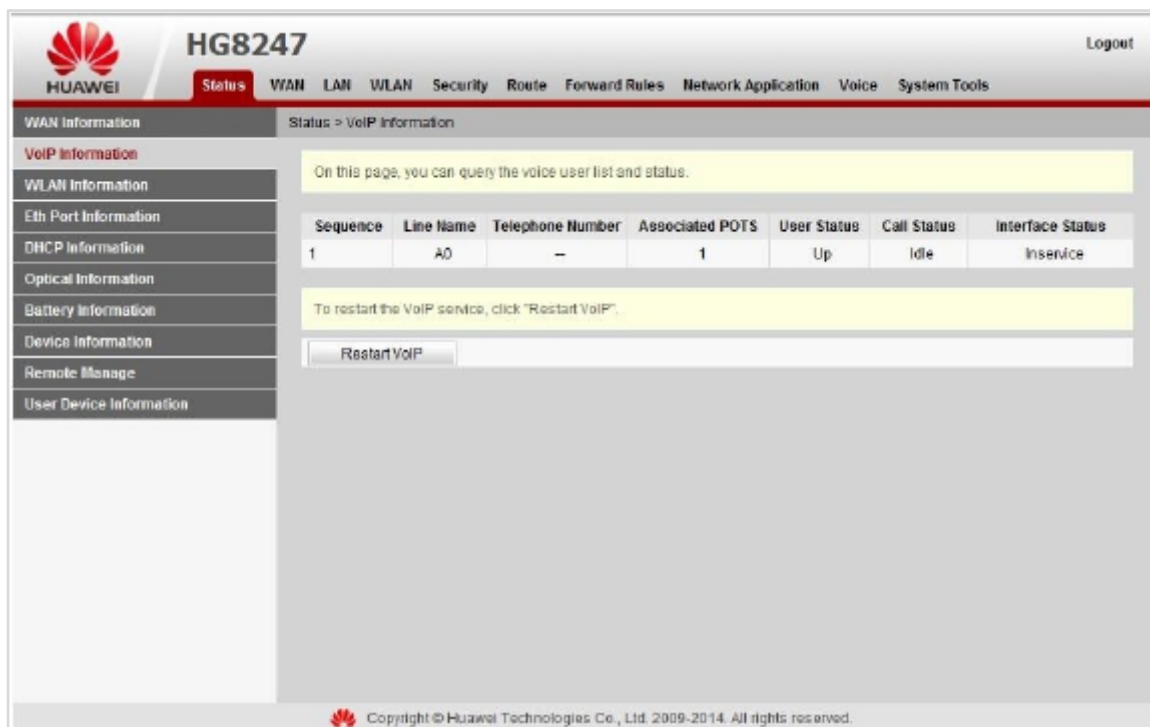
Enable Line Name: ☒

Line Name: A0 *

Associated POTS: 1

Apply Cancel

Step 6 View the configuration result.



3.2.4 Experiment Verification

Verification method: Connect two phones to different POTS ports on the ONT and make calls to each other.

3.3 Configuration Reference

3.3.1 OLT Data Configuration

```
dba-profile add profile-id 50 type1 fix 2048
```

```
ont-lineprofile gpon profile-id 50 profile-name 5hg82471
tcont 1 dba-profile-id 50
gem add 1 eth tcont 1
gem mapping 1 1 vlan 172
commit
quit
```

```
ont-srvprofile gpon profile-id 50 profile-name 5hg8247
ont-port eth 4 pots 2catv 1
commit
quit
```

```
interface gpon 0/3
port 0 ont-auto-find enable
```

```
ont confirm 0 ontid 5 sn-auth 48575443B203E407 omci ont-lineprofile-id 50 ont-srvprofile-id 50
quit
```

```
vlan 172 smart
port vlan 172 0/20 0
interface vlanif 172
ip address 17.1.1.100 8
quit
service-port 1 vlan 172 gpon 0/3/0 ont 5 gemport 1 multi-service user-vlan 172

arp proxy enable
interface vlanif 172
arp proxy enable
quit

ip route-static 200.200.200.0 255.255.255.0 17.0.0.1
```

3.4 Questions

1. Where can I define the mapping between user numbers and terminal IDs?

4 GPON ONU H.248 Voice Service Configuration

4.1 Experiment Introduction

4.1.1 About This Experiment

In this experiment, the data configuration of the OLT and the MDU is used to implement the voice call between two users of different MDUs connected to the same PON port.

4.1.2 Objectives

- Understand the basic principles of GPON.
- Understand the basic principles of VoIP.
- Understand the GPON data profile.
- Master the configuration of the MDU data.
- Configure voice service profiles.
- Understand the overall configuration process of the voice service.

4.1.3 Experiment Networking

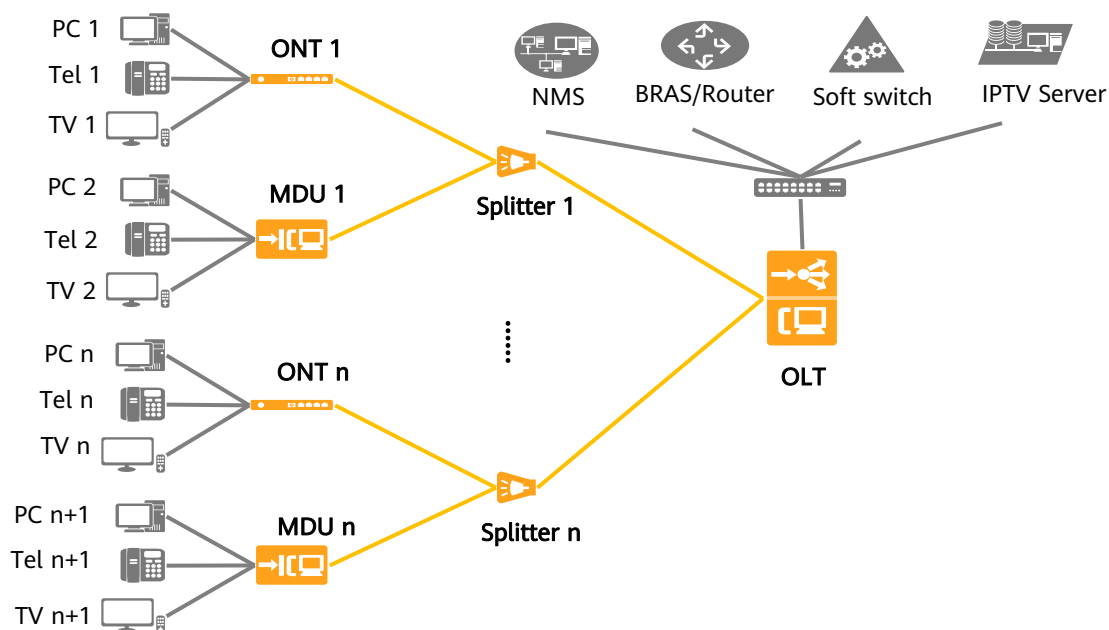


Figure 4-1 Topology for configuring the GPON ONU VoIP service

4.1.4 Experiment Planning

According to the VoIP configuration scenario of the access network, phone 1 and phone 2 are common phones, and PC1 and PC2 run the Windows operating system.

Table 4-1 GPON ONU VoIP data plan

Configuration Item	Data	Configuration Item	Data
OLT service VLAN	VLAN:172 Type: Smart IP:17.1.1.100/8	OLT service VLAN	VLAN:172 Type: Smart IP:17.1.1.100/8
OLT management VLAN	VLAN:500 Type: Smart IP:10.1.1.1/24	OLT management VLAN	VLAN:500 Type: Smart IP:10.1.1.1/24
OLT upstream port	0/20/0	OLT upstream port	0/20/0
OLT downstream port	0/3/0	OLT downstream port	0/3/0
MDU	ID:7 SN:4857544329FC9507	MDU	ID:8 SN:48575443F2154B04

	IP:10.1.1.7/24		IP:10.1.1.8/24
TCONT,DBA	0 (management) 3 (service) DBA profile ID: 70	TCONT,DBA	0 (management) 4 (service) DBA profile ID: 80
GEM Port	0 (management) 3 (service)	GEM Port	0 (management) 4 (service)
ONT user VLAN	500 (management) 172 (service)	ONT user VLAN	500 (management) 172 (service)

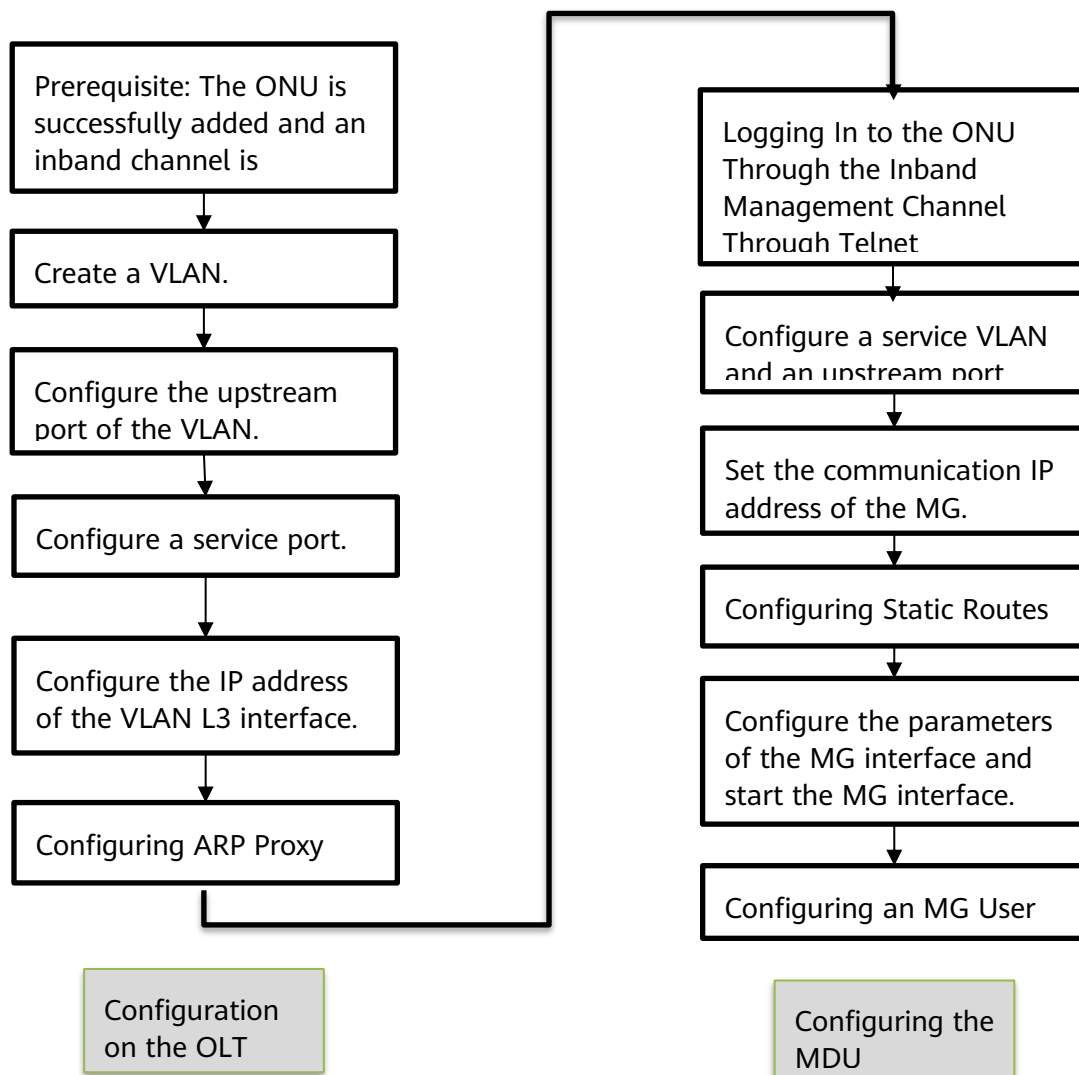
Table 4-2 Data plan for the GPON ONU H.248 voice server

Configuration Item	Data	Configuration Item	Data
H.248 server	200.200.200.200	Signaling IP address	17.248.48.1(MDU1) 17.248.48.2(MDU2)
H.248 server port	2944	Media IP address	17.248.48.1(MDU1) 17.248.48.2(MDU2)
Transmission Mode	UDP	Encoding	Text
Voice gateway	17.0.0.1/8	Phone Number	18980100/18980200
VLANIF IP	17.248.48.1/8(MDU1) 17.248.48.2/8(MDU2)	Terminal ID	0 (MDU1 and MDU2)

4.2 Experiment Task Configuration

4.2.1 Configuration Roadmap

Figure 4-2 Flowchart for configuring the GPON ONU VoIP service



4.2.2 Configuration Procedure (OLT Side)

Step 1 DBA Profile Configuration

Run the display dba-profile command to query the existing DBA profiles in the system. If the existing DBA profile in the system cannot meet the requirements, run the dba-profile add command to add a DBA profile. Set the profile ID to 70, bandwidth type to type1, and bandwidth value to 1024 kbit/s.

```

Huawei_OLT(config)#dba-profile
{add<K>|delete<K>|modify<K>}:add
{profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K>}:profile-id
{profile-id<U><10,512>}:70
    
```

```
{profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K>}:type1
{fix<K>}:fix
{fix-bandwidth<U><128,10000000>}:1024
{<cr>|bandwidth_compensate<K>}:
Command:
dba-profile add profile-id 70 type1 fix 1024
Adding a DBA profile succeeded
Profile ID: 70
Profile name: dba-profile_70
```

Display the status of a DBA profile.

```
Huawei_OLT(config)#display dba-profile profile-id 70
-----
Profile-name: dba-profile_30
Profile-ID: 30
type: 1
Bandwidth compensation: No
Fix-delay: No
Fix(kbps): 1024
Assure(kbps): 0
Max(kbps): 0
Additional-bandwidth: -
Best-effort-priority: -
Best-effort-weight: -
bind-times: 0
-----
```

Step 2 Configuring a Line Profile

Set the line profile ID to 70 and profile name to EA5822.

```
Huawei_OLT(config)#ont-lineprofile
{epon<K>|gpon<K>}:gpon
{<cr>|profile-id<K>|profile-name<K>}:profile-id
{profile-id<U><0,8192>}:70
{<cr>|profile-name<K>}:profile-name
{profile-name<S><Length 1-32>}:EA5822
Command:
ont-lineprofile gpon profile-id 70 profile-name EA5822
```

Bind T-CONT 1 to DBA profile 30.

```
HUAWEI(config-gpon-lineprofile-70)#tcont
{tcont-list<S><Length 1-13>}:1
{<cr>|dba-profile-id<K>|dba-profile-name<K>}:dba-profile-id
{profile-id<U><0,512>}:70
Command:
tcont 1 dba-profile-id 70
```

Add GEM port 0 to carry ETH service flows, and bind GEM port 0 to T-CONT 0.

```
HUAWEI(config-gpon-lineprofile-70)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:add
{gem-index<U><0,1023>}:0
```

```
{service-type<E><eth,tdm>}:eth
{tcont<K>}:tcont
{tcont-id<U><0,127>}:0
{<cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K>}:
Command:
gem add 0 eth tcont 0
```

Add GEM port 3 to carry ETH service flows, and bind GEM port 3 to T-CONT 1.

```
HUAWEI(config-gpon-lineprofile-70)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:add
{gem-index<U><0,1023>}:3
{service-type<E><eth,tdm>}:eth
{tcont<K>}:tcont
{tcont-id<U><0,127>}:1
{<cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K>}:
Command:
gem add 3 eth tcont 1
```

Map the management flow of user-side VLAN 500 to GEM port 0. The mapping index is 0.

```
HUAWEI(config-gpon-lineprofile-70)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:mapping
{gem-index<U><0,1023>}:0
{mapping-index<U><0,7>}:0
{e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K>}:vlan
{untag<K>|vlan-id<U><0,4095>}:500
{<cr>|flow-car<K>|priority<K>|transparent<K>}:
Command:
gem mapping 0 0 vlan 500
```

Map the service flow of CVLAN 172 to GEM port 3 with the mapping index of 1.

```
HUAWEI(config-gpon-lineprofile-70)#gem
{add<K>|delete<K>|mapping<K>|modify<K>}:mapping
{gem-index<U><0,1023>}:3
{mapping-index<U><0,7>}:1
{e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K>}:vlan
{untag<K>|vlan-id<U><0,4095>}:172
{<cr>|flow-car<K>|priority<K>|transparent<K>}:
Command:
gem mapping 3 1 vlan 172
```

After the configuration is complete, run the commit command to make the configuration take effect.

```
HUAWEI(config-gpon-lineprofile-70)#commit
HUAWEI(config-gpon-lineprofile-70)#quit
```

Display information about the line profile.

```
Huawei_OLT(config)#display ont-lineprofile gpon profile-id 70
```

```
-----
Profile-ID :70
Profile-name :EA5822
Access-type :GPON
-----
FEC upstream switch :Disable
OMCC encrypt switch :Off
Qos mode :PQ
Mapping mode :VLAN
TR069 management :Disable
TR069 IP index :0
-----
<T-CONT 0> DBA Profile-ID:1
<Gem Index 0>
-----
|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
|Upstream-priority-queue:0 |Downstream-priority-queue:- |
-----
Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR
-----
0 500 - - - - -
-----
<T-CONT 1> DBA Profile-ID:70
<Gem Index 3>
-----
|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
|Upstream-priority-queue:0 |Downstream-priority-queue:- |
-----
Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR
-----
1 172 - - - - -
-----
Notes: Run the display traffic table ip command to query
traffic table configuration
-----
Binding times :0
-----
```

Step 3 Enable the auto-discovery function of the PON port.

Enter the PON board view and enable the auto-discovery function of the PON port in the network topology.

```
Huawei_OLT(config)#interface gpon
{frameid/slotid<S><Length 3-15>}:0/3
Command:
interface gpon 0/3
HUAWEI(config-if-gpon-0/3)#port
{dba<K>|optical-alarm-profile<K>|portid<U><0,7>}:0
{fec<K>|laser-switch<K>|ont-auto-find<K>|ont-password-renew<K>|range<K>}:ont-auto-find
{switch<E><enable,disable>}:enable
Command:
```

port 0 ont-auto-find enable

Step 4 Querying the MDUs that are automatically discovered by a PON port

```
HUAWEI(config-if-gpon-0/3)#display ont autofind 0
```

```
-----
Number: 1
F/S/P: 0/3/0
Ont SN: 4857544329FC9507 (HWTC-F2154B04)
Password: 0x30303030303030303030(0000000000)
Loid: 0000000000
Checkcode:
VendorID: HWTC
Ont Version: EA5822 VER.B
Ont SoftwareVersion: V8R312 C00
Ont EquipmentID: SmartAX EA5822
Ont autofind time: 2018-04-13 15:56:02+08:00
-----
```

Step 5 MDU registration

Register the MDU, set the authentication mode to SN authentication, set the management type to SNMP, and bind line profile 70 to the MDU.

```
HUAWEI(config-if-gpon-0/3)#ont
{activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K>}:confirm
{portid<U><0,7>}:0
{all<K>|loid-auth<K>|ontid<K>|password-auth<K>|sn-auth<K>}:ontid
{ontid<U><0,127>}:7
{loid-auth<K>|password-auth<K>|sn-auth<K>}:sn-auth
{sn-value<S><Length 12-16>}:4857544329FC9507
{omci<K>|password-auth<K>|snmp<K>}:snmp
{<cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K>}:ont-lineprofile-id
{profile-id<U><0,8192>}:70
{<cr>|desc<K>}:
Command:
Ont confirm 0 ontid 7 sn-auth 48575443F43FF304 snmp ont-lineprofile-id 70
Number of ONTs that can be added: 1, success: 1
PortID :0, ONTID :7
```

Display the ONT status.

```
HUAWEI(config-if-gpon-0/3)#display ont info 0 all
```

```
-----
F/S/P ONT SN Control Run Config Match Protect
ID flag state state state side
-----
0/ 3/0 7 4857544329FC9507 active online normal match no
-----
F/S/P ONT-ID Description
-----
```

```
0/ 3/0 7 ONT_NO_DESCRIPTION
```

```
-----
In port 0/ 3/0, the total of ONTs are: 1, online: 1
```

Step 6 Configure management VLAN 500.

Configure management VLAN 500, add the upstream interface to VLAN 500, and set the IP address of the interface in VLAN 500 to 10.1.1.1/24.

```
Huawei_OLT(config)#vlan
{attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-
policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093>}:500
{<cr>|to<K>|vlantype<E><mux,standard,smart,super>}:smart
Command:
vlan 500 smart
Huawei_OLT(config)#port
{desc<K>|mode<K>|vlan<K>}:vlan
{name<K>|vlan-list<S><Length 1-256>|vlanid<U><1,4093>}:500
{frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K>}:0/20
{portlist<S><Length 1-256>}:0
Command:
port vlan 500 0/20 0
Huawei_OLT(config)#interface
{adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|h
248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vdsl<K>|vlanif<K>|vrgw<K>}:vlanif
{<1-4093>}:500
Command:
interface vlanif 500
HUAWEI(config-if-vlanif500)#ip address
{ip_addr<I><X.X.X.X>}:10.1.1.1
{integer<U><0,32>|ip_addr<I><X.X.X.X>}:24
{<cr>|description<K>|sub<K>}:
Command:
ip address 10.1.1.1 24
```

Step 7 Configure a service channel for management VLAN 500.

Set the VLAN to which the service port belongs to 500, PON port to 0/3/0, ONT ID to 7, GEM port ID to 0, and C-VLAN ID to 500.

```
Huawei_OLT(config)#service-port
{desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K>}:vlan
{aoe<K>|name<K>|vlanid<U><1,4093>}:500
{adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K>}:gpon
{frameid/slotid/portid<S><Length 5-18>}:0/3/0
{ont<K>}:ont
{ontid<U><0,127>}:7
{eth<K>|gemport<K>|iphos<K>}:gemport
{gemindex<U><0,1023>}:0
{<cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K>}:multi-service
{user-8021p<K>|user-encap<K>|user-vlan<K>}:user-vlan
{other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095>}:500
{<cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K>}:

```

Command:

service-port vlan 500 gpon 0/3/0 ont 7 gemport 0 multi-service user-vlan 500

Display the status of a service port.

Huawei_OLT(config)#**display service-port all**

{<cr>|sort-by<K>||<K>}:</p>
</div>
<div data-bbox="194 203 276 217" data-label="Text">
<p>Command:</p>
</div>
<div data-bbox="194 217 356 231" data-label="Text">
<p>display service-port all</p>
</div>
<div data-bbox="194 230 379 245" data-label="Text">
<p>Switch-Oriented Flow List</p>
</div>
<div data-bbox="194 259 666 287" data-label="Text">
<p>-----</p>
<table>
<tr>
<th>INDEX</th>
<th>VLAN</th>
<th>VLAN</th>
<th>PORT</th>
<th>F/</th>
<th>S/</th>
<th>P</th>
<th>VPI</th>
<th>VCI</th>
<th>FLOW</th>
<th>FLOW</th>
<th>RX</th>
<th>TX</th>
<th>STATE</th>
</tr>
<tr>
<th>ID</th>
<th>ATTR</th>
<th>TYPE</th>
<th>TYPE</th>
<th>PARA</th>
<th></th>
<th></th>
<th></th>
<th></th>
<th></th>
<th></th>
<th></th>
<th></th>
<th></th>
</tr>
</table>
<p>-----</p>
</div>
<div data-bbox="194 301 524 316" data-label="Text">
<p>1 500 common gpon 0/3 /0 7 0 vlan 500 - - up</p>
<p>-----</p>
</div>
<div data-bbox="194 329 366 345" data-label="Text">
<p>Total: 1 (Up/Down: 1/0)</p>
</div>
<div data-bbox="161 359 802 377" data-label="Text">
<p>Step 8 Configure the inband management VLAN and IP address of the MDU.</p>
</div>
<div data-bbox="175 386 912 418" data-label="Text">
<p># On the OLT, issue the IP address 10.1.1.3 24, gateway 10.1.1.1, and VLAN 500 to the MDU.</p>
</div>
<div data-bbox="194 433 481 448" data-label="Text">
<p>Huawei_OLT(config)#interface gpon 0/3</p>
</div>
<div data-bbox="194 447 434 462" data-label="Text">
<p>HUAWEI(config-if-gpon-0/3)#ont</p>
</div>
<div data-bbox="194 462 883 533" data-label="Text">
<p>{activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-profile<K>|deactivate<K>|delete<K>|factory-setting-restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K>}:ipconfig</p>
</div>
<div data-bbox="194 532 333 547" data-label="Text">
<p>{portid<U><0,7>}:0</p>
</div>
<div data-bbox="194 547 343 561" data-label="Text">
<p>{ontid<U><0,127>}:7</p>
</div>
<div data-bbox="194 561 707 576" data-label="Text">
<p>{dhcp<K>|dscp-mapping-table<K>|ip-index<K>|pppoe<K>|static<K>}:static</p>
</div>
<div data-bbox="194 575 740 590" data-label="Text">
<p>{<cr>|gateway<K>|ip-address<K>|pri-dns<K>|slave-dns<K>|vlan<K>}:ip-address</p>
</div>
<div data-bbox="194 590 425 604" data-label="Text">
<p>{ip-address<I><X.X.X.X>}:10.1.1.7</p>
</div>
<div data-bbox="194 604 318 618" data-label="Text">
<p>{mask<K>}:mask</p>
</div>
<div data-bbox="194 618 443 633" data-label="Text">
<p>{mask<M><X.X.X.X>}:255.255.255.0</p>
</div>
<div data-bbox="194 633 627 647" data-label="Text">
<p>{<cr>|gateway<K>|pri-dns<K>|slave-dns<K>|vlan<K>}:gateway</p>
</div>
<div data-bbox="194 647 411 661" data-label="Text">
<p>{gateway<I><X.X.X.X>}:10.1.1.1</p>
</div>
<div data-bbox="194 661 508 676" data-label="Text">
<p>{<cr>|pri-dns<K>|slave-dns<K>|vlan<K>}:vlan</p>
</div>
<div data-bbox="194 676 376 690" data-label="Text">
<p>{vlanid<U><0,4095>}:500</p>
</div>
<div data-bbox="194 690 500 704" data-label="Text">
<p>{<cr>|pri-dns<K>|priority<K>|slave-dns<K>}:</p>
</div>
<div data-bbox="194 704 276 718" data-label="Text">
<p>Command:</p>
</div>
<div data-bbox="194 718 810 733" data-label="Text">
<p>ont ipconfig 0 7 static ip-address 10.1.1.7 mask 255.255.255.0 gateway 10.1.1.1 vlan 500</p>
</div>
<div data-bbox="161 748 667 765" data-label="Text">
<p>Step 9 Test the connectivity between the OLT and the MDU.</p>
</div>
<div data-bbox="194 778 441 794" data-label="Text">
<p>Huawei_OLT(config)#ping 10.1.1.7</p>
</div>
<div data-bbox="194 793 552 808" data-label="Text">
<p>PING 10.1.1.7:56 data bytes, press CTRL_C to break</p>
</div>
<div data-bbox="194 808 623 822" data-label="Text">
<p>Reply from 10.1.1.7: bytes=56 Sequence=1 ttl=255 time=1 ms</p>
</div>
<div data-bbox="194 822 631 837" data-label="Text">
<p>Reply from 10.1.1.7: bytes=56 Sequence=2 ttl=255 time=10 ms</p>
</div>
<div data-bbox="194 837 623 851" data-label="Text">
<p>Reply from 10.1.1.7: bytes=56 Sequence=3 ttl=255 time=1 ms</p>
</div>
<div data-bbox="194 851 631 865" data-label="Text">
<p>Reply from 10.1.1.7: bytes=56 Sequence=4 ttl=255 time=10 ms</p>
</div>
<div data-bbox="194 865 631 880" data-label="Text">
<p>Reply from 10.1.1.7: bytes=56 Sequence=5 ttl=255 time=10 ms</p>
</div>
</div>

```

--- 10.1.1.7 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/6/10 ms

```

Step 10 Configure a voice service VLAN.

Configure voice service VLAN 172 and add an upstream port to the voice service VLAN.

```

Huawei_OLT(config)#vlan
{attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-
policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><1,4093>}:172
{<cr>|to<K>|vlantype<E><mux,standard,smart,super>}:smart
Command:
vlan 172 smart
Huawei_OLT(config)#port
{desc<K>|mode<K>|vlan<K>}:vlan
{name<K>|vlan-list<S><Length 1-256>|vlanid<U><1,4093>}:172
{frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K>}:0/20
{portlist<S><Length 1-256>}:0
Command:
port vlan 172 0/20 0

```

Step 11 Configure the Layer 3 interface of the service VLAN.

Set the IP address of the Layer 3 interface to 17.1.1.100 /8.

```

Huawei_OLT(config)#interface
{adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|h
248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vdsl<K>|vlanif<K>|vrgw<K>}:vlanif
{<1-4093>}:172
Command:
interface vlanif 172
HUAWEI(config-if-vlanif172)#ip address 17.1.1.100 8
{<cr>|description<K>|sub<K>}:
Command:
ip address 17.1.1.100 8

```

Step 12 Configure a service port in the voice VLAN.

```

Huawei_OLT(config)#service-port
{desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K>}:vlan
{aoe<K>|name<K>|vlanid<U><1,4093>}:172
{adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K>}:gpon
{frameid/slotid/portid<S><Length 5-18>}:0/3/0
{ont<K>}:ont
{ontid<U><0,127>}:7
{eth<K>|gemport<K>|iphost<K>}: gemport
{gemindex<U><0,1023>}:3
{<cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K>}:multi-service
{user-8021p<K>|user-encap<K>|user-vlan<K>}:user-vlan
{other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095>}:172

```

```
{<cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K>}:
Command:
service-port vlan 172 gpon 0/3/0 ont 7 gemport 3 multi-servi user-vlan 172
Huawei_OLT(config)#display service-port all
{<cr>|sort-by<K>||<K>}:
Command:
display service-port all
Switch-Oriented Flow List
-----
INDEX VLAN VLAN PORT F/ S/ P VPI VCI FLOW FLOW RX TX STATE
ID ATTR TYPE TYPE PARA
-----
1 500 common gpon 0/3 /0 7 0 vlan 500 - - up
2 172 common gpon 0/3 /0 7 3 vlan 172 - - up
-----
Total: 2 (Up/Down: 2/0)
```

Step 13 Check the information about service VLAN 172.

```
Huawei_OLT(config)#display vlan 172
{<cr>|inner-vlan<K>|to<K>}:
Command:
display vlan 172
VLAN ID: 172
VLAN name: VLAN_0172
VLAN type: smart
VLAN attribute: common
VLAN description:
VLAN forwarding mode in control board: VLAN-MAC
VLAN forwarding mode: VLAN-MAC
VLAN broadcast packet forwarding policy: forward
VLAN unknown multicast packet forwarding policy: forward
VLAN unknown unicast packet forwarding policy: forward
VLAN bind service profile ID: -
VLAN bind RAIO profile index: -
VLAN priority: -
-----
F /S /P Native VLAN State
-----
0 /20/0 1 up
-----
Standard port number: 1
-----
INDEX TYPE STATE F /S /P VPI VCI FLOWTYPE FLOWPARA
-----
3 gpon up 0 /3 /0 7 3 vlan 172
-----
Service virtual port number: 2
Note: F--Frame, S--Slot, P--Port,
VPI indicates ONT ID for PON, VCI indicates GEM index for GPON,
v/e--vlan/encap, pritag--priority-tagged
```

Step 14 Enable global ARP proxy.

```
Huawei_OLT(config)#arp
```

```
{aging-mode<K>|aging-time<K>|conflict<K>|gratuitous-
arp<K>|ip_addr<I><X.X.X.X>|learning<K>|proxy<K>}:proxy
{disable<K>|enable<K>}:enable
Command:
arp proxy enable
```

Step 15 Enable ARP proxy on the Layer 3 interface.

```
Huawei_OLT(config)#interface vlanif 172
HUAWEI(config-if-vlanif172)#arp
{proxy<K>}:proxy
{disable<K>|enable<K>}:enable
{<cr>|subvlan<K>}:
Command:
arp proxy enable
```

Step 16 Configuration of MDU 2

The configuration of MDU2 is similar to that of MDU1.

4.2.3 Configuration Procedure on the MDU Side

Step 1 Log in to the MDU.

```
Huawei_OLT(config)#telnet 10.1.1.7
{<cr>|service-port<U><1,65535>}:
Command:
telnet 10.1.1.7
Press CTRL_] to quit telnet mode
Trying 10.1.1.7...
Connected to 10.1.1.7...
>>User name:root
>>User password:mduadmin
Huawei Integrated Access Software (EA5822).
Copyright(C) Huawei Technologies Co., Ltd. 2002-2012. All rights reserved.
EA5822>
User root has used a default password. Change the password in time.
```

Step 2 Create SVLAN 172 and add the upstream port to VLAN 172.

```
EA5822(config)#vlan
{attrib<K>|bind<K>|desc<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093>}:172
{<cr>|to<K>|vlantype<E><mux,standard,smart>}:smart
Command:
vlan 172 smart
EA5822(config)#port
{desc<K>|mode<K>|vlan<K>}:vlan
{vlan-list<S><Length 1-256>|vlanid<U><1,4093>}:172
{frame/slot<S><Length 1-15>|to<K>}:0/0
{portlist<S><Length 1-256>}:1
Command:
port vlan 172 0/0 1
```

Step 3 Configure the Layer 3 interface of the service VLAN.

```
EA5822(config)#interface
{emu<K>|eponnni<K>|eth<K>|gponnni<K>|loopback<K>|meth<K>|null<K>|sip<K>|vlanif<K>}:vlanif
{<1-4093>}:172
EA5822(config-if-vlanif172)#ip
{address<K>|verify<K>}:address
{ip_addr<I><X.X.X.X>}:17.248.48.1 8
{<cr>|description<K>|sub<K>}:
Command:
ip address 17.248.48.1 8
EA5822(config-if-vlanif172)#
```

Step 4 Test the connectivity of the voice service between the MDU and the OLT.

```
EA5822(config)#ping 17.1.1.100
PING 17.1.1.100:56 data bytes, press CTRL_C to break
Reply from 17.1.1.100: bytes=56 Sequence=1 ttl=255 time=10 ms
Reply from 17.1.1.100: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 17.1.1.100: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 17.1.1.100: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 17.1.1.100: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 17.1.1.100 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/2/10 ms
```

Step 5 Configure the default route from the MDU to the softswitch.

```
EA5822(config)#ip route-static 200.200.200.0 24 17.0.0.1
{<cr>|description<K>|permanent<K>|preference<K>|tag<K>}:
Command:
ip route-static 200.200.200.0 24 17.0.0.1
```

Step 6 Add an address pool in VoIP mode.

```
EA5822(config)#voip
EA5822(config-voip)#ip
{address<K>}:address
{media<K>|signaling<K>}:media
{media-ip-address<I><X.X.X.X>}:17.248.48.1
{gateway<I><X.X.X.X>}:17.0.0.1
Command:
ip address media 17.248.48.1 17.0.0.1
EA5822(config-voip)#ip
{address<K>}:address
{media<K>|signaling<K>}:signaling
{signaling-ip-address<I><X.X.X.X>}:17.248.48.1
Command:
ip address signaling 17.248.48.1
```

Step 7 Add a SIP interface.

EA5822(config)#**interface**

```
{emu<K>|eponnni<K>|eth<K>|gponnni<K>|h248<K>|loopback<K>|meth<K>|null<K>|vlanif<K>}:h248
{mgid<U><0,16777215>}:70
```

Command:

interface h248 70

Are you sure to add the MG interface?(y/n)[n]:y

Step 8 Configuring the Attributes of a SIP Interface

EA5822(config-if-h248-70)#**if-h248 attribute**

```
{2833encrypt<K>|code<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|heartBe
atRetransTimer<K>|heartBeatRetransTimes<K>|mg-media-ip1<K>|mg-media-ip2<K>|mgc-domain-
name1<K>|mgc-domain-name2<K>|mgip<K>|mgport<K>|MIDType<K>|primary-mgc-ip1<K>|primary-
mgc-ip2<K>|primary-mgc-port<K>|profile-index<K>|profile-negotiation<K>|retrans<K>|secondary-
mgc-ip1<K>|secondary-mgc-ip2<K>|secondary-mgc-port<K>|start-negotiate-
version<K>|transfer<K>}:mgip
{mgip<I>|signal-addr-name<K>|signal-dhcp-vlan<K>}:17.248.48.11
{<cr>|2833encrypt<K>|code<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|he
artBeatRetransTimer<K>|heartBeatRetransTimes<K>|mg-media-ip1<K>|mg-media-ip2<K>|mgc-
domain-name1<K>|mgc-domain-name2<K>|mgport<K>|MIDType<K>|primary-mgc-ip1<K>|primary-
mgc-ip2<K>|primary-mgc-port<K>|profile-index<K>|profile-negotiation<K>|retrans<K>|secondary-
mgc-ip1<K>|secondary-mgc-ip2<K>|secondary-mgc-port<K>|start-negotiate-
version<K>|transfer<K>}:mgport 2944
{<cr>|2833encrypt<K>|code<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|he
artBeatRetransTimer<K>|heartBeatRetransTimes<K>|mg-media-ip1<K>|mg-media-ip2<K>|mgc-
domain-name1<K>|mgc-domain-name2<K>|MIDType<K>|primary-mgc-ip1<K>|primary-mgc-
ip2<K>|primary-mgc-port<K>|profile-index<K>|profile-negotiation<K>|retrans<K>|secondary-mgc-
ip1<K>|secondary-mgc-ip2<K>|secondary-mgc-port<K>|start-negotiate-version<K>|transfer<K>}:mg-
media-ip1
{media-addr-name<K>|media-dhcp-vlan<K>|mg-media-ip<I>}:17.248.48.11
{<cr>|2833encrypt<K>|code<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|he
artBeatRetransTimer<K>|heartBeatRetransTimes<K>|mg-media-ip2<K>|mgc-domain-name1<K>|mgc-
domain-name2<K>|MIDType<K>|primary-mgc-ip1<K>|primary-mgc-ip2<K>|primary-mgc-
port<K>|profile-index<K>|profile-negotiation<K>|retrans<K>|secondary-mgc-ip1<K>|secondary-mgc-
ip2<K>|secondary-mgc-port<K>|start-negotiate-version<K>|transfer<K>}:primary-mgc-ip1
{primary-mgc-ip1<I><X.X.X.X>}:200.200.200.200
{<cr>|2833encrypt<K>|code<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|he
artBeatRetransTimer<K>|heartBeatRetransTimes<K>|mg-media-ip2<K>|mgc-domain-name1<K>|mgc-
domain-name2<K>|MIDType<K>|primary-mgc-ip2<K>|primary-mgc-port<K>|profile-index<K>|profile-
negotiation<K>|retrans<K>|secondary-mgc-ip1<K>|secondary-mgc-ip2<K>|secondary-mgc-
port<K>|start-negotiate-version<K>|transfer<K>}:primary-mgc-port
{primary-mgc-port<U><1,65535>}:2944
{<cr>|2833encrypt<K>|code<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|he
artBeatRetransTimer<K>|heartBeatRetransTimes<K>|mg-media-ip2<K>|mgc-domain-name1<K>|mgc-
domain-name2<K>|MIDType<K>|primary-mgc-ip2<K>|profile-index<K>|profile-
negotiation<K>|retrans<K>|secondary-mgc-ip1<K>|secondary-mgc-ip2<K>|secondary-mgc-
port<K>|start-negotiate-version<K>|transfer<K>}:transfer
{transfer<E><tcp,udp,sctp>}:udp
{<cr>|2833encrypt<K>|code<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|he
artBeatRetransTimer<K>|heartBeatRetransTimes<K>|mg-media-ip2<K>|mgc-domain-name1<K>|mgc-
domain-name2<K>|MIDType<K>|primary-mgc-ip2<K>|profile-index<K>|profile-
negotiation<K>|retrans<K>|secondary-mgc-ip1<K>|secondary-mgc-ip2<K>|secondary-mgc-
port<K>|start-negotiate-version<K>}:code
{code<E><text,binary>}:text
```

```
{<cr>|2833encrypt<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|heartBeatRe
transTimer<K>|heartBeatRetransTimes<K>|mg-media-ip2<K>|mgc-domain-name1<K>|mgc-domain-
name2<K>|MIDType<K>|primary-mgc-ip2<K>|profile-index<K>|profile-
negotiation<K>|retrans<K>|secondary-mgc-ip1<K>|secondary-mgc-ip2<K>|secondary-mgc-
port<K>|start-negotiate-version<K>}:start-negotiate-version
{start-negotiate-version<U><0,3>}:0
{<cr>|2833encrypt<K>|desc<K>|deviceName<K>|domainName<K>|heartBeatGenTimer<K>|heartBeatRe
transTimer<K>|heartBeatRetransTimes<K>|mg-media-ip2<K>|mgc-domain-name1<K>|mgc-domain-
name2<K>|MIDType<K>|primary-mgc-ip2<K>|profile-index<K>|profile-
negotiation<K>|retrans<K>|secondary-mgc-ip1<K>|secondary-mgc-ip2<K>|secondary-mgc-port<K>}:

```

Command:

```
if-h248 attribute mgip 17.248.48.1 mgport 2944 mg-media-ip1 17.248.48.1 primary-mgc-ip1
200.200.200.200 primary-mgc-port 2944 transfer udp code text start-negotiate-version 0

```

Step 9 Enable the H.248 interface.

```
EA5822(config-if-h248-70)#reset coldstart
Are you sure to reset the MG interface?(y/n)[n]:y
EA5822(config-if-h248-70)#
Resetting MG interface 30 succeeded

```

Step 10 Configure the data of the PSTN subscriber in slot 0/2.

```
EA5822(config)#esl user
EA5822(config-esl-user)#mgpstnuser add
{frameid/slotid/portid<S><Length 1-15>}:0/2/1
{mgid<U><0,16777215>}:70
{<cr>|potslinetype<K>|priority<K>|telno<K>|terminalid<K>}:telno
{telno-value<S><Length 1-16>}:18980100
{<cr>|potslinetype<K>|priority<K>|terminalid<K>}:terminalid
{terminalid-value<U><0,9999>}:0
{<cr>|potslinetype<K>|priority<K>}:

```

Command:

```
mgpstnuser add 0/2/1 70 telno 18980100 terminalid 0

```

```
EA5822(config-esl-user)#display mgpstnuser
{attribute<K>|frameid/slotid/portid<S><Length 1-15>|frameid/slotid<S><Length 1-
15>|mgid<U><0,16777215>}:0/2/1
{<cr>|endframeid/slotid/portid<S><Length 1-15>}:

```

Command:

```
display mgpstnuser 0/2/1

```

```
-----
F /S /P MGID TelNo Priority PotsLineType TerminalID
-----

```

```
0 /2 /1 70 18980100 Cat3 DEL A0
-----

```

Step 11 Configuration of MDU 2

The configuration of MDU2 is similar to that of MDU1.

4.2.4 Experiment Verification

Test method: Connect two phones to different POTS ports on the MDU and make a call to each other.

4.2.5 Configuration Reference

#OLT side

```
dba-profile add profile-id 70 type1 fix 1024
ont-lineprofile gpon profile-id 70 profile-name EA5822
tcont 1 dba-profile-id 70
gem add 0 eth tcont 0
gem add 3 eth tcont 1
gem mapping 0 0 vlan 500
gem mapping 3 1 vlan 172
Commit
quit

interface gpon 0/3
port 0 ont-auto-find enable
ont confirm 0 ontid 7 sn-auth 4857544329FC9507 snmp ont-lineprofile-id 70
quit
vlan 500 smart
port vlan 500 0/20 0

interface vlanif 500
ip address 10.1.1.1 24
quit

service-port vlan 500 gpon 0/3/0 ont 7 gemport 0 multi-service user-vlan 500

interface gpon 0/3
ont ipconfig 0 7 static ip-address 10.1.1.7 mask 255.255.255.0 gateway 10.1.1.1 vlan 500

quit
vlan 172 smart
port vlan 172 0/20 0

interface vlanif 172
ip address 17.1.1.100 8
quit

service-port vlan 172 gpon 0/3/0 ont 7 gemport 3 multi-service user-vlan 172

telnet 10.1.1.7
>>User name:root
>>User password: mduadmin
```

#On the MDU side

```
vlan 172
port vlan 172 0/0 1
interface vlanif 172
ip address 17.248.48.1 8
quit
```

```
Voip
ip address media 17.248.48.1 17.0.0.1
ip address signaling 17.248.48.1
quit
```

```
ip route-static 200.200.200.0 24 17.0.0.1
interface h248 30
```

```
if-h248 attribute mgip 17.248.48.1 mgport 2944 mg-media-ip1 17.248.48.1 primary-mgc-ip1
200.200.200.200 primary-mgc-port 2944 transfer udp code text start-negotiate-version 0
reset
quit
```

```
esl user
mgpstnuser add 0/2/1 70 terminalid 0 telno 18980100
```

4.3 Questions

1. Can the phone number of a user be not configured when the user information is configured? Whether services are affected?

5 GPON ONT IPTV Service Configuration

5.1 Experiment Introduction

5.1.1 About This Experiment

In this experiment, the multicast service is configured to provision the multicast service for users in the GPON ONT IPTV networking scenario.

5.1.2 Objectives

- Understand the multicast working principle.
- Understand the functions of the multicast VLAN.
- Understand the functions of user service VLANs.
- Understand the differences between IGMP users and multicast members.
- Learn the differences between the dynamic and static program matching modes.
- Understand the control of user rights.
- Master the configuration of the multicast service.

5.1.3 Experiment Networking

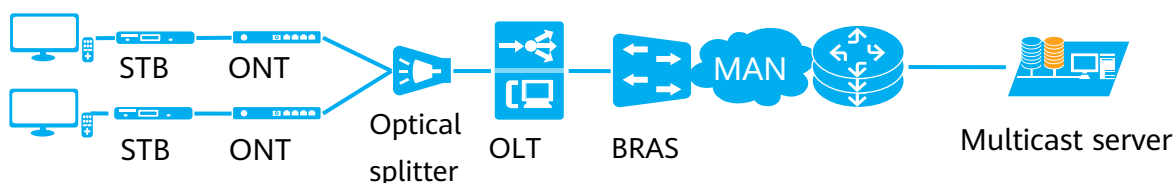


Figure 5-1 Example network of the GPON ONT IPTV service

5.1.4 Experiment Planning

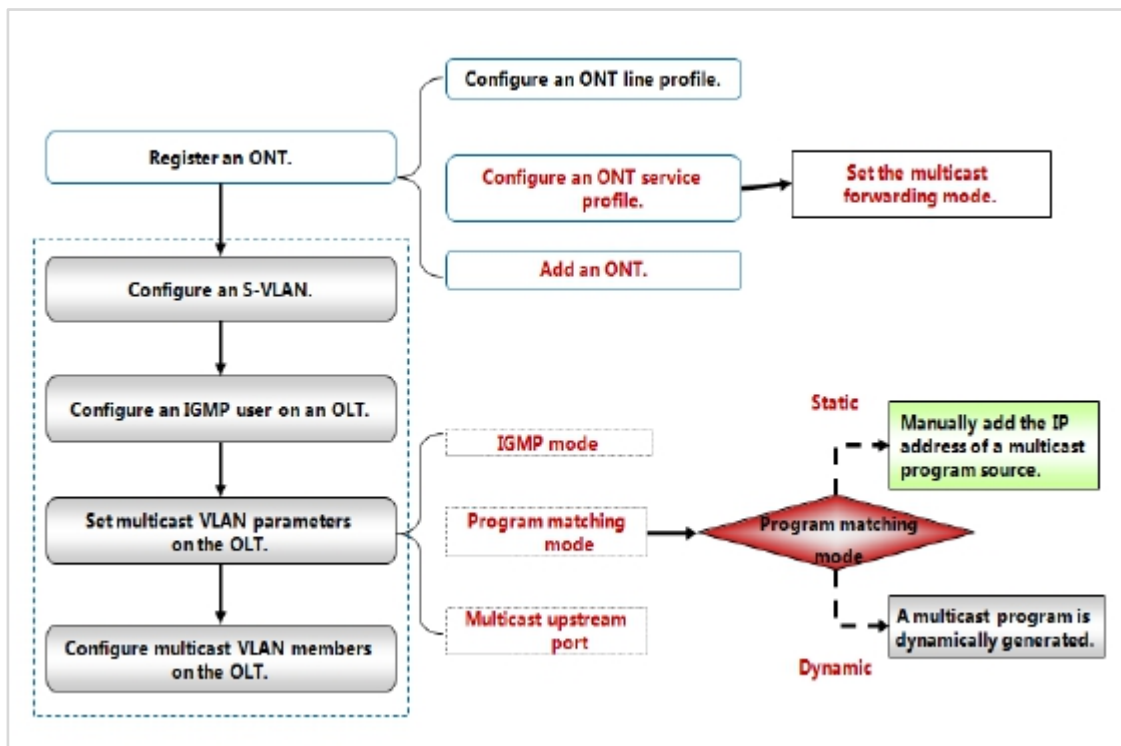
Before the configuration, pay attention to the data planning to avoid data conflict during the configuration. Please perform the experiment based on your data plan.

Collect the following information based on the actual conditions of the devices in the equipment room:

Parameter	Value
OLT upstream port ID Multicast uplink port	0/19/0
OLT GPON port ID	0/3/0
ONT ID	0
DBA profile	Profile ID: 10 Assure:10M Max:20M
Line profile	Profile ID: 11 T-CONT ID:2 Gem Index:0
Service template	Template ID: 13 Multicast forwarding mode: untag
VLAN	Service VLAN: 200 VLAN type: smart VLAN attribute: common Multicast VLAN: 200 User VLAN: 200 ONT port native-vlan: 200
Indicates the ONT downstream port ID.	ETH 2
Multicast User Authentication Mode	no-auth
IGMP	Version: V3 IGMP mode: proxy Multicast program configuration mode: disable
Multicast server	Server IP address: 192.168.46.240 Multicast group IP address: 224.1.1.1

5.2 Experiment Task Configuration

5.2.1 Service Configuration Process



5.2.2 Configuration Procedure

5.2.2.1 OLT Basic Service Configuration

1. Add an ONT to a GPON port.
2. Configure the OLT service VLAN, upstream port, and downstream port.

5.2.2.2 OLT Multicast Service Configuration

1. Create a multicast VLAN.

In the global config mode, the multicast VLAN is created. The VLAN type and VLAN ID must be consistent with those of the upper layer device according to the networking environment.

2. Configure the upstream port of the multicast VLAN.

In the global config mode, configure the upstream port of the multicast VLAN according to the actual networking and the data plan.

3. Configure the multicast upstream port mode.

In BTV mode, configure the upstream port mode of the multicast VLAN according to the actual networking and data plan.

4. Configure an IGMP user.

In the BTV mode, configure the IGMP user, configure the added service virtual port as the IGMP user, and allow the user to send IGMP messages.

5. Configure the upstream port and default upstream port of the multicast VLAN.

In multicast VLAN mode, configure the upstream port and default upstream port of the multicast VLAN according to the actual networking and data plan.

6. Configure the program matching mode of the multicast VLAN.

In the multicast VLAN mode, you can configure the mode for generating the multicast program. The mode can be automatic or manual. If the manual mode is used, you need to manually add multicast programs.

7. Configure the IGMP packet version of the multicast VLAN.

In the multicast VLAN mode, configure the IGMP version according to the actual network planning. By default, the IGMP version is V3.

8. Configure the IGMP mode of the multicast VLAN.

In multicast VLAN mode, select the working mode of the multicast VLAN according to the network planning. Generally, IGMP proxy is recommended to effectively manage IGMP messages.

9. Configure multicast VLAN members.

In the multicast VLAN mode, add a member to the multicast VLAN. Only the member can watch the programs in the multicast VLAN.

5.2.2.3 Configuration Reference

The following configuration uses an HG8245 connected to GPON port 0/3/0 as an example. The ONT ID is 0, and FE port 2 on the ONT is configured with VLAN 10 and multicast VLAN 200.

Step 1 Add a DBA profile.

```
Huawei(config)#dba-profile add profile-id 10 profile-name HG8245 type3 assure 10240 max 20240
```

Step 2 Add an ONT line profile named IPTV.

```
Huawei(config)#ont-lineprofile gpon profile-name IPTV profile-id 11
Huawei(config-gpon-lineprofile-11)#t-cont 2 dba-profile-id 10
Huawei(config-gpon-lineprofile-11)#gem add 0 eth tcont 2
Huawei(config-gpon-lineprofile-11)#mapping-mode vlan
Huawei(config-gpon-lineprofile-11)#gem mapping 0 0 vlan 200
Huawei(config-gpon-lineprofile-11)#commit
Huawei(config-gpon-lineprofile-11)#quit
```

Step 3 Configuring a Service Profile

```
Huawei(config)#ont-srvprofile gpon profile-name IPTV profile-id 13
Huawei(config-gpon-srvprofile-13)#ont port eth 4 pots 2
Huawei(config-gpon-srvprofile-13)#port vlan eth 2 200
Huawei(config-gpon-srvprofile-13)#multicast-forward untag
Huawei(config-gpon-srvprofile-13)#commit
Huawei(config-gpon-srvprofile-13)#quit
```

Step 4 Enabling the GPON port auto-discovery function

```
Huawei(config)# interface gpon 0/3
Huawei(config-if-gpon-0/3)# port 0 ont-auto-find enable
```

Step 5 Query the discovered ONTs.

```
Huawei(config)# display ont autofind
{all<K>|time<K>}:all
Command:
display ont autofind all
-----
Number F/ S/ P SN Password
-----
1 0/ 3/ 0 323031312E396341
2 0/ 3/ 0 323031312E396A41
-----
```

Step 6 Confirming the automatically discovered ONT (binding the ONT line profile and service profile)

```
Huawei(config-if-gpon-0/3)#ont confirm
{portid<U><0,7>}:0
{all<K>|ontid<K>|password-auth<K>|sn-auth<K>}:ontid
{ontid<U><0,127>}:0
{password-auth<K>|sn-auth<K>}:sn-auth
{sn<S><Length 13-16>}:323031312E396341
{omci<K>|password-auth<K>|snmp<K>}:omci
{ont-lineprofile-id<K>|ont-lineprofile-name<K>}:ont-lineprofile-id
{profile-id<U><1,4096>}:11
{ont-srvprofile-id<K>|ont-srvprofile-name<K>}:ont-srvprofile-id
{profile-id<U><1,4096>}:13
{<cr>|desc<K>}:

```

Step 7 Creating an IGMP VLAN

```
Huawei(config)#vlan 200 smart
Huawei(config)#service-port 10 vlan 200 gpon 0/3/0 ont gemport 0 multi-service user-vlan 200
```

Step 8 Configure the user VLAN of the ONT. (This step is optional. Perform this step when the STB cannot identify the VLAN or is directly connected to the PC.)

```
Huawei(config-if-gpon-0/3)#ont port native-vlan 0 0 eth 2 vlan 200
```

Step 9 Configuring an IGMP User

```
Huawei(config)#btv
Huawei(config-btv)#igmp user add service-port 10 no-auth
Huawei(config)#quit
```

Step 10 Configuring a Multicast VLAN

```
Huawei(config)#vlan 200 smart
Huawei(config)#port vlan 200 0/19 0
Huawei(config)#multicast-vlan 200
```

Step 11 Set up a multicast uplink port.

```
Huawei(config-mvlan200)#igmp uplink-port 0/19/0
```

Step 12 Configure the IGMP version of the multicast VLAN.

```
Huawei(config-mvlan200)#igmp version v3
This operation will delete all programs in current multicast vlan
Are you sure to change current IGMP version? (y/n)[n]: y
```

Step 13 Configure the program matching mode of the multicast VLAN.

```
Huawei(config-mvlan200)#igmp match mode disable
//Disable the IGMP matching mode. The OLT automatically searches for program streams based on
user requirements and does not match programs based on OLT program configurations.//
```

Step 14 Configure the IGMP mode of the multicast VLAN.

```
Huawei(config-mvlan200)#igmp mode proxy
Are you sure to change IGMP mode?(y/n)[n]:y
```

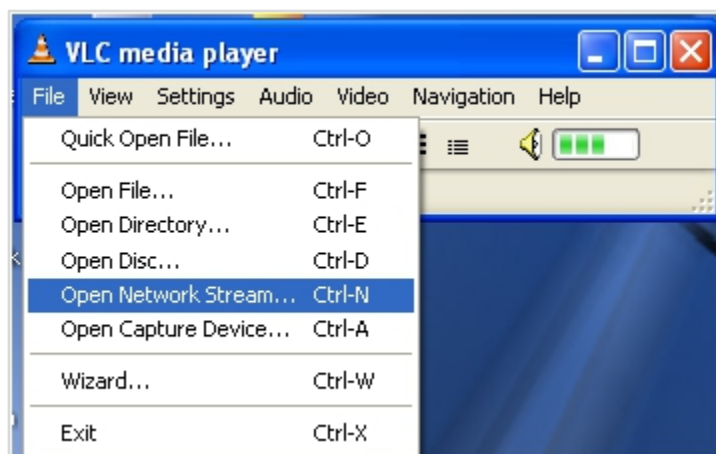
Step 15 Configure multicast VLAN members.

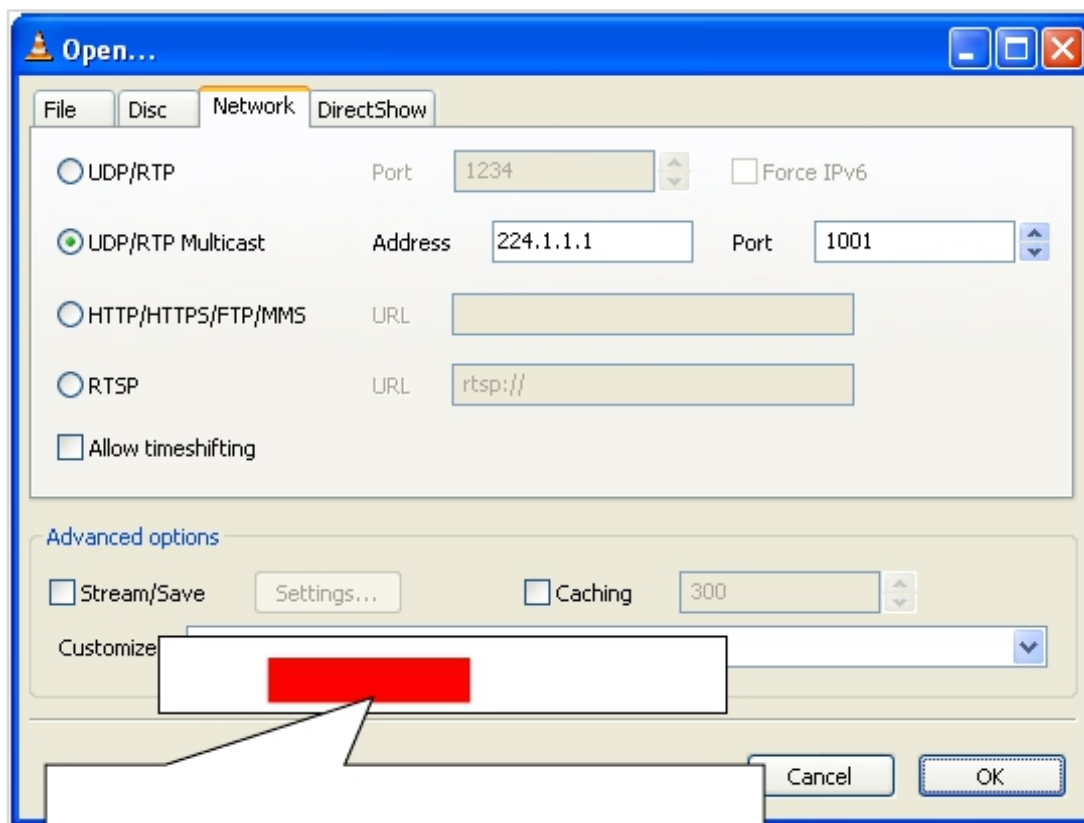
```
Huawei(config-mvlan200)#igmp multicast-vlan member service-port 10
```

5.3 Verification

Step 1 Connect the FE port of the ONT configured with the multicast service to the PC.

Step 2 The VLC is used on the user computer to simulate the IPTV, play programs, and order and watch programs according to the multicast program source provided by the actual network server.





Step 3 By using this method, you can watch all the programs on the server.



If IGMPv3 is used, the IP address of the multicast server must be added.

5.4 Questions

1. What can I do if there is no program stream after I order a program? How can I troubleshoot the fault?
2. How to check whether a user is online?
3. How to query the program bandwidth allocated to a physical port on a board?
4. How to change the program matching mode to manual?
5. How to control the user authority and specify that the user can watch only programs 224.1.1.2 and 224.1.1.4?

6 GPON ONU IPTV Service Configuration

6.1 Experiment Introduction

6.1.1 About This Experiment

In this experiment, the multicast service is configured to implement the multicast service provisioning in the GPON ONU IPTV scenario.

6.1.2 Objectives

- Understand the multicast working principle.
- Understand the functions of the multicast VLAN.
- Understand the functions of user service VLANs.
- Understand the differences between IGMP users and multicast members.
- Learn the differences between the dynamic and static program matching modes.
- Understand the control of user rights.
- Master the configuration of the multicast service.

6.1.3 Experiment Networking

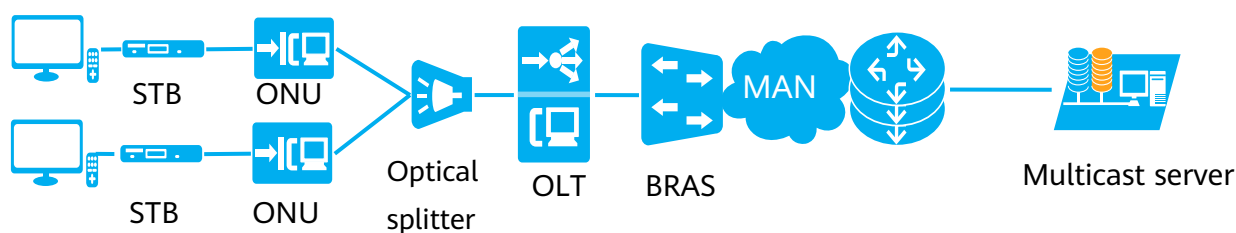


Figure 6-1 Example network of the GPON ONU IPTV service

6.1.4 Experiment Planning

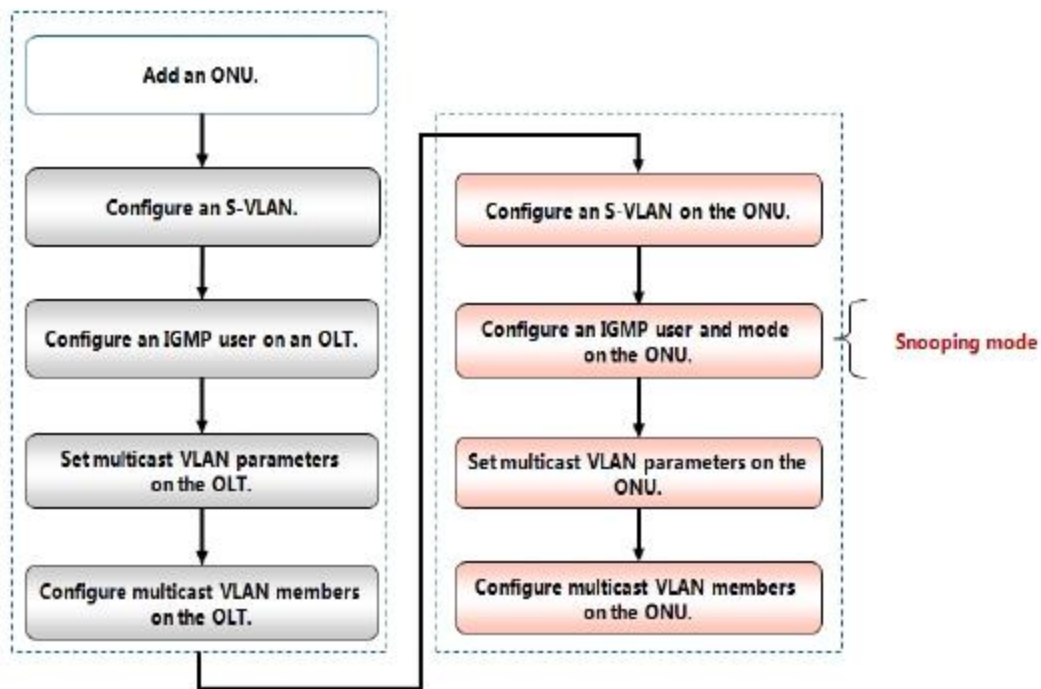
Before the configuration, pay attention to the data planning to avoid data conflict during the configuration. Please perform the experiment based on your data plan.

Collect the following information based on the actual conditions of the devices in the equipment room:

Parameter	Value
OLT upstream port ID Multicast uplink port	0/19/0
OLT GPON port ID	0/3/0
ONT ID	0
DBA profile	Profile ID: 10 Assure:10M Max:20M
Line profile	Profile ID: 11 T-CONT ID:2 Gem Index:0
VLAN	Service VLAN: 200 VLAN type: smart VLAN attribute: common Multicast VLAN: 200 User VLAN: 200 ONU user VLAN: untag
Indicates the upstream port ID of the ONU.	0/0/1
Indicates the ONU downstream port ID.	FE 0/1/2
Multicast User Authentication Mode	no-auth
IGMP	Version: V3 OLT IGMP mode: proxy ONU IGMP mode: snooping OLT and ONU multicast program configuration mode: enable (default)
Multicast server	Server IP address: 192.168.46.240 Multicast group IP address: 224.1.1.1

6.2 Experiment Task Configuration

6.2.1 Service Configuration Process



6.2.2 Configuration Procedure

6.2.2.1 OLT-side service configuration

1. Add an ONT to a GPON port.
2. Configure the OLT service VLAN, upstream port, and downstream port.
3. Create a multicast VLAN.

In the global config mode, the multicast VLAN is created. The VLAN type and VLAN ID must match the upper layer device according to the networking environment.

4. Configure the upstream port of the multicast VLAN.

In the global config mode, configure the upstream port of the multicast VLAN according to the actual networking and the data plan.

5. Configure the multicast upstream port mode.

In BTV mode, configure the upstream port mode of the multicast VLAN according to the actual networking and data plan.

6. Configure an IGMP user.

In the BTV mode, configure the IGMP user, configure the added service virtual port as the IGMP user, and allow the user to send IGMP messages.

7. Configure the upstream port and default upstream port of the multicast VLAN.

In multicast VLAN mode, configure the upstream port and default upstream port of the multicast VLAN according to the actual networking and data plan.

8. Configure the program matching mode of the multicast VLAN.

In multicast VLAN mode, you can configure the mode for generating multicast programs, which can be automatic or manual. If the manual mode is used, you need to manually add multicast programs.

9. Configure the IGMP packet version of the multicast VLAN.

In the multicast VLAN mode, configure the IGMP version according to the actual network planning. By default, the IGMP version is V3.

10. Configure the IGMP mode of the multicast VLAN.

In multicast VLAN mode, select the working mode of the multicast VLAN according to the network planning. Generally, IGMP proxy is recommended to effectively manage IGMP messages.

11. Configure multicast VLAN members.

In the multicast VLAN mode, add a member to the multicast VLAN. Only the member can watch the programs in the multicast VLAN.

6.2.2.2 ONU-side service configuration

1. Configure a service VLAN for the ONU.

In global config mode, create a VLAN. The VLAN type and VLAN ID must be the same as those on the OLT.

2. Configure the upstream port of the ONU.

In the system view, add an upstream port to the service VLAN. The upstream port of the ONU is a GPON port, which is always represented as 0/0/1 on the ONU.

3. Configure traffic entries.

In the global config mode, check the existing traffic indexes in the system. If no traffic index meets the requirement, create a traffic entry to control the user bandwidth. As the multicast service occupies a large bandwidth, configure the bandwidth to 4 Mbit/s or higher to ensure the video quality.

4. Add a service port.

In the system view, map the service virtual port to the service VLAN and create the mapping between the FE port of the ONU and the user VLAN.

5. Create a multicast VLAN.

In the global config mode, the multicast VLAN is created. The VLAN type and VLAN ID must match the upper layer device according to the networking environment.

6. Configure multicast VLAN members,

In the multicast VLAN mode, add a member to the multicast VLAN. Only the member can watch the programs in the multicast VLAN.

6.2.2.3 Configuration Reference

The following configurations are based on an ONU connected to GPON port 0/3/0, ONT ID 0, and FE port 0/1/2 on the ONU. The user service VLAN is VLAN 200, the multicast VLAN is VLAN 200, and the management VLAN of the ONU is VLAN 4000.

OLT-side service configuration

Step 1 Adding an ONU

For details, see the basic operations on the terminal. When configuring the ONT line profile, pay attention to the following points:

- (1) Add the GEM port and T-CONT of the multicast user VLAN, and configure the mapping between them.
- (2) Add the GEM port and T-CONT of the multicast VLAN, and configure the mapping between them.
- (3) Add the GEM port and T-CONT of the NMS VLAN, and configure the mapping between them.

Step 2 Configure a user VLAN and add a service port to the user VLAN.

```
Huawei(config)#vlan 200 smart
Huawei(config)#port vlan 200 0/19 0
Huawei(config)#service-port 10 vlan 200 gpon 0/3/0 ont 0 gemport 0
multi-service user-vlan 20
```

Step 3 Configuring a Multicast User

```
Huawei(config)#btv
Huawei(config-btv)#igmp user add service-port 10 no-auth
```

Step 4 Configuring a Multicast VLAN

```
Huawei(config)#multicast-vlan 200
```

Step 5 Set up a multicast uplink port.

```
Huawei(config-mvlan200)#igmp uplink-port 0/19/0
```

Step 6 Configure the multicast program and multicast source.

```
Huawei(config-mvlan200)#igmp program add name program1 ip 224.1.1.1 sourceip 192.168.46.240
```

Step 7 Configuring the IGMP Protocol Processing Mode

```
Huawei(config-mvlan200)#igmp mode proxy
```

Are you sure to change IGMP mode?(y/n)[n]:y

Step 8 Configure multicast VLAN members.

```
Huawei(config-mvlan200)#igmp multicast-vlan member service-port 10
```

ONU-side service configuration

Step 9 Create a multicast VLAN and add an upstream port and a service port to the multicast VLAN.

```
HuaweiONU(config)# vlan 200
HuaweiONU(config)# port vlan 200 0/0 1
```

Step 10 Configuring a Traffic Profile

```
HuaweiONU(config)#traffic table ip index 10 cir off priority 4 priority-policy tag-In-Packag
```

Step 11 Configure service ports.

```
HuaweiONU(config)# service-port 100 vlan 200 eth 0/1/2 multi-service user-vlan untagged rx-cttr 10
tx-cttr 10
```

Step 12 Configure the multicast mode and multicast protocol version.

```
HuaweiONU(config)# btv
HuaweiONU(config -btv)#igmp version v3
HuaweiONU(config -btv)#igmp mode snooping
```

Step 13 Adding an IGMP User

```
HuaweiONU(config -btv)# igmp user add service-port 100 no-auth
```

Step 14 Create a multicast VLAN.

```
HuaweiONU(config)#mutlcast-vlan 200
```

Step 15 Configure the multicast upstream port and multicast program.

```
HuaweiONU(config-mvlan200)#igmp uplink-port 0/0/1
HuaweiONU(config-mvlan200)#igmp program add name program1 ip 224.1.1.1 sourceip
192.168.46.240
```

Step 16 Configure multicast VLAN members.

```
HuaweiONU(config -mvlan200)# igmp multicast-vlan member service-port 100
```

6.3 Verification

- Step 1 Connect the FE port of the ONU configured with the multicast service to the PC.
- Step 2 The VLC is used on the user computer to simulate the IPTV, play programs, and order and watch programs according to the multicast program source provided by the actual network server.



- Step 3 By using this method, you can watch all the programs on the server.



If IGMPv3 is used, the IP address of the multicast server must be added.

6.4 Questions

1. What can I do if there is no program stream after I order a program? How can I troubleshoot the fault?
2. How to check whether a user is online?
3. How to query the program bandwidth allocated to a physical port on a board?

4. How to change the program matching mode to manual?
5. How to control the user authority and specify that the user can watch only programs 224.1.1.2 and 224.1.1.4?

7 POL Multicast Service Configuration

7.1 Experiment Introduction

7.1.1 About This Experiment

In this experiment, multicast services are configured to implement multicast service provisioning in the POL scenario.

7.1.2 Objectives

- Understand the multicast working principle.
- Understand the multicast routing protocol configuration process.
- Understand the multicast configuration of the access router on the POL campus network.
- Understand the functions of the multicast VLAN.
- Understand the functions of user service VLANs.
- Understand the differences between IGMP users and multicast members.
- Learn the differences between the dynamic and static program matching modes.
- Understand the control of user rights.
- Master the configuration of the multicast service.

7.1.3 Experiment Networking

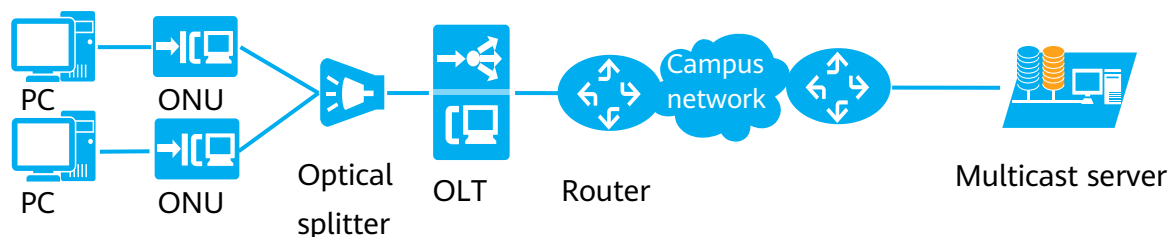


Figure 7-1 Networking diagram of the POL GPON multicast service

7.1.4 Experiment Planning

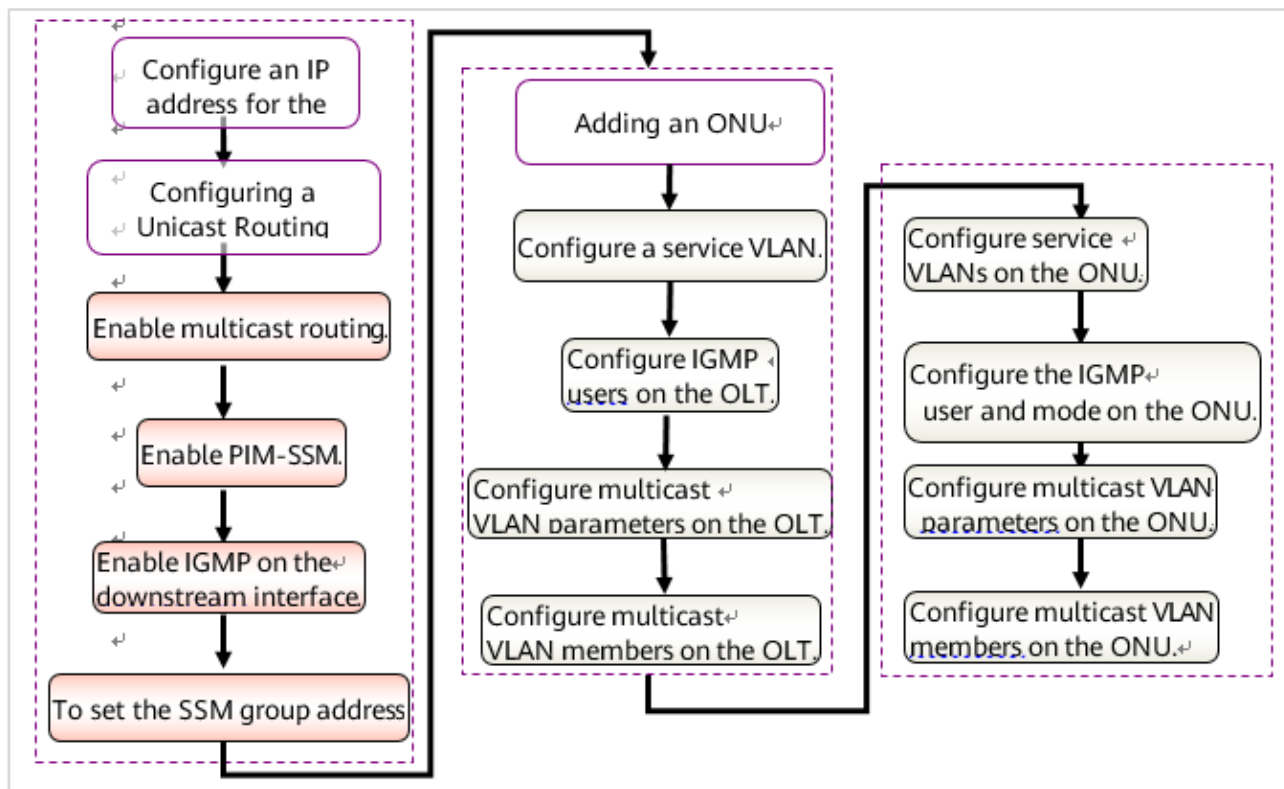
Before the configuration, pay attention to the data planning to avoid data conflict during the configuration. Please perform the experiment based on your data plan.

Collect the following information based on the actual conditions of the devices in the equipment room:

Parameter	Value
Router uplink port number	GigabitEthernet1/0/0
Downlink port number of the router	GigabitEthernet2/0/0
VLAN of the downlink sub-interface on the router	200
OLT upstream port ID	0/19/0
OLT GPON port ID	0/3/0
Indicates the upstream port ID of the ONU.	0/0/1
ONU Ethernet service port	0/1/2
PIM working mode	SSM
IGMP version	V3
Multicast server	Server IP address: 192.168.46.240 Multicast group IP address: 224.1.1.1

7.2 Experiment Task Configuration

7.2.1 Service configuration process:



7.2.2 Configuration Procedure

7.2.2.1 Service Configuration on the Access Router Side

1. Create a sub-interface on the OLT-oriented downstream interface and configure VLAN parameters.
2. Configure an IP address and a unicast routing protocol for each interface on the router.
3. Enable multicast routing on all routers that provide multicast services.
4. PIM-SM is enabled on all interfaces of the multicast router.
5. Enable IGMP on the interface of the router that is directly connected to the OLT.
6. Set the same SSM group address range on each router.

7.2.2.2 OLT-side service configuration

1. Add an ONT to a GPON port.
2. Configure the OLT service VLAN, upstream port, and downstream port.
3. Create a multicast VLAN.

In the global config mode, the multicast VLAN is created. The VLAN type and VLAN ID must match the upper layer device according to the networking environment.

4. Configure the upstream port of the multicast VLAN.

In the global config mode, configure the upstream port of the multicast VLAN according to the actual networking and the data plan.

5. Configure the multicast upstream port mode.

In BTV mode, configure the upstream port mode of the multicast VLAN according to the actual networking and data plan.

6. Configure an IGMP user.

In the BTV mode, configure the IGMP user, configure the added service virtual port as the IGMP user, and allow the user to send IGMP messages.

7. Configure the upstream port and default upstream port of the multicast VLAN.

In multicast VLAN mode, configure the upstream port and default upstream port of the multicast VLAN according to the actual networking and data plan.

8. Configure the program matching mode of the multicast VLAN.

In multicast VLAN mode, you can configure the mode for generating multicast programs, which can be automatic or manual. If the manual mode is used, you need to manually add multicast programs.

9. Configure the IGMP packet version of the multicast VLAN.

In the multicast VLAN mode, configure the IGMP version according to the actual network planning. By default, the IGMP version is V3.

10. Configure the IGMP mode of the multicast VLAN.

In multicast VLAN mode, select the working mode of the multicast VLAN according to the network planning. Generally, IGMP proxy is recommended to effectively manage IGMP messages.

11. Configure multicast VLAN members.

In the multicast VLAN mode, add a member to the multicast VLAN. Only the member can watch the programs in the multicast VLAN.

7.2.2.3 ONU-side service configuration

1. Configure a service VLAN for the ONU.

In global config mode, create a VLAN. The VLAN type and VLAN ID must be the same as those on the OLT.

2. Configure the upstream port of the ONU.

In the system view, add an upstream port to the service VLAN. The upstream port of the ONU is a GPON port, which is always represented as 0/0/1 on the ONU.

3. Configure traffic entries.

In the global config mode, check the existing traffic indexes in the system. If no traffic index meets the requirement, create a traffic entry to control the user bandwidth. As the multicast service occupies a large bandwidth, configure the bandwidth to 4 Mbit/s or higher to ensure the video quality.

4. Add a service port.

In the system view, map the service virtual port to the service VLAN and create the mapping between the FE port of the ONU and the user VLAN.

5. Create a multicast VLAN.

In the global config mode, the multicast VLAN is created. The VLAN type and VLAN ID must match the upper layer device according to the networking environment.

6. Configure multicast VLAN members,

In the multicast VLAN mode, add a member to the multicast VLAN. Only the member can watch the programs in the multicast VLAN.

7.2.2.4 Configuration Reference

In the following example, the access router GigabitEthernet1/0/0 is connected to another multicast router, GigabitEthernet2/0/0 is connected to an ONU under OLT 0/19/0, GPON 0/3/0, the ONT ID is 0, and the FE port on the ONU is 0/1/2. The user service VLAN is 200 and the multicast VLAN is 200, ONU management VLAN 4000. The single-domain PIM-SSM scenario is used as an example.

Service Configuration on the Access Router Side

- Step 1** Create a sub-interface on the OLT-oriented downstream interface and configure VLAN parameters.

```
[*Router] interface gigabitethernet 2/0/0.1
[*Router-GigabitEthernet2/0/0.1] vlan-type dot1q 200
[*Router-GigabitEthernet2/0/0.1] commit
```

- Step 2** Configure an IP address for each interface on each router and configure a unicast routing protocol. The configuration details are not provided here.

Reference:

- (1) Configure IP addresses for the interfaces according to the planning.
- (2) Enable the unicast routing protocol on the router. For details, see the configurations of OSPF and IS-IS.

- Step 3** Enable the multicast function on the access router and enable PIM-SM on each interface.

```
[~Router] multicast routing-enable
[*Router] interface gigabitethernet 2/0/0.1
[*Router-GigabitEthernet2/0/0.1] pim sm
[*Router-GigabitEthernet2/0/0.1] quit
[*Router] interface gigabitEthernet 1/0/0
[*Router-GigabitEthernet1/0/0] pim sm
[*Router-GigabitEthernet1/0/0] commit
[~Router-GigabitEthernet1/0/0] quit
```

- Step 4** Enable the IGMP function on the interface of the access router connected to the OLT. In this example, the IGMP version is set to 3.

```
[~Router] interface gigabitethernet 2/0/0.1
```

```
[~Router-GigabitEthernet2/0/0.1] igmp enable
[*Router-GigabitEthernet2/0/0.1] igmp version 3
[*Router-GigabitEthernet2/0/0.1] commit
[~Router-GigabitEthernet2/0/0.1] quit
```

Step 5 Configure the SSM group address range. In this example, the SSM group address range is set to 224.1.1.0/24 on all routers.

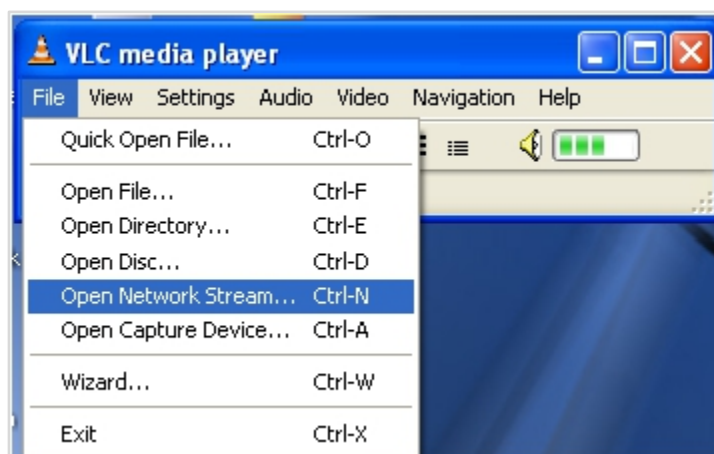
```
[~Router] acl number 2000
[*Router-acl4-basic-2000] rule permit source 224.1.1.0 0.0.0.255
[*Router-acl4-basic-2000] quit
[*Router] pim
[*Router-pim] ssm-policy 2000
[*Router-pim] commit
[~Router-pim] quit
```

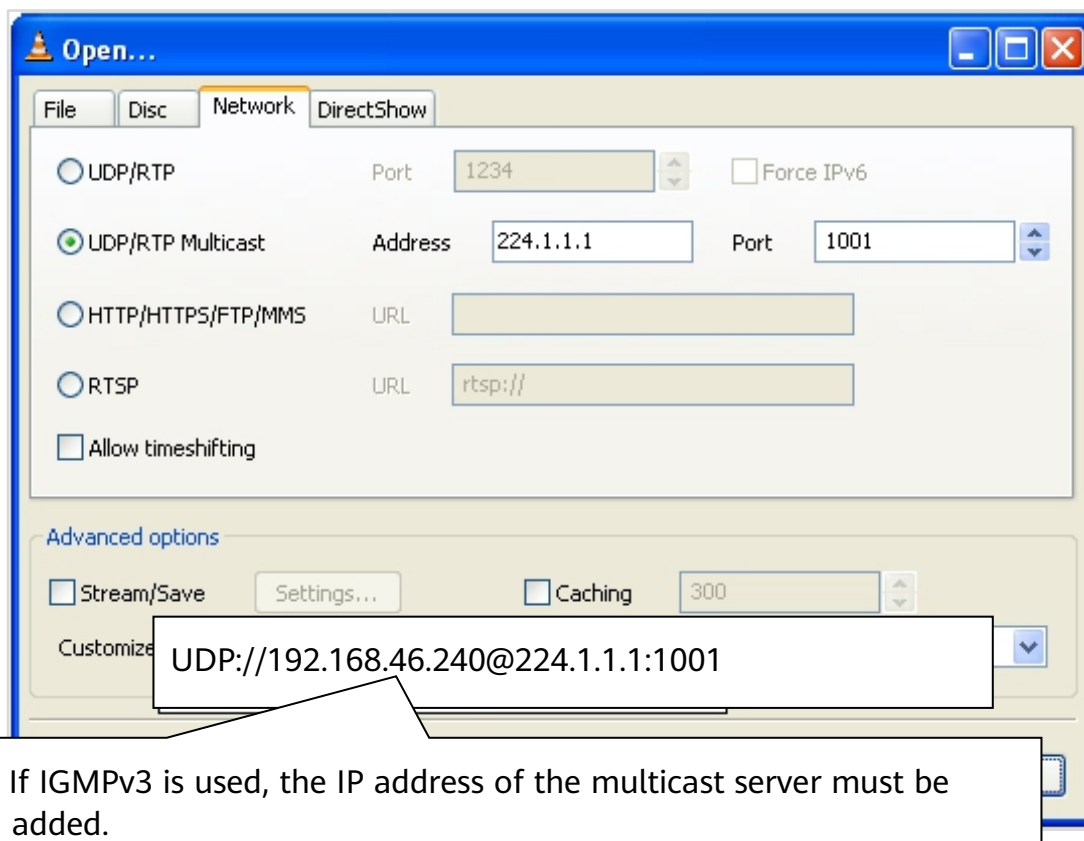
For details about service configurations on the OLT and ONU, see the multicast configurations in the GPON ONT&ONU scenario.

7.3 Verification

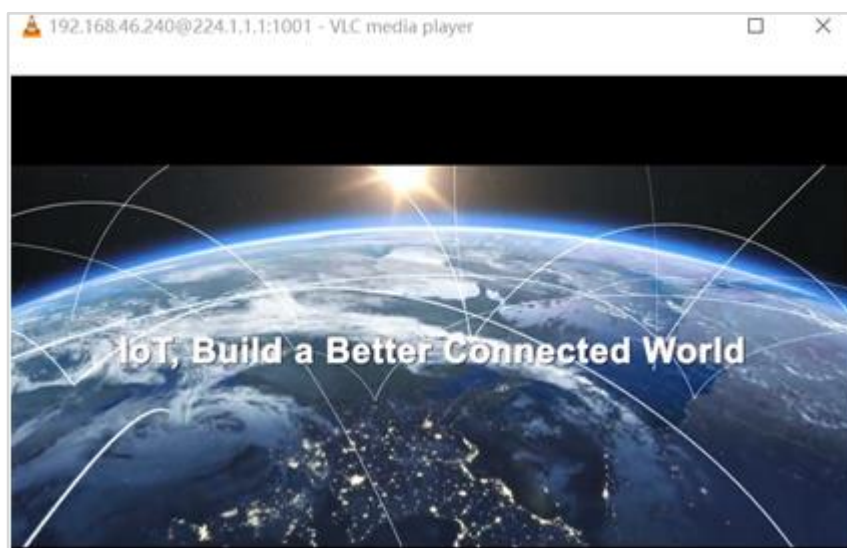
Step 1 Connect the FE port of the ONU configured with the multicast service to the PC.

Step 2 The VLC is used on the user computer to simulate the IPTV, play programs, and order and watch programs according to the multicast program source provided by the actual network server.





Step 3 By using this method, you can watch all the programs on the server.



7.4 Questions

1. After configuring the access router, how to configure the OLT, ONU, or ONT?
2. How to check whether the multicast routing protocol is correctly configured?

3. How to control the user authority and specify that the user can watch only programs 224.1.1.2 and 224.1.1.4?

8 QoS Application and Configuration

8.1 Experiment Introduction

8.1.1 About This Experiment

This experiment describes the QoS application scenario, data configuration, and verification on the OLT.

8.1.2 Objectives

- Describe the application of QoS in different scenarios.
- Have a basic command of PON QoS implementation mechanisms.
- Configure PON QoS data based on different applications.
- To verify the function of implementing PON QoS.

8.1.3 Experiment Networking

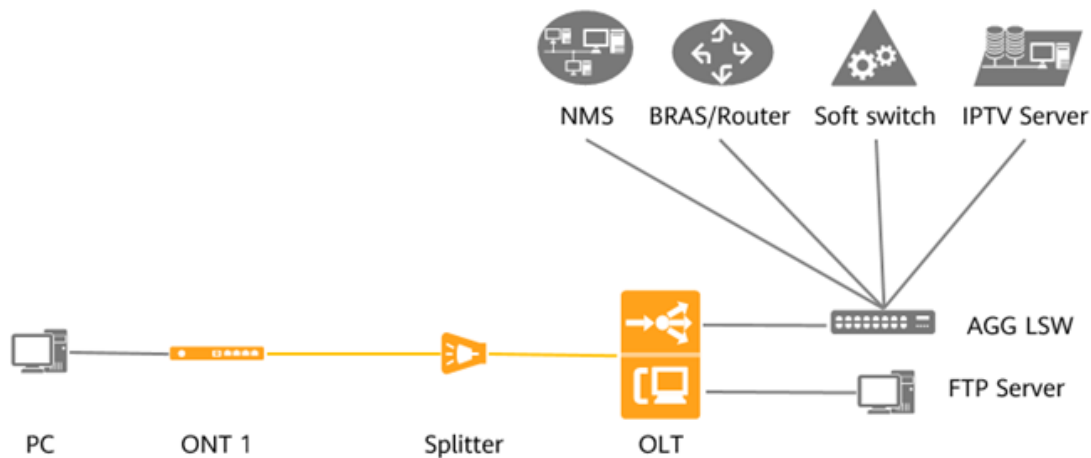


Figure 8-1 Networking diagram of QoS application configuration

8.1.4 Experiment Planning

According to the networking scenario in the lab, the optical splitter uses the 1:32 optical splitting ratio. The upstream GICG board 0/19/0 of the OLT is connected to the convergence switching device. The aggregation switch is interconnected with the softswitch, BRAS, and multicast server. In addition, related service data is configured on the interconnected NEs, connect upstream port 0/19/1 of the OLT to the FTP server (a host) and use the port as the host for packet capture and analysis.

Table 8-1 Experiment Data Planning

Type	Parameter
OLT	GIU:0/19/0
	GPON:0/3/0
VLAN	Service VLAN: 2000; inner VLAN: 35 VLAN type: smart; attribute: qinq
	Service VLAN: 1000; inner VLAN: 35 VLAN type: smart; attribute: common
HG8245Q	ETH 1
PC	IP address 192.168.100.5/24
FTP server/ Packet capture host	IP address 192.168.100.105/24

8.2 Experiment 1: Priority Application Configuration

8.2.1 Configuration Roadmap

The following uses the IP traffic profile as an example to describe how to configure the priority.

1. Configuring a Double-VLAN Service Flow
2. Setting the service VLAN priority in the traffic profile Copying the inner VLAN priority
3. Use the PC to initiate a dial-up connection, and use the Wireshark to capture packets on the PC.
4. Modify the local configuration of the service VLAN priority in the traffic profile. Set the priority to 3.
5. Use the PC to initiate a dial-up connection, and use the Wireshark to capture packets on the PC.
6. Configure a single-tagged VLAN for the service flow.
7. Setting the service VLAN priority in the traffic profile Copying the inner VLAN priority
8. Use the PC to initiate a dial-up connection, and use the Wireshark to capture packets on the PC.
9. Modify the local configuration of the service VLAN priority in the traffic profile. Set the priority to 3.
10. Use the PC to initiate a dial-up connection and use the Wireshark to capture packets and analyze the VLAN ID.

8.2.2 Configuration Procedure

Step 1 Create a line profile, a service profile, and an ONT.

```
HUAWEI(config)# dba-profile add profile-name QoS type2 assure 2048
HUAWEI(config)#ont-lineprofile gpon profile-name QoS
HUAWEI(config-gpon-lineprofile-2)#tcont 1 dba-profile-name QoS
HUAWEI(config-gpon-lineprofile-2)#gem add 1 eth tcont 1
HUAWEI(config-gpon-lineprofile-2)#gem mapping 1 0 vlan 35
HUAWEI(config-gpon-lineprofile-2)#commit
HUAWEI(config)#ont-srvprofile gpon profile-name QoS
HUAWEI(config-gpon-srvprofile-2)#ont-port eth adaptive pots adaptive
HUAWEI(config-gpon-srvprofile-2)#port vlan eth 1 35
HUAWEI(config-gpon-srvprofile-2)#commit
HUAWEI(config-if-gpon-0/3)#ont confirm 0 ontid 1 sn-auth 485754434D23C27B omci ont-lineprofile-name QoS ont-srvprofile-name QoS
```

Step 2 Sets the priority of the inner VLAN tag on the ONT side to 5.

```
HUAWEI(config-if-gpon-0/3)#ont port native-vlan 0 1 eth 1 vlan 35 priority 5
```

Step 3 Copying the inner VLAN priority as the service VLAN priority in the traffic profile QoS1

```
HUAWEI(config)#traffic table ip name QoS1 cir off priority user-cos 3 priority-policy tag-ln-package
Create traffic descriptor record successfully
```

```
-----
TD Index: 7
TD Name: QoS1
Priority: 3
Copy Priority: user-cos
Mapping Index: 0
CTAG Mapping Priority: -
CTAG Mapping Index: -
CTAG Default Priority: 0
Priority Policy: tag-pri
CIR: off
CBS: off
PIR: off
PBS: off
Fix: off
CAR Threshold Profile: -
Color Mode: color-blind
Color Policy: dei
Referenced Status: not used
-----
```

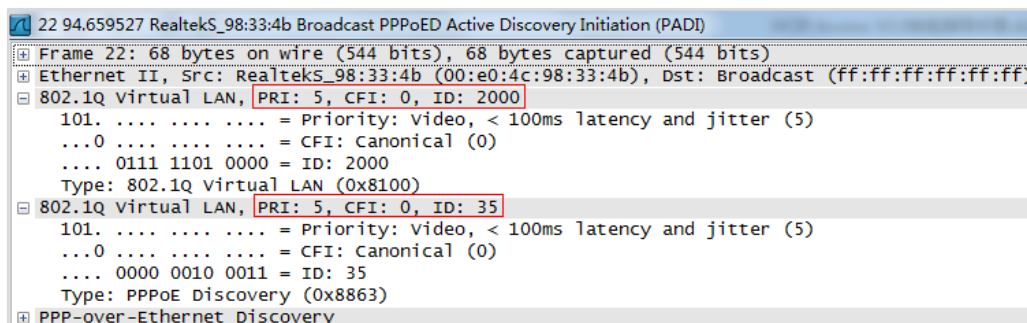
Step 4 Binding an IP Traffic Profile to a Service Port

```
HUAWEI(config)#vlan 2000 smart
HUAWEI(config)#vlan attrib 2000 q-in-q
HUAWEI(config)#port vlan 2000 0/19 0
HUAWEI(config)#service-port 1 vlan 2000 gpon 0/3/0 ont 1 gempport 1 multi-service user-vlan 35 rx-
cttr 7 tx-cttr 7
```

Step 5 Port mirroring is configured on the upstream port of the OLT and the traffic on port 0/19/0 is duplicated to port 0/19/1.

```
HUAWEI(config)#interface giu 0/19
HUAWEI(config-if-giu-0/19)#mirror port 0 1 egress
```

Step 6 Use the PC to initiate a dial-up connection, and use the Wireshark to capture packets on the PC.

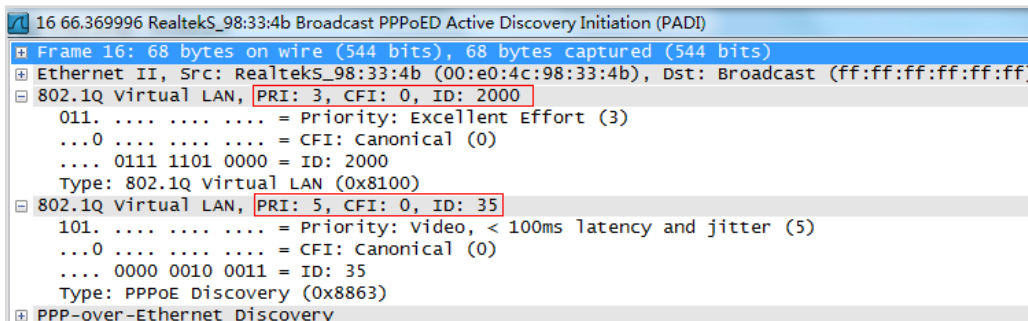


Result analysis: The outer VLAN 2000 copies the priority 5 of the inner VLAN 35, and the priority of SVLAN 2000 is 5.

Step 7 Modify the local VLAN priority of QoS1 in the traffic profile to 3.

HUAWEI(config)#**traffic table ip modify name QoS1 priority 3 priority-policy local-Setting**

Step 8 Use the PC to initiate a dial-up connection, and use the Wireshark to capture packets on the PC.



Result analysis: The priority of outer VLAN 2000 is 3 (locally configured) and the priority of inner VLAN 2000 is 5 (configured on the ONT side).

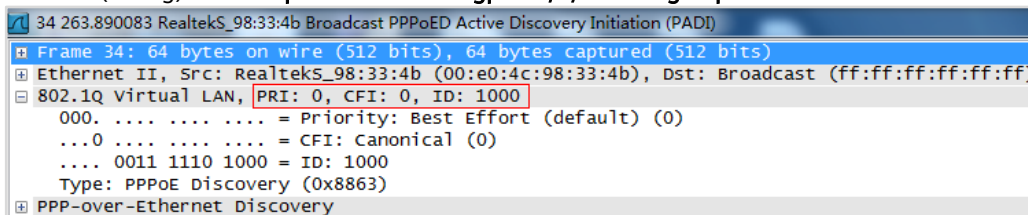
Step 9 A double-tagged VLAN service flow is deleted, a single-tagged VLAN service flow is created, and a specified traffic profile is not bound.

HUAWEI(config)#**undo service-port 1**

HUAWEI(config)#**vlan 1000 smart**

HUAWEI(config)#**port vlan 1000 0/19 0**

HUAWEI(config)#**service-port 2 vlan 1000 gpon 0/3/0 ont 1 gempport 1 multi-service user-vlan 35**



Result analysis: The specified traffic profile is not bound, and the priority of service VLAN 1000 is the default value 0.

Step 10 Create traffic profile QoS2 for the traffic stream with a single VLAN tag, and set the priority policy to copy from the inner VLAN tag.

HUAWEI(config)#**traffic table ip name QoS2 cir off priority user-cos 3 priority-policy tag-In-package**

Create traffic descriptor record successfully

TD Index: 9

TD Name: QoS2

Priority: 3

Copy Priority: user-cos

Mapping Index: 0

CTAG Mapping Priority: -

CTAG Mapping Index: -

CTAG Default Priority: 0

Priority Policy: tag-pri

CIR: off

CBS: off

PIR: off

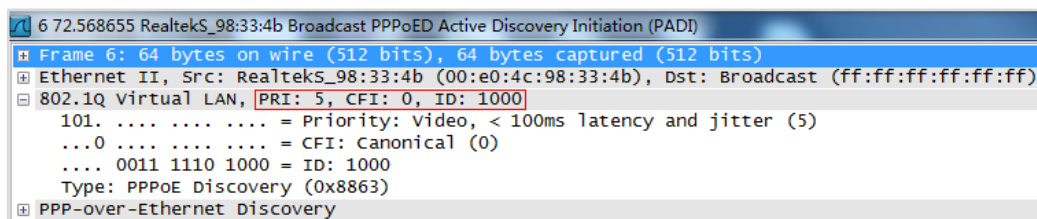
PBS: off

Fix: off
 CAR Threshold Profile: -
 Color Mode: color-blind
 Color Policy: dei
 Referenced Status: not used

Step 11 Re-create a service flow and bind the traffic profile QoS2 to the service flow. The SVLAN priority is copied from the inner VLAN priority.

```
HUAWEI(config)#undo service-port 2
HUAWEI(config)#service-port 2 vlan 1000 gpon 0/3/0 ont 1 gemport 1 multi-service user-vlan 35 rx-
ctr 9 tx-ctr 9
```

Step 12 Use the PC to initiate a dial-up connection, and use the Wireshark to capture packets on the PC.

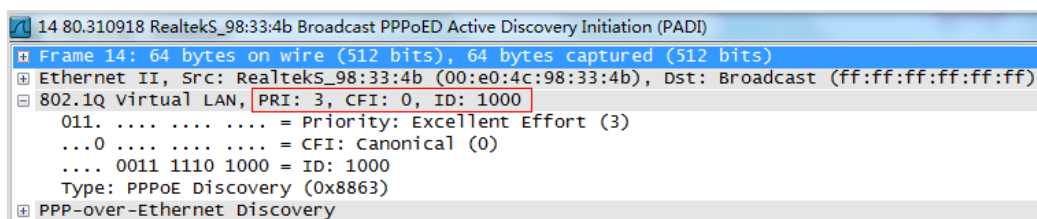


Result analysis: The priority of service VLAN 1000 is copied from the inner VLAN 35 side and the priority is 5.

Step 13 Modify the local VLAN priority of QoS2 in the traffic profile to 3.

```
HUAWEI(config)#traffic table ip modify name QoS2 priority 3 priority-policy local-Setting
```

Step 14 Use the PC to initiate a dial-up connection, and use the Wireshark to capture packets on the PC.



Result analysis: The priority of service VLAN 1000 is priority 3 set locally.

8.3 Experiment 2: Traffic Policing Configuration

8.3.1 Configuration Roadmap

1. The configuration of traffic policing is based on the priority application experiment.

2. Download rate test when the downstream rate is not limited
3. Test the download rate after the downstream rate is limited.
4. Testing the upload rate when the upstream traffic profile limits the rate higher than the DBA rate
5. Testing the upload rate when the DBA rate in the upstream direction is higher than the rate in the traffic profile

8.3.2 Configuration Procedure

Assume that PON port 0/3/0 is connected to an ONT. The ONT ETH1 port is connected to a PC, the service upstream port is 0/19/0, the FTP server is connected to port 0/19/1 on the OLT, and the IP address of the FTP server is 192.168.100.105/24. The PC is configured with a static IP address 192.168.100.x/24.

Step 1 Create a service flow without binding a traffic profile.

```
HUAWEI(config)#service-port 3 vlan 1000 gpon 0/3/0 ont 1 gempport 1 multi-service user-vlan 35
```

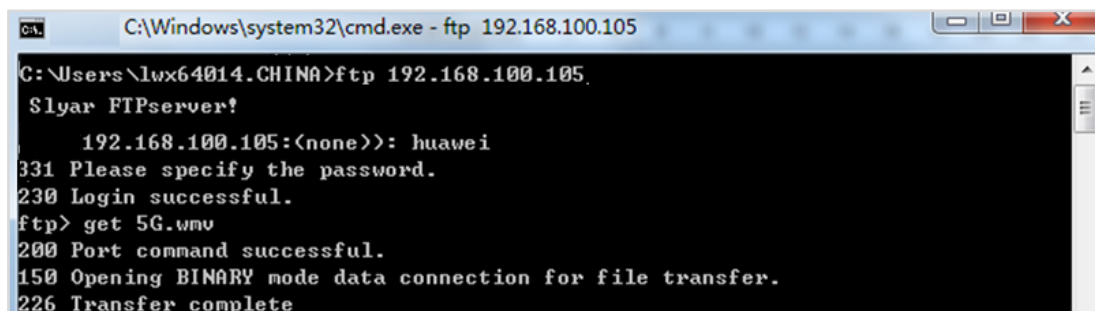
Step 2 Change the native VLAN of port 0/19/1. The PC can ping the FTP server.

```
HUAWEI(config)#port vlan 1000 0/19 1
HUAWEI(config)#interface giu 0/19
HUAWEI(config-if-giu-0/19)#native-vlan 1 vlan 1000
```



```
C:\Users\lux64014.CHINA>PING 192.168.100.105
```

Step 3 Use the PC to download files from the FTP server and test the download rate.



```
C:\Windows\system32\cmd.exe - ftp 192.168.100.105
C:\Users\lux64014.CHINA>ftp 192.168.100.105
Slyar FTPserver!
192.168.100.105:(none)> huawei
331 Please specify the password.
230 Login successful.
ftp> get 5G.wmv
200 Port command successful.
150 Opening BINARY mode data connection for file transfer.
226 Transfer complete
```

Result analysis: The average download speed is 7151.67 KB/s, which is about 57213.36 kbit/s.

Step 4 Create a traffic profile, set the upstream and downstream rates to 4 Mbit/s respectively, re-create service flows, and bind the traffic profile to the service flows.

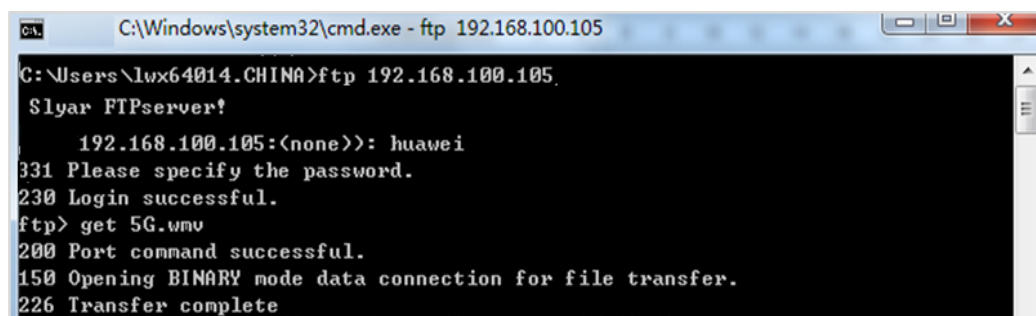
```
HUAWEI(config)#traffic table ip name QoS3 cir 4096 pir 4096 priority 0 priority-policy local-Setting
Create traffic descriptor record successfully
-----
TD Index: 10
TD Name: QoS3
Priority: 0
```

```
Copy Priority: -
Mapping Index: -
CTAG Mapping Priority: -
CTAG Mapping Index: -
CTAG Default Priority: 0
Priority Policy: local-pri
CIR: 4096 kbps
CBS: 133072 bytes
PIR: 4096 kbps
PBS: 133072 bytes
Fix: 0 kbps
CAR Threshold Profile: -
Color Mode: color-blind
Color Policy: dei
Referenced Status: not used
```

```
-----
HUAWEI(config)#undo service-port 3
```

```
HUAWEI(config)#service-port 3 vlan 1000 gpon 0/3/0 ont 1 gempport 1 multi-service user-vlan 35 rx-
cttr 10 tx-cttr 10
```

Step 5 Use the PC to download files from the FTP server and test the download rate.



```
C:\Windows\system32\cmd.exe - ftp 192.168.100.105
C:\Users\lux64014.CHINA>ftp 192.168.100.105
Slyar FTPserver!
192.168.100.105:(none): huawei
331 Please specify the password.
230 Login successful.
ftp> get 5G.wmv
200 Port command successful.
150 Opening BINARY mode data connection for file transfer.
226 Transfer complete
```

Result analysis: The average download rate is 502.98 KB/s, which is about 4023 kbit/s. This meets the rate limit result.

Step 6 Check the DBA profile bound to the service T-CONT. It is found that the rate is limited to 2048 kbit/s and the upstream rate is limited to 4096 kbit/s.

```
HUAWEI(config)#display ont info 0 3 0 1
```

```
-----
F/S/P: 0/3/0
ONT-ID: 1
Control flag: active
Run state: online
Config state: normal
Match state: match
DBA type: SR
ONT distance(m): 99
ONT last distance(m): -
ONT battery state: not support
Memory occupation: 32%
CPU occupation: 1%
Temperature: 70(C)
Authentic type: SN-auth
```

SN: 485754434D23C27B (HWTC-4D23C27B)
Management mode: OMCI
Software work mode: normal
Isolation state: normal
ONT IP 0 address/mask: 17.1.1.38/8
Description: ONT_NO_DESCRIPTION
Last down cause: -
Last up time: 2019-02-27 02:52:36+08:00
Last down time: -
Last dying gasp time: -
ONT online duration: 0 day(s), 6 hour(s), 54 minute(s), 14 second(s)
Type C support: Not support
Interoperability-mode: ITU-T
Power reduction status: -

VoIP configure method: Default

Line profile ID: 2
Line profile name: QoS

FEC upstream switch :Disable
OMCC encrypt switch :Off
Qos mode :PQ
Mapping mode :VLAN
TR069 management :Disable
TR069 IP index :0

Notes: * indicates Discrete TCONT(TCONT Unbound)

<T-CONT 0> DBA Profile-ID:1
<T-CONT 1> DBA Profile-ID:11
<Gem Index 1>

|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
|Upstream-priority-queue:0 |Downstream-priority-queue:- |

Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR

0 35 - - - - -

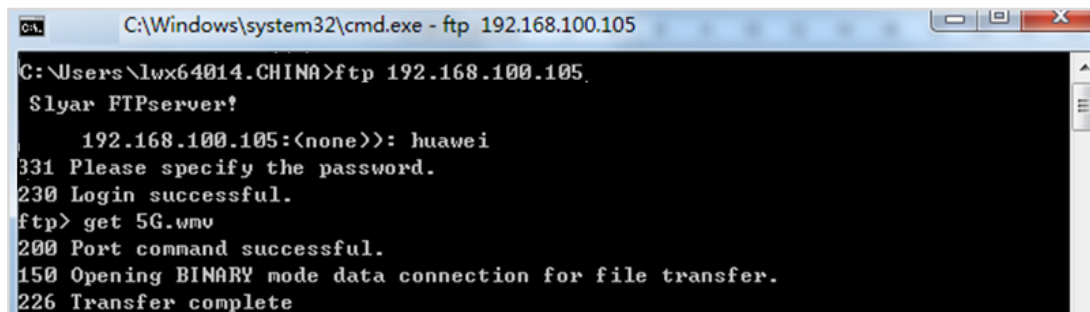
Notes: Run the display traffic table ip command to query
traffic table configuration

HUAWEI(config)#display dba-profile profile-id 11

Profile-name: QoS
Profile-ID: 11
type: 2
Bandwidth compensation: No
Fix-delay: No
Fix(kbps): 0
Assure(kbps): 2048 The value should be 2048.
Max(kbps): 0

Additional-bandwidth: -
Best-effort-priority: -
Best-effort-weight: -
bind-times: 1

Step 7 Upload files from the PC to the FTP server to test the upload rate.



```

C:\Windows\system32\cmd.exe - ftp 192.168.100.105
C:\Users\lux64014.CHINA>ftp 192.168.100.105
Slyar FTPserver!
192.168.100.105:(none)>: huawei
331 Please specify the password.
230 Login successful.
ftp> get 5G.wmv
200 Port command successful.
150 Opening BINARY mode data connection for file transfer.
226 Transfer complete

```

Result analysis: The average upload rate is 204.44 KB/s, which is about 1651 kbit/s. The upload rate meets the rate limit.

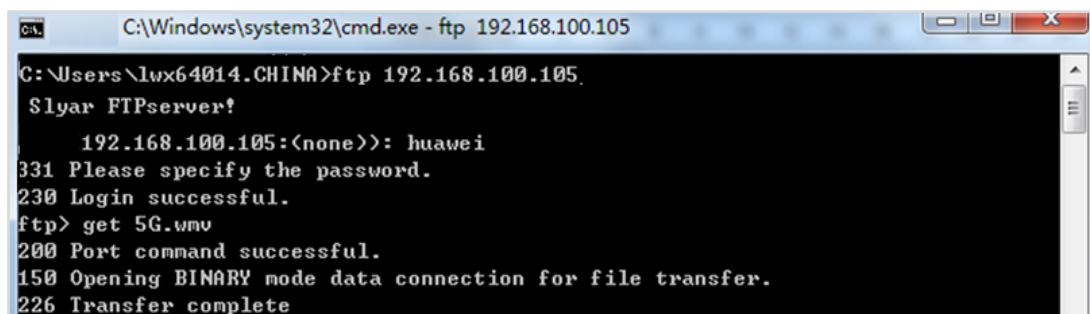
Step 8 Change the assured bandwidth to 8192 kbit/s in the DBA profile.

HUAWEI(config)#dba-profile modify profile-name QoS
> Start modifying a DBA-profile:

Step 9 > The type of the DBA-profile (1-5) [2]: 2

> The assured bandwidth of the DBA-profile(128-10000000kbps) [2048]: 8192

Step 10 Upload files from the PC to the FTP server to test the upload rate.



```

C:\Windows\system32\cmd.exe - ftp 192.168.100.105
C:\Users\lux64014.CHINA>ftp 192.168.100.105
Slyar FTPserver!
192.168.100.105:(none)>: huawei
331 Please specify the password.
230 Login successful.
ftp> get 5G.wmv
200 Port command successful.
150 Opening BINARY mode data connection for file transfer.
226 Transfer complete

```

Result analysis: The average upload rate is 496.90 KB/s, which is about 3975 kbit/s. The upload rate meets the rate limit.

9 Configuring Anti-DoS Attack

9.1 Experiment Introduction

9.1.1 About This Experiment

This section describes how to configure anti-DoS attack.

9.1.2 Objectives

- Describe the purpose of OLT anti-DoS attack.
- Use the anti-DoS attack function of the OLT.
- To verify the function of anti-DoS attack of the OLT.

9.1.3 Experiment Networking

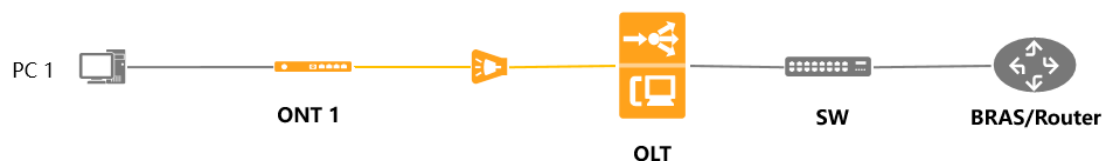


Figure 9-1 Networking diagram of anti-DoS attack

9.1.4 Experiment Planning

Table 9-1 In the live network, the optical splitter uses the 1:64 optical splitting ratio. The ONT/ONU is connected to the PON interface board of the OLT through the optical splitter, and the OLT is connected to the BRAS. In this experiment, terminal users connected to the OLT are simulated to send DoS attack packets.

Table 9-2 Experiment Data Planning

Equipment	Port	Parameter
OLT	GIU	0/10/0
	GPON	0/6/0

ONT	ETH1	VLAN:11
PC	Network port	Automatic IP address obtaining

9.2 Experiment Task Configuration

9.2.1 Configuration Roadmap

1. The OLT is connected to the IP network through the upstream port, the data configuration is complete, and the OLT works in the normal state.
2. PC1 can access the network normally.
3. PC1 simulates an unauthorized user to send a large number of protocol control packets.

9.2.2 Configuration Procedure

Step 1 Check the CPU usage of the control board or service board.

DoS attacks may cause high CPU usage. In normal cases, the CPU usage of the control board and service board does not exceed 20%. The CPU usage is high only when the board is saving data or writing data to the flash memory. This process does not last for more than 2s. Run the command to check the CPU usage of the control board or the service board.

Query the CPU usage of the main control board in slot 0/9.

```
Huawei_OLT(config)#display cpu
{frameid/slotid<S><Length 1-15>}:0/9
```

```
Command:
display cpu 0/9
CPU occupancy: 12%
```

Display the CPU usage of the control board in slot 0/10.

```
Huawei_OLT(config)#display cpu 0/10
Send message for inquiring board cpu occupancy successfully, board executing...
CPU occupancy: 7%
```

Display the CPU usage of the service board in slot 0/2.

```
Huawei_OLT(config)#display cpu 0/2
Send message for inquiring board cpu occupancy successfully, board executing...
CPU occupancy: 5%
```

Step 2 Enable anti-DoS attack on the device.

Run the command to enable the anti-DoS attack function of the device. Then, the device monitors the control packets sent from each port. When the rate of the control packets sent to the CPU exceeds the preset threshold, the system considers that a DoS attack occurs, adds the user port to the blacklist, and discards the control packets that exceed the limit.

```
Huawei_OLT(config)#security anti-dos  
{control-packet<K>|disable<K>|enable<K>}:enable
```

Command:
security anti-dos enable

Step 3 Configuring a Packet Processing Policy

Run the command to set the policy for processing protocol packets when a DoS attack occurs. The system does not check the status of the anti-DoS attack function when the policy command is executed. If the anti-DoS attack function is disabled, the system does not detect DoS attacks and sends protocol packets to the CPU for processing. If the anti-DoS attack function is enabled, the system processes protocol packets according to the preset processing policy.

```
Huawei_OLT(config)#security anti-dos control-packet policy  
{deny<K>|permit<K>}:permit
```

Command:
security anti-dos control-packet policy permit

Step 4 In step 4, the rate threshold for sending protocol packets to the CPU must be set to permit. Otherwise, step 4 is meaningless.

When the anti-DoS attack function is enabled and the protocol packets can be sent to the CPU for processing, if the rate exceeds the preset value, the system reports the DoS attack alarm and discards the excess packets.

Run the following command on the GPON port:

```
Huawei_OLT(config)#security anti-dos control-packet rate  
{frameid/slotid/portid<S><Length 1-15>}:0/5/0  
{default<K>|ont<K>|value<U><0,63>}:ont 1  
{default<K>|gempport<K>|value<U><0,254>}:gempport  
{gemindex<U><0,1023>}:1  
{default<K>|value<U><0,254>}:60
```

Command:
security anti-dos control-packet rate 0/5/0 ont 1 gempport 1 60
default Default threshold for the rate of sending protocol packets to the CPU. By default, the rate threshold for sending protocol packets to the CPU is 63 pps.

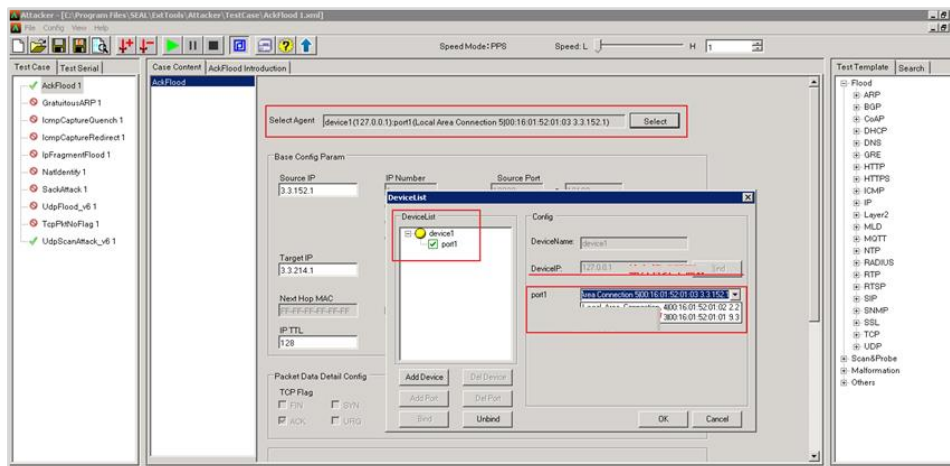
Step 5 Use the PC to simulate an unauthorized user to initiate a DoS attack.

● Introduction to Common DoS Tools

No.	Test Tool	Advantages	Restriction
1	Seal attacker (V1.0.19)	1. 181 types of DDoS attacks are provided and will be added in later versions. 2. Simple configuration 3. Attacker is a self-developed tool. You can submit a requirement and accept new attacks after the requirement is approved.	1. The performance of a single node is 30,000 pps. In delay mode, the performance can reach 100,000+ pps (depending on the performance of the executor). 2. A free license is required.
2	Hping3	1. Supports DoS attacks, firewall tests, port scanning, idle scanning, file transfer, and Trojan horses. 2. The Linux tool facilitates concurrent and distributed deployment.	1. Non-sift mode. In-depth research tools are required to construct various attack packets.
3	mz	1. Send almost all possible and impossible packets. 2. The Linux tool facilitates concurrent and distributed deployment.	1. Non-sift mode. In-depth research tools are required to construct various attack packets. 2. Non-professionals tend to crash their own networks.
4	pser	1. Excellent performance, with a peak value of 500,000 packets per second 2. Load simulation 3. The Linux tool facilitates concurrent and distributed deployment.	1. Non-sift mode. In-depth research tools are required to construct various attack packets.

● How to Use the Attacker Tool

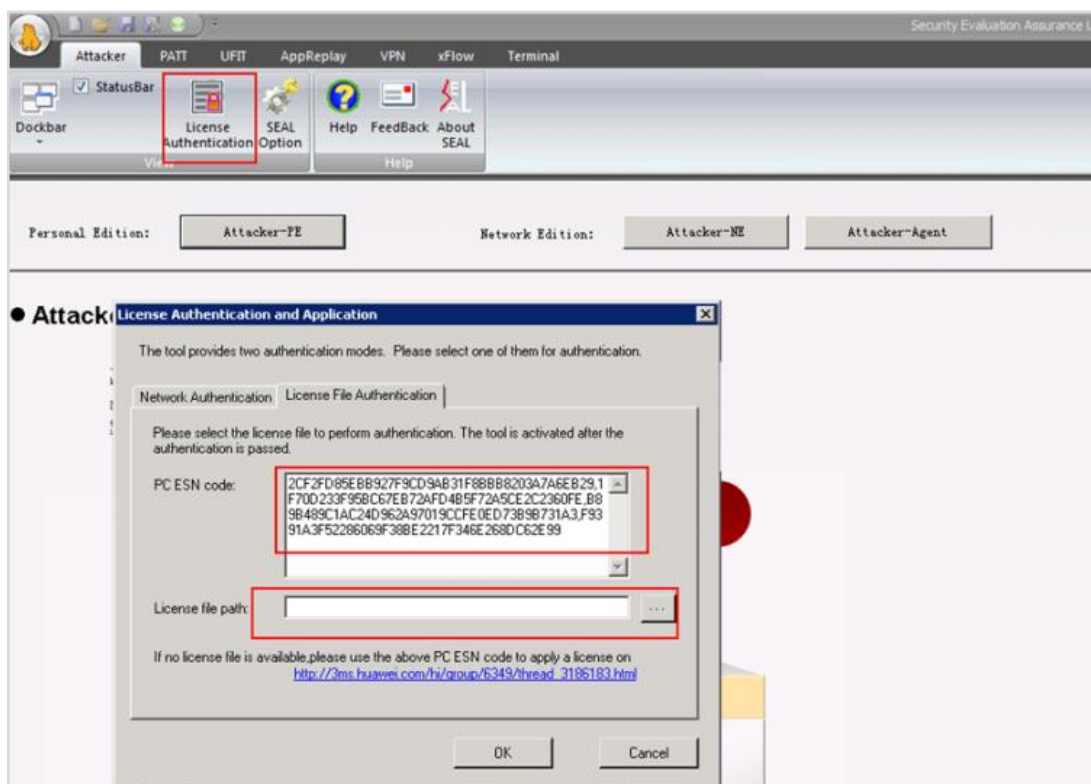
- Huawei-developed anti-DDoS tool is developed for 2012 Laboratories - R D capability improvement - Networking Security Competence Center.
- Advantage: The USG6000V series provides 181 types of DDoS attacks. 90 types of flood packets, 75 types of malformed packets, and 16 other types of packets.
- Restrictions: The attack rate of Windows-based applications is limited. The maximum attack rate is 3000 pps. Distributed deployment can be used to improve the concurrent attack bandwidth.
- Note: When deploying the test network, you need to set up at least two network planes, one for the management plane and the other for the actual attack network plane. In Attacker, set DeviceIP to the management plane IP address and AttackPort1 to the device corresponding to the private network to prevent impact on the lab network.



- Other Precautions
 - Search for seal on rTools and download and install it.



- For the license of the SEAL version, send the version number and ESN to the wuhaiting WX37700/chenchao WX337836. For details about how to obtain the ESN, see FAQ(HYPERLINK "http://3ms.huawei.com/hi/group/6349/thread_4551973.html?mapId=4262337" http://3ms.huawei.com/hi/group/6349/thread_4551973.html?mapId=4262337).



9.3 Verification

9.3.1 Check whether an alarm is generated on the OLT.

PC1 sends a large number of protocol packets. The anti-DoS attack function is enabled in the system. When the rate threshold for sending protocol packets to the CPU is set, the system reports a DoS attack alarm. Event ID: 0x29000008, indicating that a DoS attack occurs on the corresponding port.

9.4 Reference.

9.4.1 OLT Configuration

```
Huawei_OLT(config)#display current-configuration
{<cr>|include-default<K>|interface<K>|ont<K>|port<K>|section<K>|service-port<K>||<K>}:

Command:
display current-configuration
[Active: H901MPLB; Standby: H901MPLB]
[Patch Info: SPC201]
[MA5800V100R017:5800]
#
[pre-config]
<pre-config>
board add 0/6 H901GPHF
traffic table ip index 10 name "ip-traffic-table_10" cir 4096 cbs 133072 pir 8064 pbs 260048 color-
mode color-blind priority user-cos 5 priority-policy local-setting
dba-profile add profile-id 77 profile-name "shaf3y" type2 assure 2048
ont-srvprofile gpon profile-id 31 profile-name "mona"
ont-port pots adaptive 32 eth adaptive 8
port vlan eth 1 translation 11 user-vlan 11
commit
ont-lineprofile gpon profile-id 31 profile-name "ali"
tcont 2 dba-profile-id 77
gem add 2 eth tcont 2
gem mapping 2 0 vlan 11
commit
quit
#
[vlan-config]
<vlan-config>
vlan 3030 smart
vlan 4000 smart
vlan attrib 3030 q-in-q
port vlan 3030 0/10 0
port vlan 4000 0/10 0
#
```

```
<gpon-0/6>
interface gpon 0/6
port 0 ont-auto-find enable
ont add 0 1 sn-auth "48574453B6595B7B" omci ont-lineprofile-id 31 ont-srvprofile-id 31 desc
"ONT_NO_DESCRIPTION"
ont port native-vlan 0 1 eth 1 vlan 11 priority 0
ont port native-vlan 0 1 iphost vlan 34 priority 0
#
[bbs-config]
<bbs-config>
service-port 30 vlan 3030 gpon 0/6/0 ont 1 gemport 2 multi-service user-vlan 11 tag-transform
default inbound traffic-table index 10 outbound traffic-table index 10
#
[security-config]
<security-config>
security anti-dos enable
security anti-dos control-packet rate 0/6/0 ont 1 gemport 2 60
#
<Vlanif4000>
interface Vlanif4000
ip address 10.175.202.57 255.255.254.0
#
<MEth0>
interface MEth0
ip address 10.11.104.2 255.255.255.0
#
```

9.5 Questions

Stop the DoS attack initiated by the PC and check what happens on the OLT.

Recovery event: 0x29000009 The DoS attack on the user port disappears.

10

Configuring Anti-ICMP/IP Attack

10.1 Experiment Introduction

10.1.1 About This Experiment

Configure the anti-ICMP/IP attack function to meet system security requirements. In an ICMP/IP attack, a malicious user sends ICMP or IP packets whose destination IP address is the system IP address. If the malicious user sends a large number of ICMP packets (such as ping packets) and IP packets to the access system in a short period of time, the access system is overloaded and cannot process valid tasks. The anti-ICMP/IP attack feature means that the system discards the ICMP and IP packets sent from the user side to the device.

10.1.2 Objectives

- Describe the ICMP/IP attack principle.
- Huawei anti-ICMP/IP attack configuration is complete.
- To verify the function of anti-ICMP/IP attack.

10.1.3 Experiment Networking

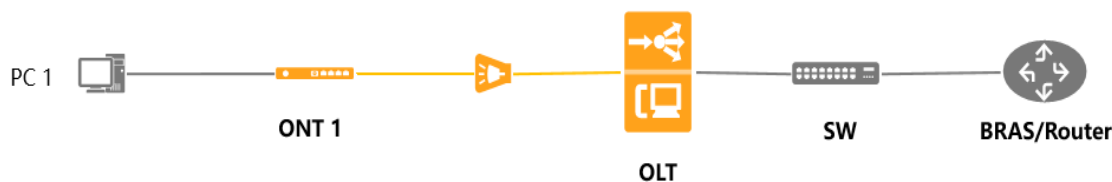


Figure 10-1 Networking diagram of anti-ICMP/IP attack

10.1.4 Experiment Planning

In the live network, the optical splitter uses the 1:64 optical splitting ratio. The ONT/ONU is connected to the PON interface board of the OLT through the optical splitter, and the OLT is connected to the BRAS. In this experiment, terminal users connected to the OLT are simulated to send DoS attack packets.

Table 10-1 Experiment Data Planning

Equipment	Port	Parameter
OLT	GIU	0/10/0
	GPON	0/6/0
ONT	ETH4	VLAN:4000
PC	Network port	Automatic IP address obtaining

10.2 Experiment Task Configuration

10.2.1 Configuration Roadmap

The following uses a PC as an example to simulate an unauthorized user to access the network:

(1) The OLT is connected to the IP network through the upstream port. The data configuration is complete. The VLAN interface IP address is configured on the OLT. The OLT works in the normal state.

The PC is connected to the ONT, and the ONT is connected to the GPBD board through optical splitter.

(3) The PC simulates an unauthorized user to send a large number of ICMP packets to the VLANIF interface.

10.2.2 Configuration Procedure

Step 1 The data service of the terminal is added on the OLT according to the data plan.

Step 2 Create a VLAN with the VLAN ID of 4000 and the IP address of the L3 interface of 10.175.202.57/23. Query VLANIF information.

```
Huawei_OLT#display interface vlanif 4000
{<cr>||<K>}:
Command:
display interface vlanif 4000
Vlanif4000 current state: UP (ifindex: 8392608)
Line protocol current state: UP
Description: HUAWEI, SmartAX Series, Vlanif4000 Interface
The Maximum Transmit Unit is 1500 bytes
Forward plane MTU: -
Internet Address is 10.175.202.57/23
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 487b-6b7d-ab37
VLAN Encap-mode: single-tag
```

Step 3 Create a virtual service channel for unauthorized users, add the ONT port to VLAN 4000, and configure a virtual port for maintenance services. In this way, the ONT can ping the interface address of VLAN 4000. For details about data configuration,

see the FTTx Service Provisioning Configuration Guide. The following information is displayed:

Query the line profile.

```
Huawei_OLT(config)#display ont-lineprofile gpon profile-id 31
{<cr>||<K>};
```

Command:

```
display ont-lineprofile gpon profile-id 31
```

```
-----
Profile-ID :31
```

```
Profile-name :ali
```

```
Access-type :GPON
-----
```

```
FEC upstream switch :Disable
```

```
OMCC encrypt switch :Off
```

```
Qos mode :PQ
```

```
Mapping mode :VLAN
```

```
TR069 management :Disable
```

```
TR069 IP index :0
-----
```

```
<T-CONT 0> DBA Profile-ID:1
```

```
<T-CONT 1> DBA Profile-ID:77
```

```
<Gem Index 1>
-----
```

```
|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
```

```
|Upstream-priority-queue:0 |Downstream-priority-queue:- |
-----
```

```
Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR
-----
```

```
0 4000 - - - - -
-----
```

```
<T-CONT 2> DBA Profile-ID:77
```

```
<Gem Index 2>
-----
```

```
|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |
```

```
|Upstream-priority-queue:0 |Downstream-priority-queue:- |
-----
```

```
Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR
-----
```

```
0 11 - - - - -
-----
```

```
-----
Notes: Run the display traffic table ip command to query
traffic table configuration
-----
```

```
Binding times :1
-----
```

Query a service template.

Huawei_OLT(config)#**display ont-lineprofile gpon profile-id 31**
{<cr>||<K>}:

Command:

display ont-lineprofile gpon profile-id 31

Profile-ID :31

Profile-name :ali

Access-type :GPON

FEC upstream switch :Disable

OMCC encrypt switch :Off

Qos mode :PQ

Mapping mode :VLAN

TR069 management :Disable

TR069 IP index :0

<T-CONT 0> DBA Profile-ID:1

<T-CONT 1> DBA Profile-ID:77

<Gem Index 1>

|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |

|Upstream-priority-queue:0 |Downstream-priority-queue:- |

Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR

0 4000 - - - - -

<T-CONT 2> DBA Profile-ID:77

<Gem Index 2>

|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:- |

|Upstream-priority-queue:0 |Downstream-priority-queue:- |

Mapping VLAN Priority Port Port Bundle Flow Transparent
index type ID ID CAR

0 11 - - - - -

Notes: Run the display traffic table ip command to query
traffic table configuration

Binding times :1

Huawei_OLT(config)#

Huawei_OLT(config)#

Huawei_OLT(config)#**display ont-srvprofile gpon profile-id 31**
{<cr>||<K>}:

Command:

display ont-srvprofile gpon profile-id 31

Profile-ID: 31
Profile-name: mona
Access-type: GPON

Port-type Port-number Max-adaptive-number

POTS adaptive 32
ETH adaptive 8
VDSL 0 -
TDM 0 -
MOCA 0 -
CATV 0 -

TDM port type: E1
TDM service type: TDMoGem
MAC learning function switch: Enable
ONT transparent function switch: Disable
Ring check switch: Disable
Ring port auto-shutdown: Enable
Ring detect frequency: 8 (pps)
Ring resume interval: 300 (s)
Multicast forward mode: Unconcern
Multicast forward VLAN: -
Multicast mode: Unconcern
Upstream IGMP packet forward mode: Unconcern
Upstream IGMP packet forward VLAN: -
Upstream IGMP packet priority: -
Native VLAN option: Concern
Upstream PQ color policy: -
Downstream PQ color policy: -

Port-type Port-ID QinQmode PriorityPolicy Inbound Outbound

ETH 1 unconcern unconcern unconcern unconcern
ETH 2 unconcern unconcern unconcern unconcern
ETH 3 unconcern unconcern unconcern unconcern
ETH 4 unconcern unconcern unconcern unconcern
ETH 5 unconcern unconcern unconcern unconcern
ETH 6 unconcern unconcern unconcern unconcern
ETH 7 unconcern unconcern unconcern unconcern
ETH 8 unconcern unconcern unconcern unconcern

Notes: Run the display traffic table ip command to query
traffic table configuration

Port-type Port-ID DownstreamMode MismatchPolicy

ETH 1 operation discard
ETH 2 operation discard
ETH 3 operation discard
ETH 4 operation discard
ETH 5 operation discard
ETH 6 operation discard

ETH 7 operation discard

ETH 8 operation discard

Port-type Port-ID Dscp-mapping-table-index

ETH 1 0

ETH 2 0

ETH 3 0

ETH 4 0

ETH 5 0

ETH 6 0

ETH 7 0

ETH 8 0

IPHOST 1 0

Port Port Service-type Index S-VLAN S-PRI C-VLAN C-PRI ENCAP S-PRI
type ID POLICY

ETH 1 Translation 1 11 - 11 - - -

ETH 4 Translation 1 4000 - 4000 - - -

Notes: * indicates transparent attribute of the vlan

Port-type Port-ID IGMP-mode IGMP-VLAN IGMP-PRI Max-MAC-Count

ETH 1 - - - Unlimited

ETH 2 - - - Unlimited

ETH 3 - - - Unlimited

ETH 4 - - - Unlimited

ETH 5 - - - Unlimited

ETH 6 - - - Unlimited

ETH 7 - - - Unlimited

ETH 8 - - - Unlimited

Binding times: 1

Modify the ONT native VLAN.

HUAWEI(config-if-gpon-0/6)#ont port native-vlan 0 1

{eth<K>|iphost<K>|moca<K>}:eth

{ont-portid<U><1,8>}:4

{priority<K>|vlan<K>}:vlan 4000

{<cr>|priority<K>}:

Command:

ont port native-vlan 0 1 eth 4 vlan 4000

Create a service virtual tunnel.

Huawei_OLT(config)#service-port vlan 4000 gpon 0/6/0 ont 1 gemport 1 multi-service user-vlan 4000

Step 4 Run the command to query the status of the security function.

```
Huawei_OLT(config)#display security config
{<cr>||<K>};
```

Command:

display security config

Anti-ipspoofing function: disable
Anti-dos function: enable
Anti-macspoofing function: disable
Anti-ipattack function: disable
Anti-icmpattack function: disable
Source-route filter function: disable
Anti-macduplicate function: disable
PPPoE overall aging time(sec): 360
PPPoE aging period(sec): 90
ARP detect mode: dummy
Anti-dos control-packet policy: deny
Packet unaffected by anti-ipspoofing: --
NS-reply function: disable
NS-reply unknown-policy: forward
ARP-reply function: disable
ARP-reply unknown-policy: forward
Anti-ipv6spoofing function: disable
IPv6 DAD proxy function: disable
IPv6 bind route and ND: disable
Packet unaffected by anti-macspoofing: IGMP
DHCP client identifier: chaddr
Packet ignored by anti-macspoofing: --
Anti-illegal-arp function: enable
Anti-illegal-nd function: enable
User delete delay(sec): 0
Anti-macduplicate alarm function: disable
Anti-ipv6attack function: disable
Anti-icmpv6attack function: disable
Anti-ipconflict refresh function: disable

Step 5 Ping the IP address of VLAN 4000 on the OLT from the ONU PC.

Step 6 Run the security anti-ipattack enable command to enable the anti-IP attack function.

```
Huawei_OLT(config)#security anti-ipattack enable
```

Step 7 Run the security anti-icmpattack enable command to enable the anti-ICMP attack function.

```
Huawei_OLT(config)#security anti-icmpattack enable
Huawei_OLT(config)#display security config
{<cr>||<K>};
```

Command:

display security config

Anti-ipspoofing function: disable
Anti-dos function: enable
Anti-macspoofing function: disable
Anti-ipattack function: enable

```

Anti-icmpattack function: enable
Source-route filter function: disable
Anti-macduplicate function: disable
PPPoE overall aging time(sec): 360
PPPoE aging period(sec): 90
ARP detect mode: dummy
Anti-dos control-packet policy: deny
Packet unaffected by anti-ipspoofing: --
NS-reply function: disable
NS-reply unknown-policy: forward
ARP-reply function: disable
ARP-reply unknown-policy: forward
Anti-ipv6spoofing function: disable
IPv6 DAD proxy function: disable
IPv6 bind route and ND: disable
Packet unaffected by anti-macspoofing: IGMP
DHCP client identifier: chaddr
Packet ignored by anti-macspoofing: --
Anti-illegal-arp function: enable
Anti-illegal-nd function: enable
User delete delay(sec): 0
Anti-macduplicate alarm function: disable
Anti-ipv6attack function: disable
Anti-icmpv6attack function: disable
Anti-ipconflict refresh function: disable

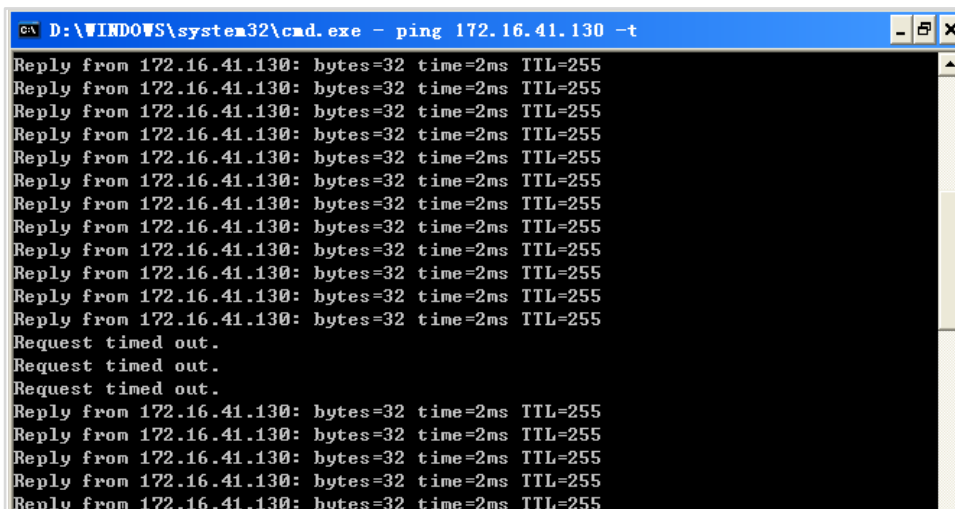
```

Step 8 Ping the IP address of VLAN 4000 on the OLT from the ONU PC again. To verify the function of anti-ICMP/IP attack.

10.3 Verification

10.3.1 Check whether the PC can ping the OLT.

When the anti-ICMP/IP attack function is enabled, the IP address of VLAN 4000 on the OLT cannot be pinged from the PC. When the anti-ICMP/IP attack function is disabled, the IP address can be pinged from the PC.



```

C:\D:\WINDOWS\system32\cmd.exe - ping 172.16.41.130 -t
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Request timed out.
Request timed out.
Request timed out.
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255
Reply from 172.16.41.130: bytes=32 time=2ms TTL=255

```

11

Configuring MAC Address Filtering

11.1 Experiment Introduction

11.1.1 About This Experiment

This chapter describes MAC address filtering to prevent malicious users from forging MAC addresses of network devices to attack the carrier network.

11.1.2 Objectives

- Describe the MAC address learning mechanism of Huawei devices.
- Use the MAC address filtering function of Huawei devices.
- The MAC address filtering function is configured on Huawei devices.

11.1.3 Experiment Networking



Figure 11-1 Networking diagram of anti-ICMP/IP attack

11.1.4 Experiment Planning

Table 11-1 Experiment Data Planning

Equipment	Port	Parameter
OLT	GIU	0/10/0
	GPON	0/6/0
ONT	ETH1	VLAN:11

PC	Network port	Automatic IP address obtaining
----	--------------	--------------------------------

11.2 Experiment Task Configuration

11.2.1 Configuration Roadmap

In this experiment, PC1 connected to the OLT sends data packets. The source MAC address of the packets is the same as the system MAC address of the OLT.

(1) The OLT is connected to the IP network through the upstream port, the data configuration is complete, and the OLT works in the normal state.

PC1 is connected to the ONT through the voice phone, and the ONT is connected to the GPBD board through optical splitter.

(3) Change the MAC address of PC1 to be the same as the MAC address of the system, and simulate an unauthorized user to send packets to the device.

11.2.2 Configuration Procedure

Step 1 The data service of the terminal is added on the OLT according to the data plan.

Step 2 Check the MAC address of the OLT.

```
Huawei_OLT#display sysman
{centralized-mgmt<K>|console<K>|firewall<K>|ip-access<K>|ip-refuse<K>|mac-address<K>|protocol-
type<E><<telnet,ssh,snmp>|service<K>|source<K>|vpn-instance<K>}: mac-address
{<cr>||<K>};
```

Command:

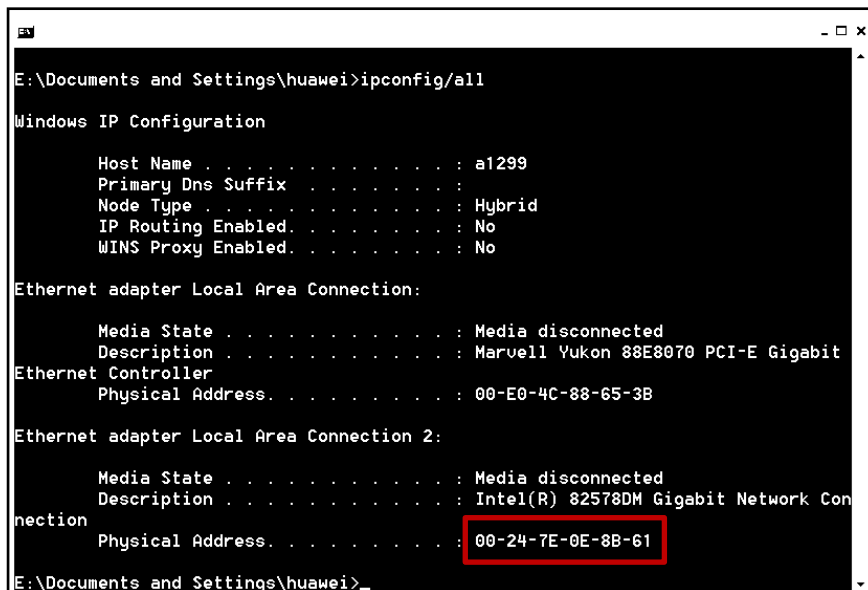
display sysman mac-address

Current MAC address of active control board: 487b-6b7d-ab36

Default MAC address of active control board: 487b-6b7d-ab36

Step 3 Check the MAC address of the network adapter that connects PC1 to the ONT port.

On the PC, choose **Start > Run**. In the Run dialog box, enter **cmd**. In the displayed DOS window, enter **ipconfig/all** under the cursor, as shown in the following figure.



```
E:\Documents and Settings\huawei>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : a1299
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Description . . . . . : Marvell Yukon 88E8070 PCI-E Gigabit
Ethernet Controller
    Physical Address. . . . . : 00-E0-4C-88-65-3B

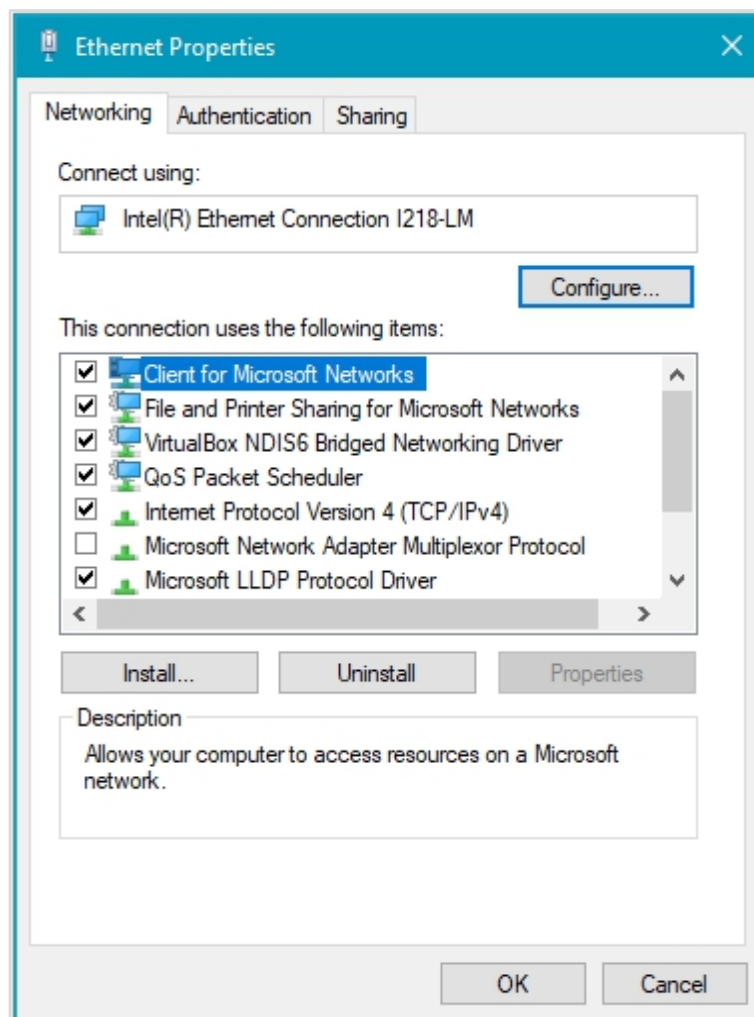
Ethernet adapter Local Area Connection 2:

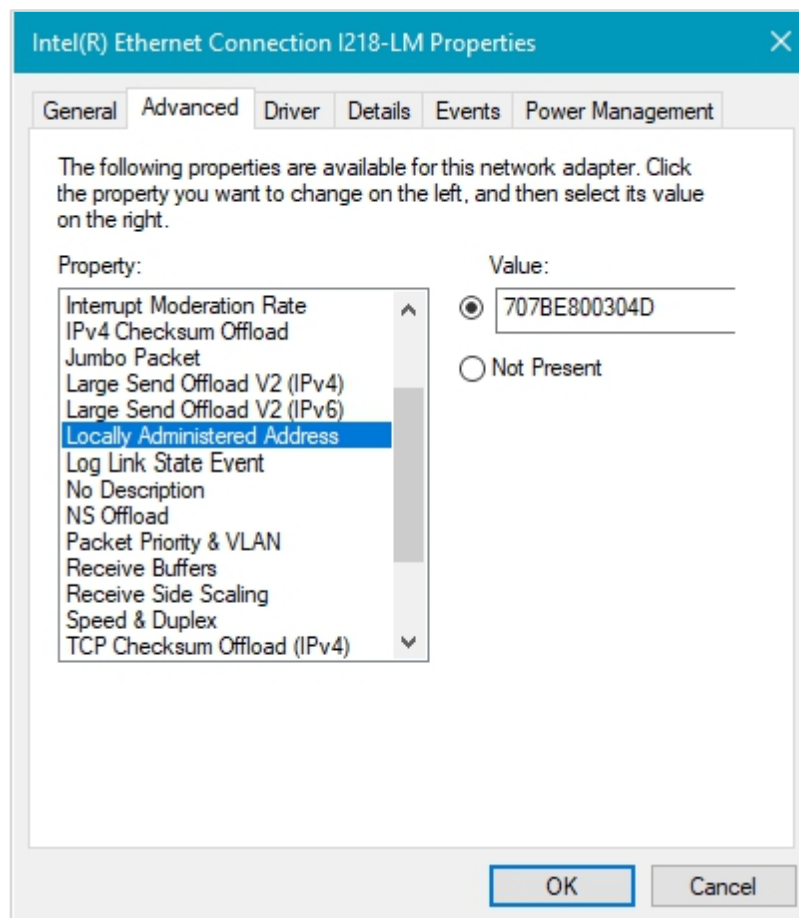
    Media State . . . . . : Media disconnected
    Description . . . . . : Intel(R) 82578DM Gigabit Network Con
nection
    Physical Address. . . . . : 00-24-7E-0E-8B-61

E:\Documents and Settings\huawei>
```

Step 4 Change the MAC address of PC1 to be the same as the MAC address of the system.

On the LAN or high-speed Ethernet card, choose **Local Area Connection > Properties > Network Card > Configuration > Advanced > Properties > Local Management Address >** Enter the MAC address to be modified. Enter the system MAC address. Record the original MAC address of the network adapter before modification because the original MAC address needs to be changed back after result verification.





Step 5 Change the MAC address of the network adapter on PC1 as follows: The MAC address is the same as the system MAC address of the OLT.



Step 6 Disable the anti-ICMP/IP attack function and ping the VLAN interface address of the OLT from PC1.

Step 7 Run the command to enable anti-MAC address filtering.

The Huawei_OLT(config)#security mac-filter source 487b-6b7d-ab36 command is repeated.

Query anti-MAC address filtering.

12

Configuring Access/DOA Address Segments

12.1 Experiment Introduction

12.1.1 About This Experiment

This chapter describes how to configure access or denial of access (DOA) address segments for a firewall to prevent users on unauthorized address segments from logging in to the system and therefore ensure system security.

12.1.2 Objectives

- Describe the function of the permitted/rejected address segment of Huawei devices.
- Use the function of allowing or denying access to the address segment on the Huawei device.
- This section describes how to maintain the firewall access control function of Huawei devices.

12.1.3 Experiment Networking

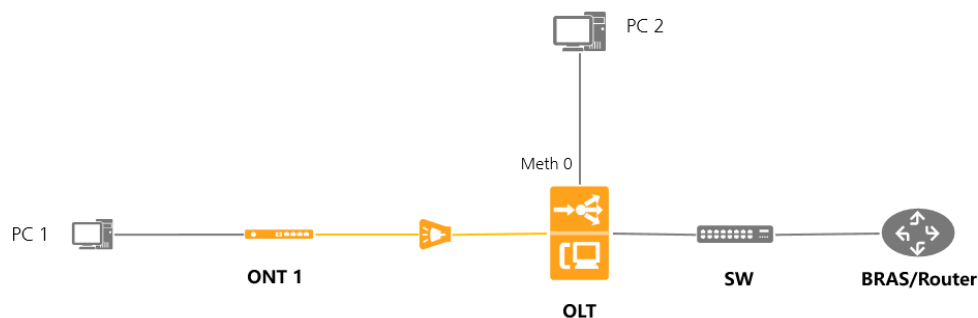


Figure 12-1 Networking for the device to access the firewall

12.1.4 Experiment Planning

Table 12-1 Experiment Data Planning

Equipment	Port	Parameter
-----------	------	-----------

OLT	GIU	0/19/0
	Meth 0	IP address:192.168.41.130/25
PC2		IP address:192.168.41.20/25

12.2 Experiment Task Configuration

12.2.1 Configuration Procedure

Step 1 Run the firewall enable command to enable the firewall function. When you need to prevent unauthorized users from intruding into the system, run this command to enable the firewall function to filter out the packets from untrusted users. The packet filtering firewall can be configured for a specified port only after the firewall function is enabled.

```
Huawei_OLT(config)#firewall enable
```

Step 2 Add an ACL to allow the terminals in the 192.168.41.0/25 network segment to access the device. The IP address of the device for outband maintenance is 192.168.41.130/25.

```
Huawei_OLT(config)#acl 3000
HUAWEI(config-acl-adv-3000)#rule 5
{deny<K>|permit<K>}:permit
{<0-255>|gre<K>|icmp<K>|ip<K>|ipinip<K>|tcp<K>|udp<K>}:ip
{<cr>|destination<K>|dscp<K>|fragment<K>|precedence<K>|source<K>|time-range<K>|tos<K>}:source
{any<K>|ip_addr<I><X.X.X.X>}:192.168.41.0
{0<K>|ip_addr<I><X.X.X.X>}:0.0.0.127
{<cr>|destination<K>|dscp<K>|fragment<K>|precedence<K>|time-range<K>|tos<K>}:destination
{any<K>|ip_addr<I><X.X.X.X>}:192.168.41.130
{0<K>|ip_addr<I><X.X.X.X>}:0.0.0.0
{<cr>|dscp<K>|fragment<K>|precedence<K>|time-range<K>|tos<K>}:

```

Command:

```
rule 5 permit ip source 192.168.41.0 0.0.0.127 destination
192.168.41.130 0.0.0.0
```

Step 3 Prevent other IP addresses from accessing the device.

```
HUAWEI(config-acl-adv-3000)#rule
{deny<K>|integer<U><0,4294967294>|permit<K>}:10
{deny<K>|permit<K>}:deny
{<0-255>|gre<K>|icmp<K>|ip<K>|ipinip<K>|tcp<K>|udp<K>}:ip
{<cr>|destination<K>|dscp<K>|fragment<K>|precedence<K>|source<K>|time-range<K>|tos<K>}:destination
{any<K>|ip_addr<I><X.X.X.X>}:192.168.41.130
{0<K>|ip_addr<I><X.X.X.X>}:0
{<cr>|dscp<K>|fragment<K>|precedence<K>|source<K>|time-range<K>|tos<K>}:

```

Command:
rule 10 deny ip destination 192.168.41.130 0

Step 4 Enable the ACL on the outband interface.

```
Huawei_OLT(config)#interface meth 0  
HUAWEI(config-if-meth0)#firewall  
{packet-filter<K>}:packet-filter  
{INTEGER<2000-2999>|INTEGER<3000-3999>}:3000  
{inbound<K>|outbound<K>}:inbound
```

Command:
firewall packet-filter 3000 inbound
ACL applied successfully

12.3 Verification

12.3.1 Check whether the PC can log in to the OLT.

Before the access control list (ACL) is enabled, terminals in different network segments from the maintenance network can access the OLT. After the ACL is enabled, only the IP addresses that are allowed to access the OLT can be used to access the OLT.

13 802.1X authentication

13.1 Experiment Introduction

13.1.1 About This Experiment

This topic describes how to configure 802.1X authentication for the OLT to improve user access security and system security.

13.1.2 Objectives

- Describe the application scenarios of 802.1X authentication.
- Understand how to configure OLT 802.1x.

13.1.3 Experiment Networking

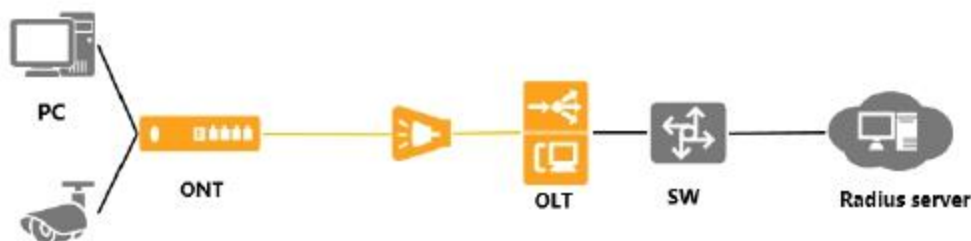


Figure 13-1 Networking diagram of 802.1x authentication

13.1.4 Experiment Planning

Table 13-1 Experiment Data Planning

Equipment	Port	Parameter
OLT	GIU	0/19/0

Radius		Authentication IP address: 10.10.66.66
--------	--	---

13.2 Experiment Task Configuration

13.2.1 Configuration Roadmap

The core switch functions as the 802.1X authentication device, and 802.1X protocol packet transparent transmission is configured on the OLT.

13.2.2 Configuration Procedure

Step 1 The ONU/OLT goes online normally and is in the normal state. By default, the ONU transparently transmits EAPOL and BPDU packets.

Step 2 Configure the OLT to transparently transmit EAPoL packets.

```
Huawei_OLT(config)#protocol permit-forwarding eapol enable
```

Step 3 Configure the OLT to transparently transmit BPDU packets.

```
Huawei_OLT(config)#vlan service-profile profile-id 10
huawei(config-vlan-srvprof-10)#bpdu tunnel enable
huawei(config-vlan-srvprof-10)#commit
huawei(config-vlan-srvprof-10)#quit
Huawei_OLT(config)#vlan bind service-profile 11 profile-id 10
```

13.3 Verification

After the configuration is complete, 802.1X-related protocol packets are transparently transmitted to the core switch for 802.1X authentication. Check whether the PC can obtain an IP address.

14

PON Networking Protection Configuration

14.1 Experiment Introduction

14.1.1 About This Experiment

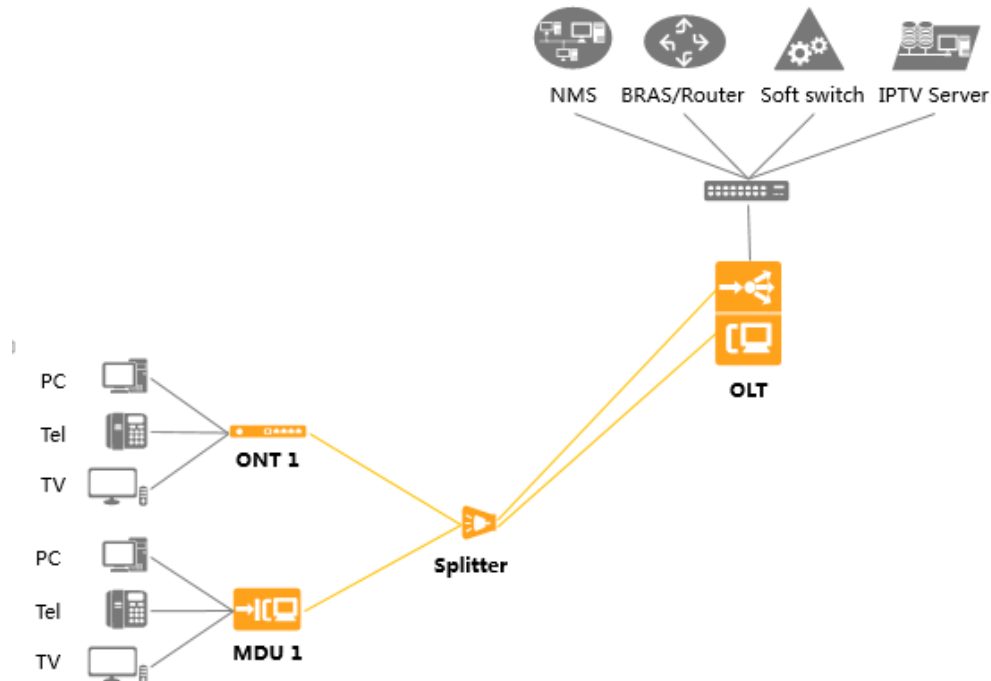
This document describes how to configure and verify GPON downstream and upstream network protection, including GPON type B protection, GPON type C protection, Ethernet link aggregation, and Ethernet port protection groups.

14.1.2 Objectives

- Describe the application scenarios of the GPON Type B/Type C protection switching solution.
- Complete the configuration of the GPON type B/type C protection switching solution.
- To verify the GPON Type B/Type C protection switching
- Learn how to configure Ethernet link aggregation protection.
- Master the working configuration of the Ethernet port protection group.

14.1.3 Configuring Type B Single-Homing Automatic Protection Switching

14.1.3.1 Experiment Networking

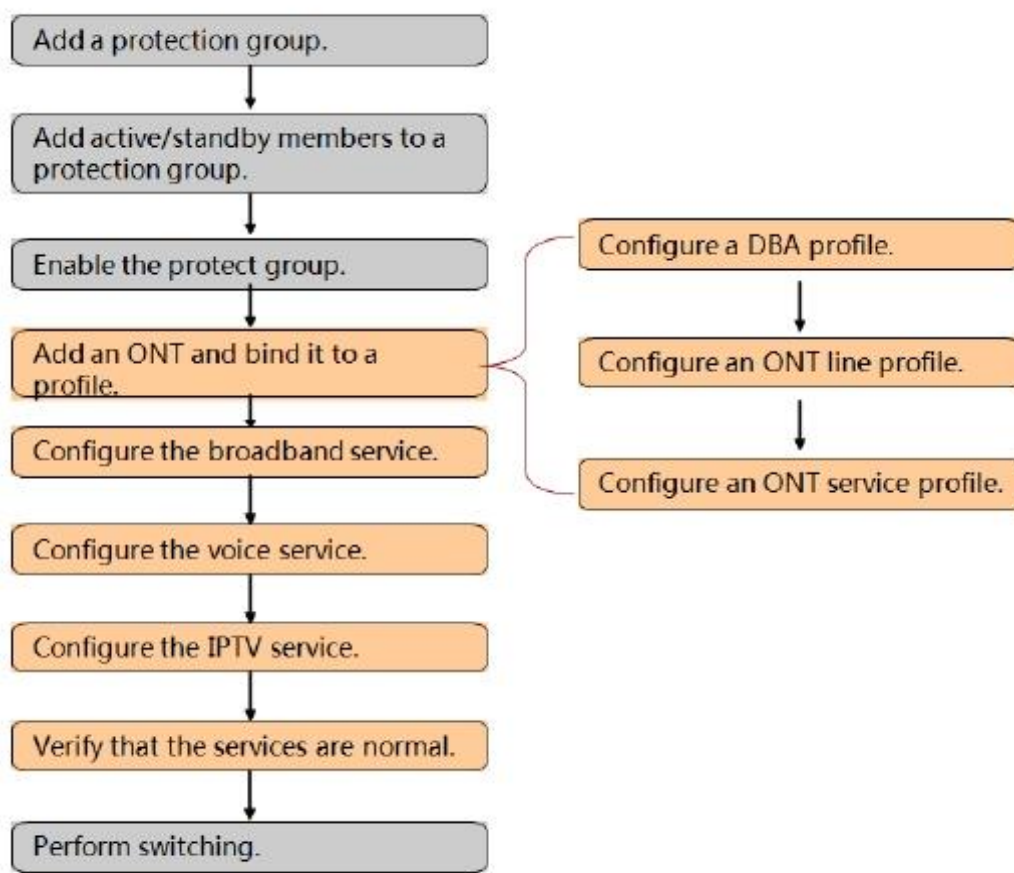


14.1.3.2 Experiment Planning

Item	Data
OLT upstream port	Upstream port: 0/19/0
Active and standby PON ports on the OLT	Master port:0/5/0 Slaver port:0/6/0

14.1.3.3 Configuration Procedure

After the type B protection is configured, the service configuration of the ONU is the same as that before the protection. That is, the service configuration is still applied to only the active GPON port.



 **NOTE** For details about the steps in orange, see the Configuration Guide to the GPON Data and Voice Multicast Service.

Step 1 You have logged in to the system as an administrator.

Step 2 Configure a protection group.

```

Huawei_OLT(config)#protect-group
{pgid<U><0,511>|protect-target<K>}:0
{<cr>|protect-target<K>}:protect-target
{as-mainboard-lag<K>|as-mainboard-port<K>|as-mainboard<K>|epon-uni-ont<K>|epon-uni-port<K>|eth-nni-lag-member<K>|eth-nni-port<K>|gpon-uni-ont<K>|gpon-uni-port<K>|service-process-board<K>|stm-nni-port<K>}:gpon-uni-port
{workmode<K>}:workmode
{timedelay<K>}:timedelay
  
```

Command:

```
protect-group 0 protect-target gpon-uni-port workmode timedelay
```

Step 3 Configure the active and standby members in the protection group.

```
huawei(protect-group-0)#protect-group member
```

```
{board<K>|port<K>}:port 0/5/0
```

```
{ont<K>|role<K>}:role
```

```
{member-role<E><work,protect>}:work
```

Command:

```
protect-group member port 0/5/0 role work huawei(protect-group-0)#protect-group member port 0/6/0 role protect
```

Step 4 Enable a protection group.

```
huawei(protect-group-0)#protect-group enable
```

```
huawei(protect-group-0)#quit
```

Step 5 Adding templates

Step 6 Add a terminal to PON port 0/5/0 and bind a profile to the terminal.

Step 7 Verify that data, voice, and FTP services are normal.

Step 8 To query the initial status of a protection group

```
Huawei_OLT(config)#display protect-group 0
```

```
-----
```

```
Group ID: 0
```

```
Protect Target: Port of GPON uni
```

```
Work Mode: timedelay
```

```
Description:
```

```
Admin State: enable
```

```
Operation: none
```

```
Reversion: disable
```

```
Reversion Time(s): 720
```

```
Anti-jitter Time(s): 720
```

```
PPPoE Single MAC: Unbind
```

```
Standard type B: no
```

```
-----
```

```
Member Role Operation State PeerMember
```

```
-----
```

```
0/5/0 work none active none
```

```
0/6/0 protect none standby none
```

```
-----
```

Step 9 Simulate the removal of the optical fiber from the active PON port 0/5/0. The system switches to the standby PON port 0/6/0 instantly.

Check the protection group status.

```
huawei(protect-group-0)#display protect-group 0
```

Group ID: 0

Protect Target: Port of GPON uni

Work Mode: timedelay

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Anti-jitter Time(s): 720

PPPoE Single MAC: Unbind

Standard type B: no

Member Role Operation State PeerMember

0/5/0 work auto switch src standby 0/5/0

0/6/0 protect auto switch dst active 0/6/0

Step 10 Restore the optical fiber of PON port 0/5/0, and then remove the optical fiber of the active PON port 0/6/0. The system switches to PON port 0/5/0 instantly.

```
huawei(protect-group-0)#display protect-group 0
```

Group ID: 0

Protect Target: Port of GPON uni

Work Mode: timedelay

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Anti-jitter Time(s): 720

PPPoE Single MAC: Unbind

Standard type B: no

Member Role Operation State PeerMember

0/5/0 work none active none

0/6/0 protect none standby none

Step 11 The system supports forced switching. For example, to forcibly switch services from the active PON port 0/5/0 to 0/6/0, check the status of the protection group.

huawei(protect-group-0)# **force-switch port 0/5/0 to 0/6/0**

huawei(protect-group-0)#**display protect-group 0**

Group ID: 0

Protect Target: Port of GPON uni

Work Mode: timedelay

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Anti-jitter Time(s): 720

PPPoE Single MAC: Unbind

Standard type B: no

Member Role Operation State PeerMember

0/5/0 work force switch src standby 0/6/0

0/6/0 protect force switch dst active 0/5/0

Step 12 After a forced switchover, the automatic switchover fails and services are interrupted if the optical fiber connected to the active PON port 0/6/0 is removed. Check the protection group status.

huawei(protect-group-0)#display protect-group 0

Group ID: 0

Protect Target: Port of GPON uni

Work Mode: timedelay

Description:

Admin State: enable
Operation: none
Reversion: disable
Reversion Time(s): 720
Anti-jitter Time(s): 720
PPPoE Single MAC: Unbind
Standard type B: no

Member Role Operation State PeerMember

0/5/0 work force switch src standby 0/6/0
0/6/0 protect force switch dst active 0/5/0

Step 13 Cancel Forced Switching

```
huawei(protect-group-0)# undo force-switch port 0/5/0 to 0/7/0
```

Step 14 Check the status change of the protection group when all ONTs connected to the PON port go offline. The initial configuration is as follows:

```
huawei(protect-group-0)#display protect-group 0
```

Group ID: 0
Protect Target: Port of GPON uni
Work Mode: timedelay
Description:
Admin State: enable
Operation: none
Reversion: disable
Reversion Time(s): 720
Anti-jitter Time(s): 720
PPPoE Single MAC: Unbind
Standard type B: no

Member Role Operation State PeerMember

0/5/0 work none active none
0/6/0 protect none standby none

Step 15 Shut down all the ONTs connected to the PON port or remove the optical fibers connected to all the ONTs. Then, the system automatically performs the switchover.

```
huawei(protect-group-0)#display protect-group 0
```

```
-----
```

```
Group ID: 0
```

```
Protect Target: Port of GPON uni
```

```
Work Mode: timedelay
```

```
Description:
```

```
Admin State: enable
```

```
Operation: none
```

```
Reversion: disable
```

```
Reversion Time(s): 720
```

```
Anti-jitter Time(s): 720
```

```
PPPoE Single MAC: Unbind
```

```
Standard type B: no
```

```
-----
```

```
Member Role Operation State PeerMember
```

```
-----
```

```
0/5/0 work auto switch src standby 0/6/0
```

```
0/6/0 protect auto switch dst active 0/5/0
```

```
-----
```

Step 16 Deleting a PON Protection Pair

```
Huawei_OLT(config)#protect-group 0
```

```
huawei(protect-group-0)#protect-group disable
```

```
huawei(protect-group-0)#undo protect-group member port 0/6/0
```

```
huawei(protect-group-0)#undo protect-group member port 0/5/0
```

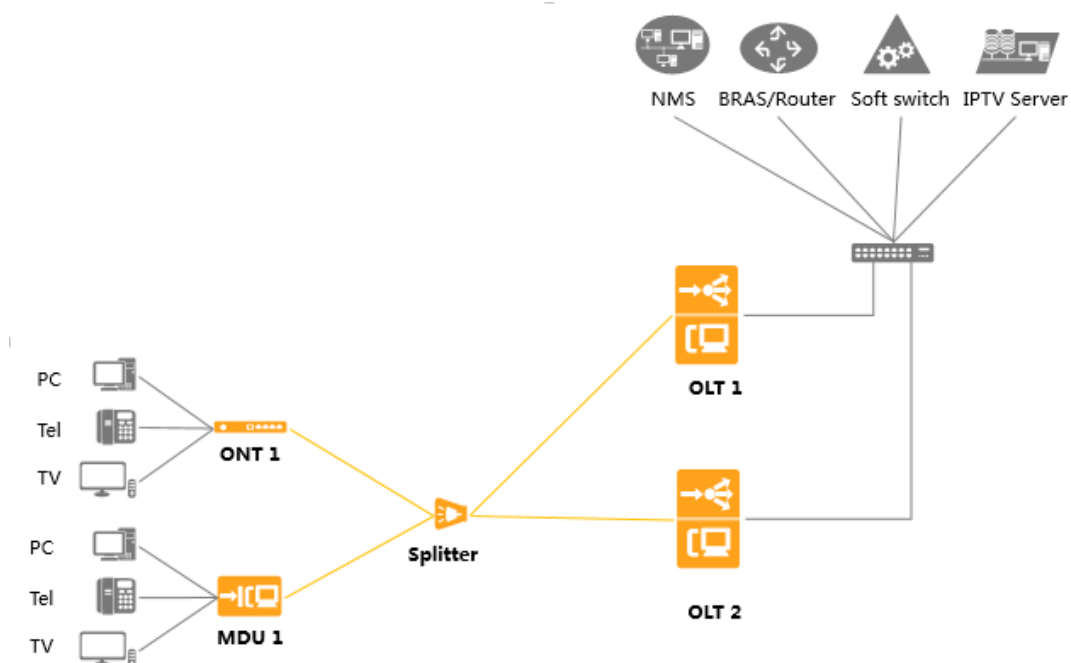
```
huawei(protect-group-0)#quit
```

```
Huawei_OLT(config)#undo protect-group 0
```

To delete a protect group, disable the protect group, delete the member port (standby port), and then delete the protect group.  **NOTE**

14.1.4 Configuring Type B Dual-Homing Automatic Switching Protection

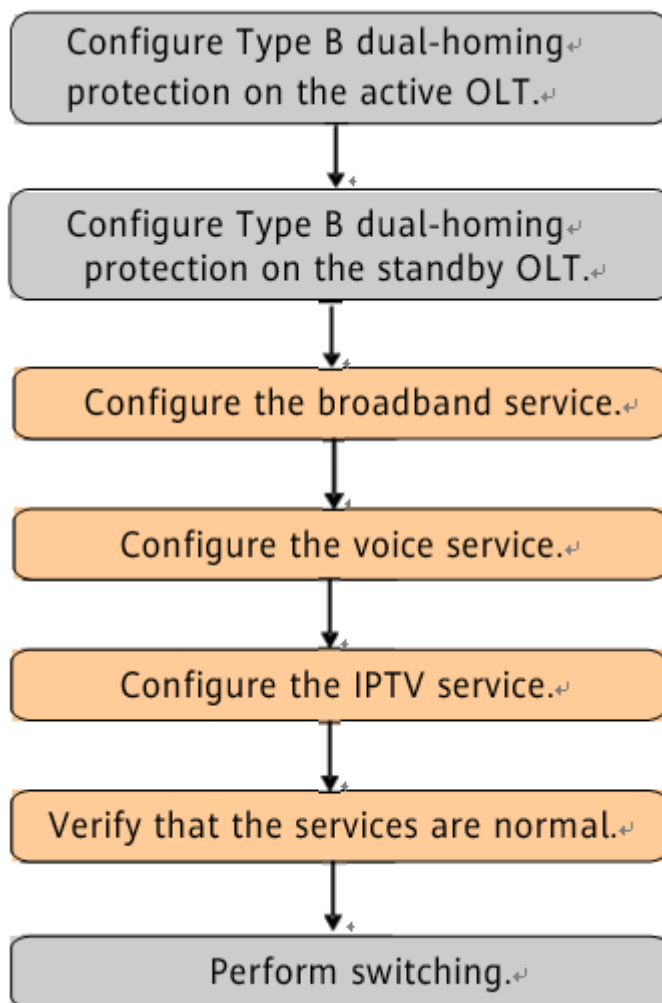
14.1.4.1 Experiment Networking



14.1.4.2 Experiment Planning

Item	Data
IP addresses of the active and standby OLTs	IP address of the active OLT: 10.175.192.35/22 Standby OLT IP address: 10.175.192.36/22
Active and standby OLT PON ports	0/5/0

14.1.4.3 Configuration Procedure



Step 1 You have logged in to the system as an administrator.

Step 2 Configure IP interworking between the active and standby OLTs.

Step 3 Configure type B dual-homing protection on the active OLT huawei.

```
Huawei_OLT(config)#dual-parenting local-node ip-address 10.175.192.35 port 6076 key work_4234
```

```
Huawei_OLT(config)#dual-parenting peer-node standby ip-address 10.175.192.36 port 6076 key protect_4234
```

```
Huawei_OLT(config)#dual-parenting sync enable
```

```
Huawei_OLT(config)#protect-group 1 protect-target gpon-uni-port workmode dual-parenting
```

```
huawei(protect-group-1)#protect-group member port 0/5/0 role work
```

```
huawei(protect-group-1)#peer-group-member peer-node standby peer-port 0/5/0
```

```
huawei(protect-group-1)#uplink-monitor port 0/9/1
```

```
huawei(protect-group-1)#quit:
```

Step 4 Configure type B dual-homing protection on the standby OLT huawei.

```
Huawei_OLT(config)#dual-parenting local-node ip-address 10.175.192.36 port 6076 key
protect_4234
Huawei_OLT(config)#dual-parenting peer-node active ip-address 10.175.192.35 port 6076
key work_4234
Huawei_OLT(config)#dual-parenting sync enable
Huawei_OLT(config)#protect-group 1 protect-target gpon-uni-port workmode dual-
parenting
huawei(protect-group-1)#protect-group member port 0/5/0 role protect
huawei(protect-group-1)#peer-group-member peer-node active peer-port 0/5/0
huawei(protect-group-1)#protect-group enable
huawei(protect-group-1)#uplink-monitor port 0/9/1
huawei(protect-group-1)#quit
```

Step 5 To query the information about the working OLT protection group

```
Huawei_OLT(config)#display protect-group 1
{<cr>||<K>}:

```

Command:

```
display protect-group 1
```

```
-----
Group ID: 1
Protect Target: Port of GPON uni
Work Mode: dual-parenting
Description:
Admin State: enable
Operation: none
Reversion: disable
Reversion Time(s): 720
Anti-jitter Time(s): 720
Monitor uplink: port 0/9/0
Uplink status: normal
Standard type B: no
Peer node name: standby
Peer port: 0/5/0
Handshake state: normal
```

```
-----
Member Role Operation State PeerMember
```

```
-----
0/5/0 work none active none
```

Standby Member Status: ready

Step 6 Query the information about the dual-homing peer node.

huawei#**display dual-parenting peer-node**

{<cr>|node-name<S><Length 1-15>}:

Command:

display dual-parenting peer-node

Node name IP address TCP port State

standby 10.175.192.36 6076 normal

Step 7 Disconnect the backbone optical fiber of the working OLT and query the protection group information of the working OLT.

Huawei_OLT(config)#**display protect-group 1**

{<cr>||<K>}:

Command:

display protect-group 1

Group ID: 1

Protect Target: Port of GPON uni

Work Mode: dual-parenting

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Anti-jitter Time(s): 720

Monitor uplink: port 0/9/0

Uplink status: normal

Standard type B: no

Peer node name: standby

Peer port: 0/5/0

Handshake state: normal

Member Role Operation State PeerMember

0/5/0 work auto switch src standby none

Standby Member Status: ready

Step 8 Query information about the protection group on the standby OLT.

Huawei_OLT(config)#**display protect-group 1**

{<cr>||<K>}:

Command:

display protect-group 1

Group ID: 1

Protect Target: Port of GPON uni

Work Mode: dual-parenting

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Anti-jitter Time(s): 720

Monitor uplink: port 0/9/0

Uplink status: normal

Standard type B: no

Peer node name: active

Peer port: 0/5/0

Handshake state: normal

Member Role Operation State PeerMember

0/5/0 protect auto switch src active none

Standby Member Status: ready

Step 9 Restore the fiber connection of the working OLT, disconnect the fiber connection of the protection OLT, restore the working status of the working OLT, disconnect the upstream port 0/9/1 of the OLT, and check the status of the protection group on the working OLT.

Huawei_OLT(config)#**display protect-group 1**

{<cr>||<K>}:

Command:

display protect-group 1

Group ID: 1

Protect Target: Port of GPON uni

Work Mode: dual-parenting

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Anti-jitter Time(s): 720

Monitor uplink: port 0/9/1

Uplink status: fault

Standard type B: no

Peer node name: standby

Peer port: 0/5/0

Handshake state: normal

Member Role Operation State PeerMember

0/5/0 work auto switch src standby none

Standby Member Status: ready

Step 10 Delete a protection group.

huawei(protect-group-1)#**undo protect-group member port 0/5/0**

huawei(protect-group-1)#**undo peer-group-member**

Huawei_OLT(config)#**undo protect-group 1**

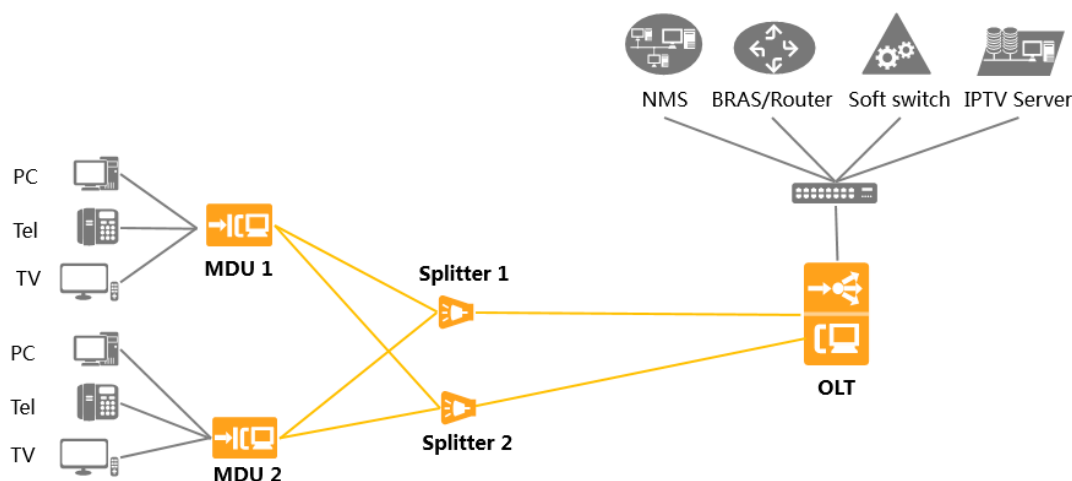
Huawei_OLT(config)#**dual-parenting sync disable**

Huawei_OLT(config)#**undo dual-parenting local-node**

Huawei_OLT(config)#**undo dual-parenting peer-node standby**

14.1.5 Configuring Type C Single-Homing Automatic Protection Switching

14.1.5.1 Experiment Networking

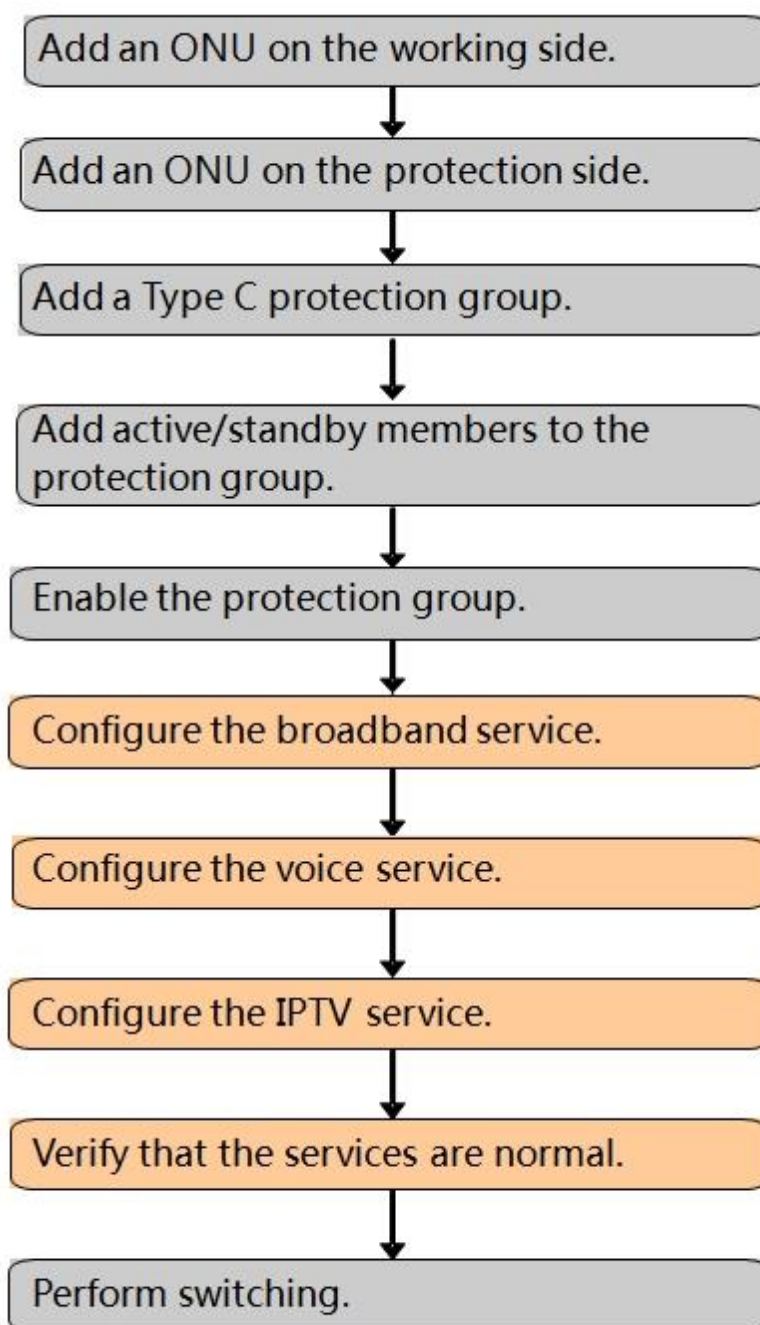


14.1.5.2 Experiment Planning

Item	Data
OLT upstream port	Upstream port: 0/19/0
OLT PON port	Master port:0/5/0 Slaver port:0/5/1

14.1.5.3 Configuration Procedure

After GPON type C single-homing protection is configured, service configurations for ONU access are the same as those before the protection. That is, service configurations are still applied only to the active PON port of the OLT and the active PON upstream port of the ONU.



NOTE For details about the steps in orange, see the Configuration Guide to the GPON Data and Voice Multicast Service.

Step 1 You have logged in to the system as an administrator.

Step 2 Add a working ONU.

```

huawei(config-if-gpon-0/5)#ont add 0 1 sn-auth 48575443F4400D04 snmp ont-
lineprofile-id 23
  
```

```
{<cr>|desc<K>}:

```

Command:

```
ont add 0 1 sn-auth 48575443F4400D04 snmp ont-lineprofile-id 23

```

Number of ONTs that can be added: 1, success: 1

PortID :0, ONTID :1

Step 3 Adding a Protection-Side ONU

```
huawei(config-if-gpon-0/5)#ont add 1 1 protect-side

```

Number of ONTs that can be added: 1, success: 1

PortID :1, ONTID :1

Step 4 Add a GPON type C protection group, add working members to the protection group, and enable the protection group.

```
Huawei_OLT(config)#protect-group protect-target gpon-uni-ont workmode portstate

```

```
huawei(protect-group-0)#protect-group member port 0/5/0 ont 1 role work

```

```
huawei(protect-group-0)#protect-group member port 0/5/1 ont 1 role protect

```

```
huawei(protect-group-0)#protect-group enable

```

Step 5 Configure the ONU service and verify the data. It is found that the voice service is normal.

Step 6 To query the initial status of a protection group

```
Huawei_OLT(config)#display protect-group 0

```

```
-----
Group ID: 0

```

```
Protect Target: ONT of GPON

```

```
Work Mode: portstate

```

```
Description:

```

```
Admin State: enable

```

```
Operation: none

```

```
Reversion: disable

```

```
Reversion Time(s): 720

```

```
-----
Member Role Operation State PeerMember

```

```
-----
0/5/0/1 work none active none

```

```
0/5/1/1 protect none standby none

```

Step 7 Ping the management IP address of the ONU. During this process, remove the optical fiber of the active PON port 0/5/0. The ONU switches to the standby PON port and goes online immediately. The ping test is not interrupted.

```
Huawei_OLT(config)#ping -c 100 10.175.192.41
PING 10.175.192.41:56 data bytes, press CTRL_C to break
Reply from 10.175.192.41: bytes=56 Sequence=1 ttl=255 time=10 ms
Reply from 10.175.192.41: bytes=56 Sequence=2 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=3 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=4 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=5 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=6 ttl=255 time=6 ms
Reply from 10.175.192.41: bytes=56 Sequence=7 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=8 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=9 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=10 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=11 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=12 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=13 ttl=255 time=18 ms
Reply from 10.175.192.41: bytes=56 Sequence=14 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=15 ttl=255 time=7 ms
Reply from 10.175.192.41: bytes=56 Sequence=16 ttl=255 time=13 ms
Reply from 10.175.192.41: bytes=56 Sequence=17 ttl=255 time=7 ms
```

Step 8 To query the status of a protection group

```
Huawei_OLT(config)#display protect-group 0
```

```
-----
Group ID: 0
Protect Target: ONT of GPON
Work Mode: portstate
Description:
Admin State: enable
Operation: none
Reversion: disable
Reversion Time(s): 720
```

```
-----
Member Role Operation State PeerMember
```

```
-----
0/5/0/1 work auto switch src standby 0/5/1/1
```

0/5/1/1 protect auto switch dst active 0/5/0/1

Step 9 Reconnect the optical fiber to port 0/5/0, and then disconnect the optical fiber from the active PON port 0/5/1. The ONU switches to PON port 0/5/0 and goes online. Query the protection group status.

Huawei_OLT(config)#**display protect-group 0**

Group ID: 0

Protect Target: ONT of GPON

Work Mode: portstate

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Member Role Operation State PeerMember

0/5/0/1 work none active none

0/5/1/1 protect none standby none

Step 10 The system supports the forced switching. For example, the ONU is forcibly switched from the active PON port 0/5/0 to 0/5/1. Query the protection group status.

huawei(protect-group-0)#**force-switch port 0/5/0 ont 1 to 0/5/1 ont 1**

huawei(protect-group-0)#**display protect-group 0**

Group ID: 0

Protect Target: ONT of GPON

Work Mode: portstate

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Member Role Operation State PeerMember

```
-----  
0/5/0/1 work force switch src standby 0/5/1/1  
0/5/1/1 protect force switch dst active 0/5/0/1  
-----
```

Step 11 After a forced switchover, the optical fiber connected to the active PON port 0/5/1 cannot be automatically switched, and services are interrupted. Query the status of the 0/5/1 port and the protection group.

```
huawei(config-if-gpon-0/5)#display port state 1  
-----
```

```
F/S/P 0/5/1  
Optical Module status Online  
Port state Offline  
Port energy-saving flag No  
xPON MAC chipset state Normal  
Last down cause LOS  
Last up time 2018-07-10 13:57:15+08:00  
Last down time 2018-07-10 14:18:39+08:00  
Laser state Normal  
Available bandwidth(Kbps) 1241088  
Temperature(C) 44  
TX Bias current(mA) 40  
Supply Voltage(V) 3.39  
TX power(dBm) 3.90  
Illegal rogue ONT Inexistent  
Max rate(Kbps) 2500000  
Max Distance(Km) 20  
Wave length(nm) 1490  
Fiber type Single Mode  
Signal detect Failed  
TX fault Normal  
Identifier SFP  
Ext.Identifier GBIC/SFP function is defined by serial ID only  
Connector SC  
Encoding NRZ  
Length(9μm)(km) 20.0  
.....  
Huawei_OLT(config)#display protect-group 0
```

Group ID: 0
Protect Target: ONT of GPON
Work Mode: portstate
Description:
Admin State: enable
Operation: none
Reversion: disable
Reversion Time(s): 720

Member Role Operation State PeerMember

0/5/0/1 work force switch src standby 0/5/1/1
0/5/1/1 protect force switch dst active 0/5/0/1

Step 12 Cancel the forced switching and query the protection group status.

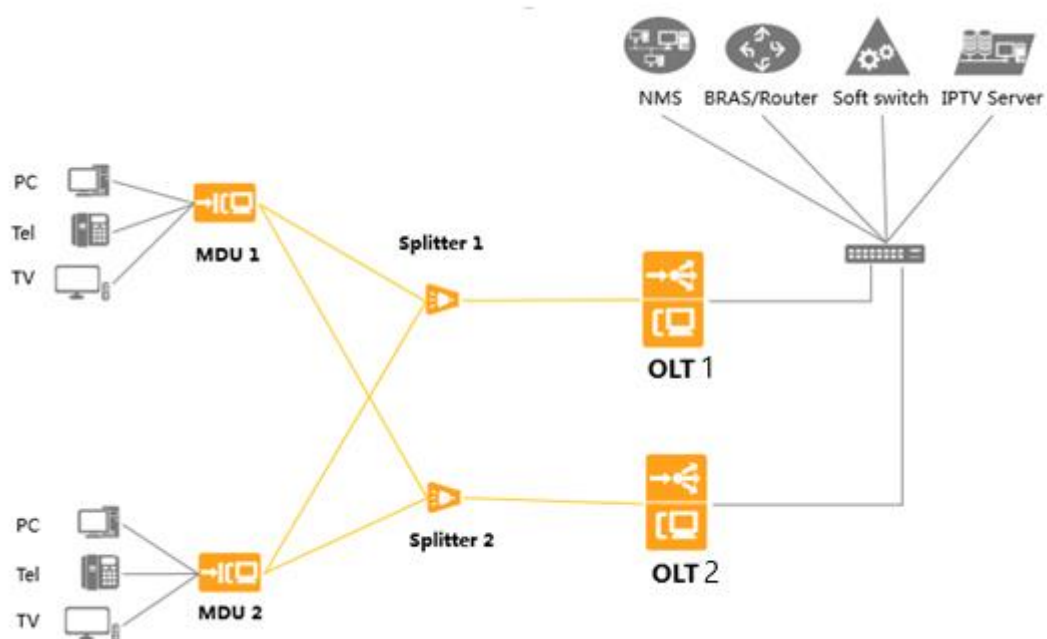
```
huawei(protect-group-0)#undo force-switch port 0/5/0 ont 1 to 0/5/1 ont 1
```

Step 13 Deleting a PON Protection Pair

```
huawei(protect-group-0)#protect-group disable  
huawei(protect-group-0)#undo protect-group member port 0/5/1 ont 1  
huawei(protect-group-0)#undo protect-group member port 0/5/0 ont 1  
huawei(protect-group-0)#quit  
Huawei_OLT(config)#undo protect-group 0
```

14.1.6 Configuring Type C Dual-Homing Automatic Switching Protection

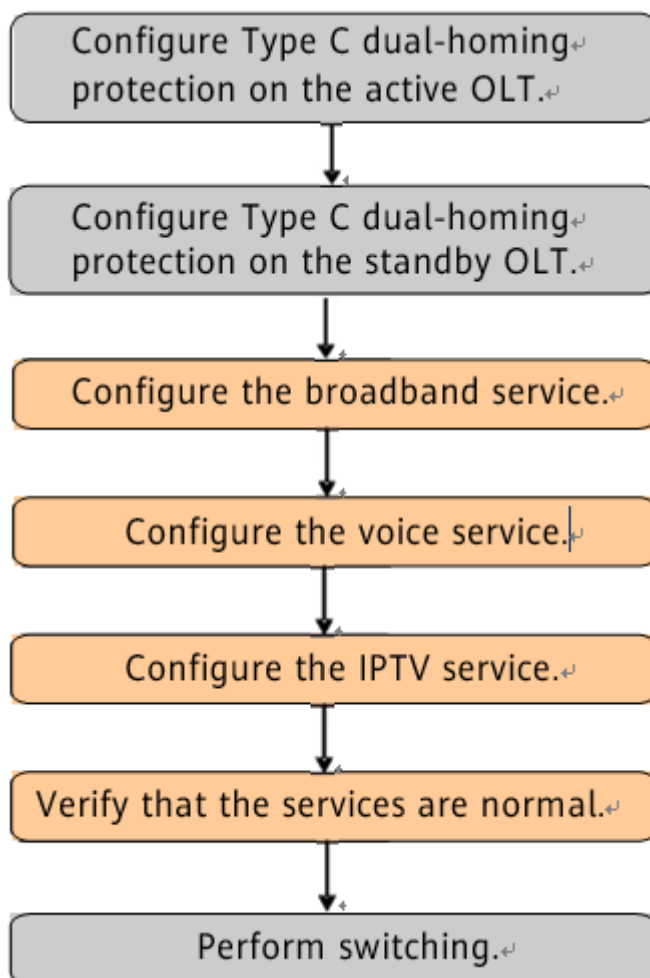
14.1.6.1 Experiment Networking



14.1.6.2 Experiment Planning

Item	Data
IP addresses of the active and standby OLTs	IP address of the active OLT: 10.175.192.35/22 Standby OLT IP address: 10.175.192.36/22
Active and standby OLT PON ports	0/5/0

14.1.6.3 Configuration Procedure



Step 1 You have logged in to the system as an administrator.

Step 2 Configure GPON type C dual-homing protection on the active OLT.

```

huawei(config-if-gpon-0/5)#ont add 0 1 sn-auth 48575443F4400D04 snmp ont-lineprofile-id 23

```

```

{<cr>|desc<K>}:

```

Command:

```

ont add 0 1 sn-auth 48575443F4400D04 snmp ont-lineprofile-id 23

```

Number of ONTs that can be added: 1, success: 1

PortID :0, ONTID :1

```

Huawei_OLT(config)#protect-group 1 protect-target gpon-uni-ont workmode dual-parenting

```

```

huawei(protect-group-1)#protect-group member port 0/5/0 ont 1 role work

```

After a working member is added to a dual-homing protection group, the protection group is automatically enabled.

Step 3 Configure GPON type C dual-homing protection on the standby OLT.

```
huawei(config-if-gpon-0/5)#ont add 0 1 sn-auth 48575443F4400D04 snmp ont-  
lineprofile-id 23  
{<cr>|desc<K>}:  
Command:
```

Command:

```
ont add 0 1 sn-auth 48575443F4400D04 snmp ont-lineprofile-id 23
```

Number of ONTs that can be added: 1, success: 1

PortID :0, ONTID :1

```
Huawei_OLT(config)#protect-group 1 protect-target gpon-uni-ont workmode dual-  
parenting
```

```
huawei(protect-group-1)#protect-group member port 0/5/0 ont 1 role protect
```

```
huawei(protect-group-1)#protect-group enable
```

All profiles (such as the DBA profile and line profile) of the active and standby ONUs must be the same.

Step 4 Configure the ONU service and verify the data. It is found that the voice service is normal.

Step 5 Query the initial status of the protection group on the active OLT.

```
Huawei_OLT(config)#display protect-group 1
```

Group ID: 1

Protect Target: ONT of GPON

Work Mode: dual-parenting

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Monitor uplink: -

Uplink status: -

Member Role Operation State PeerMember

0/5/0/1 work none active none

Step 6 The standby OLT is queried as follows:

```
Huawei_OLT(config)#display protect-group 1
```

Group ID: 1
Protect Target: ONT of GPON
Work Mode: dual-parenting
Description:
Admin State: enable
Operation: none
Reversion: disable
Reversion Time(s): 720
Monitor uplink: -
Uplink status: -

Member Role Operation State PeerMember

0/5/0/1 protect none standby none

Step 7 Remove the optical fiber from PON port 0/5/0 of the active OLT. The ONU switches to the standby OLT immediately. Check the status of the protection group on the active OLT after the switchover. The following figure shows the query result on the active OLT.

Huawei_OLT(config)#**display protect-group 1**

Group ID: 1
Protect Target: ONT of GPON
Work Mode: dual-parenting
Description:
Admin State: enable
Operation: none
Reversion: disable
Reversion Time(s): 720
Monitor uplink: -
Uplink status: -

Member Role Operation State PeerMember

0/5/0/1 work auto switch src standby none

Step 8 The standby OLT is queried as follows:

```
Huawei_OLT(config)#display protect-group 1
```

```
-----  
Group ID: 1  
Protect Target: ONT of GPON  
Work Mode: dual-parenting  
Description:  
Admin State: enable  
Operation: none  
Reversion: disable  
Reversion Time(s): 720  
Monitor uplink: -  
Uplink status: -  
-----
```

```
Member Role Operation State PeerMember
```

```
-----  
0/5/0/1 protect auto switch dst active none  
-----
```

Step 9 Reconnect the optical fiber of the active OLT 0/5/0, and then remove the optical fiber of the active PON port on the standby OLT 0/5/0. The ONU switches to the active OLT and goes online immediately. Query the protection group status of the standby OLT.

```
Huawei_OLT(config)#display protect-group 1
```

```
-----  
Group ID: 1  
Protect Target: ONT of GPON  
Work Mode: dual-parenting  
Description:  
Admin State: enable  
Operation: none  
Reversion: disable  
Reversion Time(s): 720  
Monitor uplink: -  
Uplink status: -  
-----
```

```
Member Role Operation State PeerMember
```

```
0/5/0/1 protect auto switch src standby none
```

Step 10 Delete the active and standby OLT type C protection configurations.

Active OLT:

```
huawei(protect-group-1)#undo protect-group member port 0/5/0 ont 1
```

```
Huawei_OLT(config)#undo protect-group 1
```

Standby OLT:

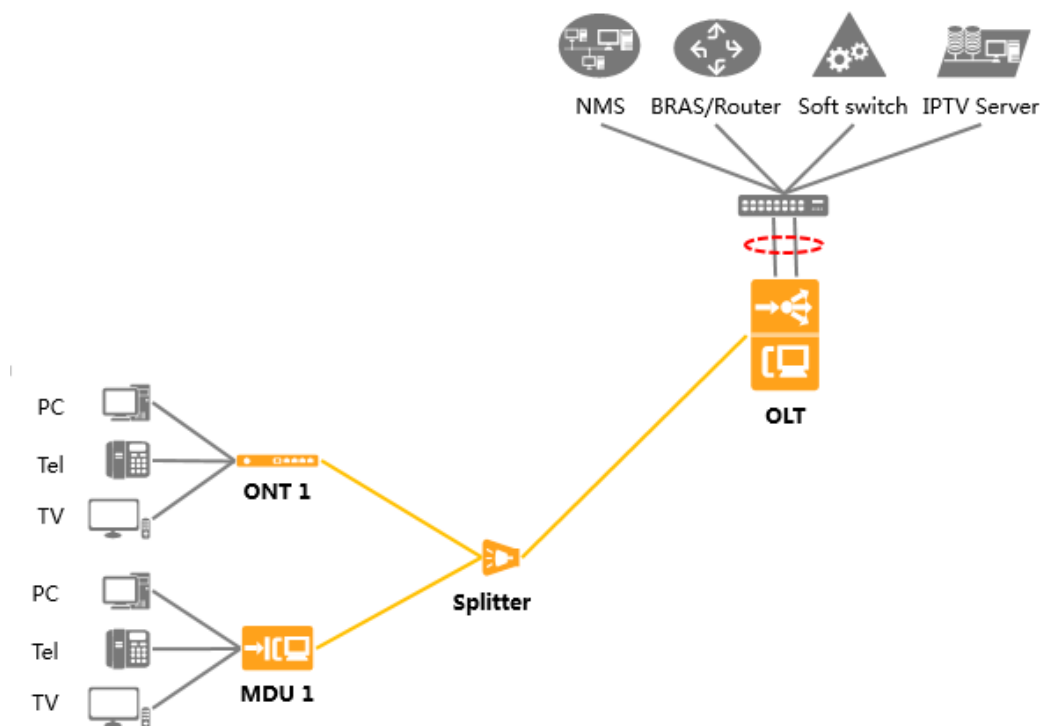
```
huawei(protect-group-1)# protect-group disable
```

```
huawei(protect-group-1)#undo protect-group member port 0/5/0 ont 1
```

```
Huawei_OLT(config)#undo protect-group 1
```

14.1.7 Configuring Ethernet Link Aggregation

14.1.7.1 Networking Topology



14.1.7.2 Experiment Planning

OLT upstream port	L3 Switch Interface	VLAN Planning
0/19/0	0/0/3	Data: 2000
0/20/0	0/0/4	Voice: 172 Video: 200

14.1.7.3 Configuration Procedure

Step 1 Create an inter-board Ethernet aggregation group. The OLT uses the GIU board for upstream transmission. Upstream ports 0/19/0 and 0/20/0 on the active and standby GIU boards are configured as an inter-board aggregation group. Each member port in the aggregation group distributes packets based on the source and destination MAC addresses and works in LACP static aggregation mode.

```
huawei (config)#link-aggregation 0/19 0 0/20 0 egress-ingress workmode lacp-static
```

Step 2 Query the configuration.

```
Huawei_OLT(config)#display link-aggregation all
```

```
-----
```

```
Master port Link aggregation mode Port NUM Work mode Max link number
```

```
-----
```

```
0/19/0 egress-ingress 2 lacp-static -
```

```
-----
```

```
Total: 1 link aggregation(s)
```

Step 3 Queries the configuration of an LACP static aggregation group.

```
Huawei_OLT(config)#display lacp link-aggregation summary
```

```
Actor
```

```
Priority: 32768 MAC Address: 707b-e800-304d
```

```
Short Period: 1 s Long Period: 30 s
```

```
Priority Preemption: Disable
```

```
-----
```

```
Agg Partner Partner Select Standby Load Backup Master
```

```
ID Pri MAC Ports Ports Sharing Protect Port
```

```
-----
```

```
1 32768 0819-a6ec-c4ed 2 0 YES NO 0 /19/0
```

```
-----
```

```
Huawei_OLT(config)#display lacp link-aggregation verbose
```

{INTERGER<1-512>}:1

Command:

display lacp link-aggregation verbose 1

System Priority: 32768

MAC Address: 707b-e800-304d

Protect Mode: Non-BackupProtect

Master Port: 0/19/0

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/0

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Step 4 Connects to the L3 switch. The following uses Huawei S3300 as an example to describe how to configure data.

```
//Add VLANs that are allowed to pass the LAG to the LAG.
```

```
//<Quidway>system-view
```

```
//[Quidway]vlan 2000 172 200
```

```
//[Quidway]interface Eth-Trunk 1
```

```
//[Quidway-Eth-Trunk1]port link-type trunk
```

```
//[Quidway-Eth-Trunk1]port trunk allow-pass vlan 2000 172 200
```

```
//Add Eth-trunk interfaces GE0/0/3 and GE0/0/4.
```

```
//[Quidway]interface eth-trunk 1
```

```
//[Quidway-Eth-Trunk1]bpdu enable
```

```
//[Quidway-Eth-Trunk1]mode lacp-static
```

```
//[Quidway]interface GigabitEthernet 0/0/3
```

```
//[Quidway-GigabitEthernet0/0/3]eth-trunk 1
```

```
//[Quidway-GigabitEthernet0/0/3]quit
```

```
//[Quidway]interface GigabitEthernet 0/0/4
```

```
//[Quidway-GigabitEthernet0/0/4]eth-trunk 1
```

```
//[Quidway-GigabitEthernet0/0/4]quit
```

//Configure a trunk interface on the upstream Layer 3 interface to transparently transmit the VLAN tags and send the VLAN tags to the upper-layer network.

//Add GE0/0/1 to Eth-trunk 1.

//[Quidway-GigabitEthernet0/0/1]eth-trunk 1 0

Step 5 Queries the configuration of an LACP static aggregation group.

Huawei_OLT(config)# **display lacp link-aggregation port 0/19/0**

Port Number: 1

Selected AggID: 1

System ID: 32768, 707b-e800-304d

Received LACPDU: 301

Illegal LACPDU: 0

Sent LACPDU: 38

Actor

Port-priority: 16384

Oper-key: 1

Flag: 3F

Type: Master port

Role: SELECTED

Status: C&D

Partner

System ID: 32768, 0819-a6ec-c4ed

Port Number: 3

Port-priority: 32768

Oper-key: 305

Flag: 3D

huawei (config)# display lacp link-aggregation port 0/20/0

Port Number: 2

Selected AggID: 1

System ID: 32768, 707b-e800-304d

Received LACPDU: 312

Illegal LACPDU: 0

Sent LACPDU: 17

Actor

Port-priority: 16384

Oper-key: 1
Flag: 3F
Type: Sub port
Role: SELECTED
Status: C&D
Partner
System ID: 32768, 0819-a6ec-c4ed
Port Number: 4
Port-priority: 32768
Oper-key: 305
Flag: 3D

Step 6 To add an upstream port to a VLAN, you only need to add a member port to the aggregation group. Example:

```
Huawei_OLT(config)#vlan 200  
{<cr>|to<K>|vlantype<E><mux,standard,smart,super>}:smart  
command:  
vlan 200 smart
```

```
MA5600T(config)#port vlan 200 0/19 0  
The port list includes link aggregation or protect group port(s),  
so the other link aggregation or protect group port(s)  
not being included by the port list will be processed  
Query VLAN 20.  
MA5600T(config)#display vlan 20  
{<cr>|to<K>}:  
  
Command:  
display vlan 200  
VLAN ID: 200  
VLAN type: smart  
VLAN attribute: common  
VLAN description:  
VLAN forwarding mode: VLAN-MAC  
VLAN broadcast packet forwarding policy: forward  
VLAN unknown multicast packet forwarding policy: forward  
VLAN unknown unicast packet forwarding policy: forward
```

VLAN bind service profile ID: -
VLAN bind RAIO profile index: -
VLAN priority: -

F/S /P Native VLAN State

0/19/0 1 up

0/20/0 1 up

Standard port number: 2

Service virtual port number: 0

Step 7 Port mirroring is used to capture LACP packets. Mirror the traffic on port 0/19/0 to port 0/19/1.

Huawei_OLT(config)#**interface giu 0/19**

MA5600T(config-if-giu-0/19)#mirror port

{src-portid<U><0,1>}:0

{dst-portid<U><0,1>}:1

{mirror-direction<E><ingress,egress,all>}:all

Command:

mirror port 0 1 all

MA5600T(config-if-giu-0/19)#display mirror

Source port Direction Destination port

(F/ S/ P) (F/ S/ P)

0/19/ 0 all 0/19/ 1

Format and content of LACPDU

Destination MAC address: 0x0180-c200-0002,

Source MAC address,

Protocol type: The value is 0x8809, and the subtype value is 0x01, indicating LACP packets,

Version: The current version is 0x01,

Actor information: carries information about the local system and port, such as the system ID, port priority, and key,

Partner information: includes the information about the peer system that is currently stored in the local system. Other information is reserved.

The captured packets are as follows:

No.	Time	Source	Destination	Protocol	Info
1	0.000000	70:7b:e8:00:30:52	Broadcast	ARP	who has 17.1.1.90? Gratuitous ARP
2	0.000000	08:19:a6:ec:c4:ed	Slow-Protocols	LACP	Link Aggregation Control ProtocolVersion 1. Actor Port = 3
3	1.284422	08:19:a6:ec:c4:ed	Slow-Protocols	LACP	Link Aggregation Control ProtocolVersion 1. Actor Port = 3
4	2.289234	08:19:a6:ec:c4:ed	Slow-Protocols	LACP	Link Aggregation Control ProtocolVersion 1. Actor Port = 3
5	3.317682	08:19:a6:ec:c4:ed	Slow-Protocols	LACP	Link Aggregation Control ProtocolVersion 1. Actor Port = 3

Frame 2 (124 bytes on wire, 124 bytes captured)					
Ethernet II, Src: 08:19:a6:ec:c4:ed (08:19:a6:ec:c4:ed), Dst: Slow-Protocols (01:80:c2:00:00:02)					
Destination: Slow-Protocols (01:80:c2:00:00:02)					
Source: 08:19:a6:ec:c4:ed (08:19:a6:ec:c4:ed)					
Type: Slow Protocols (0x8809)					
Link Aggregation Control Protocol					
Slow Protocols subtype: LACP (0x01)					
LACP Version Number: 0x01					
Actor Information: 0x01					
Actor Information Length: 0x14					
Actor System Priority: 32768					
Actor System: 08:19:a6:ec:c4:ed (08:19:a6:ec:c4:ed)					
Actor Key: 305					
Actor Port Priority: 32768					
Actor Port: 3					
Actor State: 0x3d (Activity, Aggregation, Synchronization, collecting, Distributing)					
Reserved: 000000					
Partner Information: 0x02					
Partner Information Length: 0x14					
Partner System Priority: 32768					
Partner System: 70:7b:e8:00:30:4d (70:7b:e8:00:30:4d)					
Partner Key: 1					
Partner Port Priority: 16384					
Partner Port: 1					
Partner State: 0x3f (Activity, Timeout, Aggregation, Synchronization, Collecting, Distributing)					
Reserved: 000000					
Collector Information: 0x02					

Step 8 Run the link-aggregation add-member command to add upstream port 0/20/1 to the existing aggregation group to further improve the bandwidth or reliability of the existing aggregation group. After a port is added to an aggregation group, the bandwidth of the aggregation group is increased and the link reliability is improved.

MA5600T(config)#link-aggregation add-member

Shelf/slot/port of the master port in the {frame/slot/port<S><Length 5-7>}:0/19/0 // aggregation group

{frame/slot<S><Length 3-4>}:0/20 //: subrack ID and slot ID of the port to be added to the aggregation group

{port-list<S><Length 1-255>}:1

{<cr>|frame/slot<S><Length 3-4>}:<cr>

Command:

link-aggregation add-member 0/19/0 0/20 1

Query the status of the added port 0/20/1.

MA5600T(config)#display lacp link-aggregation verbose 1

System Priority: 32768

MAC Address: 707b-e800-304d

Protect Mode: Non-BackupProtect

Master Port: 0/19/0

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/0

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/1

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Run the following command to delete a member port:

Huawei_OLT(config)#link-aggregation delete-member 0/19/0 0/20 1

- Step 9** Run the link-aggregation max-link-number command to configure the maximum number of links in an Ethernet port aggregation group. The maximum number of links in an aggregation group is the maximum number of ports that can forward packets in the aggregation group. This implements the active/standby protection of the links in the LAG. If the port for forwarding packets in the aggregation group is faulty, the standby port is selected to forward packets.

Huawei_OLT(config)#**link-aggregation max-link-number**

{frame/slot/port<S><Length 5-7>}:**0/19/0**

{link-number<U><1,8>}:**2**

Command:

link-aggregation max-link-number 0/19/0 2

Query the status of the aggregation group.

MA5600T(config)#display lacp link-aggregation verbose 1

System Priority: 32768

MAC Address: 707b-e800-304d

Protect Mode: Non-BackupProtect

Master Port: 0/19/0

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/1

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/0

Actor Port Priority: 16384 Oper-key: 1

Port Role: STANDBY Port Status: Non-C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

To query the summary of LACP static aggregation, run the following command:

Huawei_OLT(config)#display lacp link-aggregation summary

Actor

Priority: 32768 MAC Address: 707b-e800-304d

Short Period: 1 s Long Period: 30 s

Priority Preemption: Disable

```
-----
Agg Partner Partner Select Standby Load Backup Master
ID Pri MAC Ports Ports Sharing Protect Port
-----
```

```
1 32768 0819-a6ec-c4ed 2 1 YES NO 0 /19/0
-----
```

Run the undo link-aggregation max-link-number command to restore the maximum number of links in an aggregation group to the default value. That is, all the ports in the aggregation group can forward packets for protection. The active/standby protection function of the links in the aggregation group is disabled.

Step 10 Working principle of LACP:

The OLT and Layer 3 switch send protocol packets to each other.

The devices at both ends determine the Actor based on the system priority.

Finally, the active interfaces on both ends are determined according to the upper threshold of active interfaces and the maximum number of active interfaces on the Actor.

If an active link fails, the Actor selects the link with the highest priority from the N backup links to take over the services.

Step 11 Determine the active end, active interface, and standby interface between the OLT and the interconnected Layer 3 device based on the LACP working principle.

Step 12 Verify the LAG function based on step 8.

Remove the network cable from network port 0/19/0 and check the LAG status.

huawei (config)#**display lacp link-aggregation verbose 1**

System Priority: 32768

MAC Address: 707b-e800-304d

Protect Mode: Non-BackupProtect

Master Port: 0/19/0

Actor Port Priority: 16384 Oper-key: 0

Port Role: STANDBY Port Status: Non-C&D

Sub Port: 0/20/1

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/0

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Huawei_OLT(config)# display lacp link-aggregation summary

Actor

Priority: 32768 MAC Address: 707b-e800-304d

Short Period: 1 s Long Period: 30 s

Priority Preemption: Disable

Agg	Partner	Partner Select	Standby	Load	Backup	Master
ID	Pri	MAC	Ports	Ports	Sharing	Protect Port
1	32768	0819-a6ec-c4ed	2	1	YES	NO 0 /19/0

Step 13 Reconnect the network cable to network port 0/19/0 and check the LAG status.
Remove the network cable from network port 0/20/0 and check the LAG status.

Huawei_OLT(config)#display lacp link-aggregation verbose 1

System Priority: 32768

MAC Address: 707b-e800-304d

Protect Mode: Non-BackupProtect

Master Port: 0/19/0

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/1

Actor Port Priority: 16384 Oper-key: 1

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/0

Actor Port Priority: 16384 Oper-key: 0

Port Role: STANDBY Port Status: Non-C&D

Huawei_OLT(config)#display lacp link-aggregation summary

Actor

Priority: 32768 MAC Address: 707b-e800-304d

Short Period: 1 s Long Period: 30 s

Priority Preemption: Disable

Agg Partner Partner Select Standby Load Backup Master

ID Pri MAC Ports Ports Sharing Protect Port

1 32768 0819-a6ec-c4ed 2 1 YES NO 0 /19/0

- Step 14** Reconnect the network cable to network port 0/20/0 and check the LAG status.
Remove the network cable from network port 0/20/1 and check the LAG status.

Huawei_OLT(config)#**display lacp link-aggregation summary**

Actor

Priority: 32768 MAC Address: 707b-e800-304d

Short Period: 1 s Long Period: 30 s

Priority Preemption: Disable

```
-----
Agg Partner Partner Select Standby Load Backup Master
ID Pri MAC Ports Ports Sharing Protect Port
-----
```

```
1 32768 0819-a6ec-c4ed 2 1 YES NO 0 /19/0
-----
```

```
Huawei_OLT(config)#display lacp link-aggregation verbose 1
System Priority: 32768
MAC Address: 707b-e800-304d
Protect Mode: Non-BackupProtect
```

```
Master Port: 0/19/0
Actor Port Priority: 16384 Oper-key: 1
Port Role: SELECTED Port Status: C&D
Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed
Port Priority: 32768 Oper-key: 305
```

```
Sub Port: 0/20/0
Actor Port Priority: 16384 Oper-key: 1
Port Role: SELECTED Port Status: C&D
Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed
Port Priority: 32768 Oper-key: 305
```

```
Sub Port: 0/20/1
Actor Port Priority: 16384 Oper-key: 0
Port Role: STANDBY Port Status: Non-C&D
```

Step 15 Reconnect the network cable to network port 0/20/1 and check the LAG status.
Remove the network cables from any two network ports on 0/19/0 0/20/0 0/20/1,
and check the status.

```
Huawei_OLT(config)#display lacp link-aggregation verbose 1
System Priority: 32768
MAC Address: 707b-e800-304d
Protect Mode: Non-BackupProtect
```

```
Master Port: 0/19/0
Actor Port Priority: 16384 Oper-key: 1
```

Port Role: SELECTED Port Status: C&D

Partner System Priority: 32768 MAC Address: 0819-a6ec-c4ed

Port Priority: 32768 Oper-key: 305

Sub Port: 0/20/0

Actor Port Priority: 16384 Oper-key: 0

Port Role: STANDBY Port Status: Non-C&D

Sub Port: 0/20/1

Actor Port Priority: 16384 Oper-key: 0

Port Role: STANDBY Port Status: Non-C&D

Huawei_OLT(config)#display lacp link-aggregation summary

Actor

Priority: 32768 MAC Address: 707b-e800-304d

Short Period: 1 s Long Period: 30 s

Priority Preemption: Disable

```
-----
Agg Partner Partner Select Standby Load Backup Master
ID Pri MAC Ports Ports Sharing Protect Port
-----
```

```
1 32768 0819-a6ec-c4ed 1 2 YES NO 0 /19/0
-----
```

Step 16 To verify the impact on services when the network cable is removed from a network port in a LAG.

Perform the following operations in sequence:

Remove the network cable from port 0/19/0 and then connect the network cable to port 0/19/0.

Remove the network cable from port 0/20/0.

Remove the network cable from port 0/21/0.

Remove the network cables 0/20/0 and 0/21/0 at the same time, and then the fault is rectified.

Remove the network cables from ports 0/19/0 and 0/20/0 at the same time.

Perform a ping test to check the impact on services.

Step 17 Create an Ethernet aggregation group - inter-board aggregation. Example:

The MA5600T/MA5603T uses the GIU board for upstream transmission. Upstream ports 0/19/0 and 0/20/0 on the active and standby GIU boards are configured as an inter-board

aggregation group. Each member port in the aggregation group distributes packets based on the source and destination MAC addresses and works in LACP static aggregation mode.

```
huawei (config)#link-aggregation 0/19 0 0/20 0 egress-ingress workmode lacp-static
```

Step 18 Create an aggregation group (with the control board and upstream board). For example:

When the MA5600T/MA5603T is configured with only one SCUN control board, the control board and the GIU slot are configured as an inter-board aggregation group. The member ports in the aggregation group distribute packets according to the source MAC address and work in LACP static aggregation mode.

```
Huawei_OLT(config)#link-aggregation 0/9 0-3 0/20 0-1 ingress workmode lacp-static
```

```
----End
```

Verification

(1) Because the LAG works in load sharing mode, the link is set up in egress-ingress mode according to the packet distribution mode of the LAG in step 1. The member ports in the LAG select a link to distribute packets according to the source MAC address, destination MAC address, and internal algorithm of the system. Therefore, the link selection is unique. If there are N active links in a LAG, the impact of link interruption on services is 1/N. In this case, some data flows forwarded on the link may be lost.

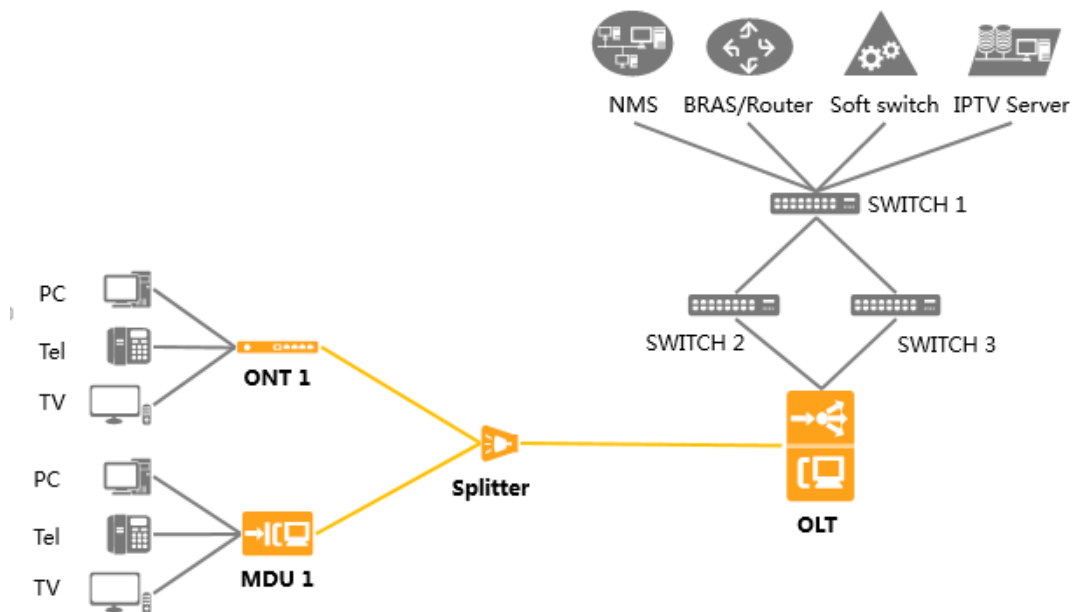
(2) Services are not interrupted when any link is removed during a voice service call, but the service quality is not affected. If the data service is not obviously perceived, perform the ping test and obtain the packet header on the port for analysis.

Ping test result: Each time the network cable is removed and inserted, packet loss may occur.

14.1.8 Configuring an Ethernet Port Protection Group

14.1.8.1 Experiment Networking

Networking structure description: When the active uplink works normally, the connects to the active PE, and the standby uplink is shut down. When the uplink of the interface connected to the active PE is faulty, the standby uplink is enabled immediately and services are switched to the standby uplink. This ensures the backup of the uplink. In the experiment, the dual GIU upstream networking is used to describe the upstream protection group function.



14.1.8.2 Experiment Planning

OLT upstream port	L3 Switch Interface	VLAN Planning
0/19/0	0/0/3	Data: 2000
0/20/0	0/0/4	Voice: 172 Video: 200

14.1.8.3 Configuration Procedure

- Step 1 Add a terminal to the OLT according to the data plan, and enable the voice and data services.
- Step 2 The service ports of the active and standby switches transparently transmit service VLANs and management VLANs.
- Step 3 Create a protection group and select the protection between upstream Ethernet ports.

```

Huawei_OLT(config)#protect-group
{pgid<U><0,63>|protect-target<K>}:1
{<cr>|protect-target<K>}:protect-target
{as-mainboard-lag<K>|as-mainboard-port<K>|as-mainboard<K>|epon-uni-ont<K>|epon-uni-port<K>|eth-nni-lag-member<K>|eth-nni-port<K>|gpon-uni-port<K>|service-process-board<K>|stm-nni-port<K>}:eth-nni-port
{workmode<K>}:workmode
{smart-link<K>|timedelay<K>}:timedelay
  
```

Command:

protect-group 1 protect-target eth-nni-port workmode timedelay

- Step 4** Add member ports to the protection group. The OLT uses the GICG board for upstream transmission. Configure upstream ports 0/19/0 and 0/20/0 on the active and standby GIU boards as a protection group.

```
huawei(protect-group-1)#protect-group member
{board<K>|port<K>}:port
{frameid/slotid/portid<S><Length 1-15>}:0/19/0
{ont<K>|role<K>}:role
{member-role<E><work,protect>}:work
```

command:

protect-group member port 0/19/0 role work

```
huawei(protect-group-1)#protect-group member port 0/20/0
{ont<K>|role<K>}:role
{member-role<E><work,protect>}:protect
```

commmand:

protect-group member port 0/20/0 role protect

- Step 5** Enable a protection group. After a protection group is created, you need to make the protection group take effect. After this command is run, the protection switching function can be implemented for the members in the protection group.

```
huawei(protect-group-1)#protect-group enable
```

- Step 6** The two links between the directly connected switches are configured as trunk links to transparently transmit packets of all VLANs.

- Step 7** Check the protection group status.

```
huawei(protect-group-1)#display protect-group 1
{<cr>|pgid<U><0,63>}:1
```

Group ID: 1

Protect Target: Port of Ethernet nni

Work Mode: timedelay

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Member Role Operation State PeerMember

0/19/0 work none active none

0/20/0 protect none standby none

Step 8 Query the VLAN status. After the protection group is created, the VLAN is added to the member port of the protection group.

Huawei_OLT(config)#**display vlan 2000**

{<cr>|to<K>}:

Command:

display vlan 2000

VLAN ID: 2000

VLAN type: smart

VLAN attribute: QinQ

VLAN description:

VLAN forwarding mode: VLAN-MAC

VLAN broadcast packet forwarding policy: forward

VLAN unknown multicast packet forwarding policy: forward

VLAN unknown unicast packet forwarding policy: forward

VLAN bind service profile ID: -

VLAN bind RAIO profile index: -

VLAN priority: -

F/S /P Native VLAN State

0/19/0 1 up

0/20/0 1 down

Standard port number: 2

INDEX TYPE STATE F/ S/ P VPI VCI FLOWTYPE FLOWPARA

0 gpon up 0/ 5/ 0 1 1 vlan 2

Service virtual port number: 1

Note: F--Frame, S--Slot, P--Port,

VPI indicates ONT ID for PON, VCI indicates GEM index for GPON,

v/e--vlan/encap, pri-tag--priority-tagged

Step 9 Remove the network cable from the active member port 0/19/0 and check the status of the protection group.

huawei(protect-group-1)#**display protect-group 1**

Group ID: 1

Protect Target: Port of Ethernet nni

Work Mode: timedelay

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Member Role Operation State PeerMember

0/19/0 work auto switch src standby 0/20/0

0/20/0 protect auto switch dst active 0/19/0

Step 10 Reconnect the network cable to member port 0/19/0 and check the status of the protection group.

huawei(protect-group-1)#**display protect-group 1**

Group ID: 1

Protect Target: Port of Ethernet nni

Work Mode: timedelay

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Member Role Operation State PeerMember

0/19/0 work auto switch src standby 0/20/0

0/20/0 protect auto switch dst active 0/19/0

Step 11 Remove the network cable from member port 0/20/0 and check the protection group status.

huawei(protect-group-1)#**display protect-group 1**

Group ID: 1

Protect Target: Port of Ethernet nni

Work Mode: timedelay

Description:

Admin State: enable

Operation: none

Reversion: disable

Reversion Time(s): 720

Member Role Operation State PeerMember

0/19/0 work none active none

0/20/0 protect none standby none

----End

Verification

During a voice call, services are not interrupted when the network cable connected to the active member port is removed, but users are aware of the service interruption. If the data service is not obviously perceived, perform the ping test and obtain the packet header on the port for analysis.

Ping test result: Packet loss occurs every time the network cable of the active member port is removed and inserted.

